Spring 2008

ISA Books and Standards

Your unbiased source for technical information from industry experts



- Books and Standards from Leading Automation Experts
- Your Trusted Source for Applications Solutions
- Find the Latest Technical Papers, CDs, and Training

Setting the Standard for Automation™

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

Table of Contents

ISA—your unbiased source for technical information from industry experts

ISA Books

- Basic Continuous Control 4
- Basic Discrete, Sequencing, and Manufacturing Control 28
- Advanced Control 40
- Reliability, Safety, and Electrical 48
- Integration and Software....56
- Deployment and Maintenance 64
- Work Structure....72

ISA Standards and User Resources....80

Order Information and Form....90

Index....92



© iStock International Inc. / PeterClose

Publishing

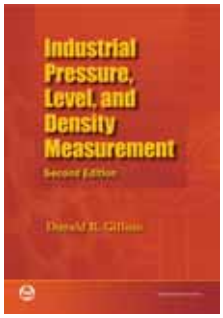
ISA is the authoritative publisher of books, magazines, and technical journals covering the automation profession. Written and reviewed by experts, these publications help keep automation professionals fully informed about the latest technical developments, applications, trends, and standards; and ISA magazines feature the latest industry and business news impacting their career.

ISA resources can

As an automation professional, you need the technical know-how to do your job well. ISA resources offer authoritative and thorough coverage of the world of automation. They are written and reviewed by experts in their fields. The resources are organized by the following technical categories defining automation.

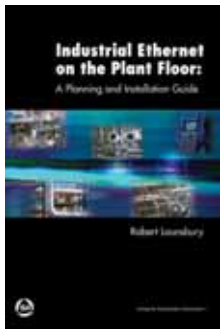
1 Basic Continuous Control

If you're looking for a fundamental approach and overview, the *Basic Continuous Control* technical area includes easy-to-understand resources on analytical instrumentation, process instrumentation, continuous control, control valves, analog communications, documentation, and control equipment. This category begins on page 4.



2 Basic Discrete, Sequencing and Manufacturing Control

As industry needs and technologies merge, the resources found in the *Basic Discrete, Sequencing and Manufacturing Control* technical area will help you understand this ever-changing arena and provide valuable insight into the function and application of discrete input and output devices, discrete and sequencing control, as well as motor and drive control. This category begins on page 28.



help you on your path to success!

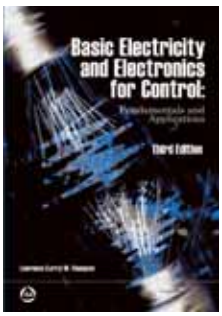
3 Advanced Control

Apply and expand your existing technical knowledge of process modeling, batch control, building automation, environmental issues, and the like using the resources listed in the *Advanced Control* technical area. This category begins on page 40.



4 Reliability, Safety and Electrical

Concerned about plant and personnel safety? The *Reliability, Safety, and Electrical* technical area contains resources that are essential if you are working in the areas of alarm management, process safety, electrical installations and electrical safety. This category begins on page 48.



5 Integration and Software

The *Integration and Software* technical area provides easy-to-read fundamental resources on digital communications, industrial networks, MES integration, network security, software, data management, and operator interfaces. This category begins on page 56.



6 Deployment and Maintenance

The resources found in the *Deployment and Maintenance* technical area will help you understand the work involved in operator training, checkout, system testing and startup, troubleshooting, long-term support and system management. This category begins on page 64.



7 Work Structure

Need to identify the scope of a project and justify its costs? The *Work Structure* technical area will help get you started with topics such as automation benefits, project justification, project management, and interpersonal skills. This category begins on page 72.



Basic Continuous Control

Some might call this category "process control" or "instrumentation and control," because the material in these topics is closest to the historical scope of ISA and is consistent with ISA's original name, Instrument Society of America. But today, proportional, integral, derivative (PID) and other continuous control techniques are used in many applications outside the process industries. For example, they are used in areas such as automotive paint shop controls, motion controls, electrical equipment controls, building automation, ship stabilization controls, and many, many other areas.

PID is so pervasive that one could not be considered a well-educated automation professional if he or she does not know the basic concepts of PID and PID tuning. In fact, the CAP® Steering Team decided early on that while many people today work with manufacturing automation information technology, that does not involve basic plant floor control, it did not make sense for them to be called automation professionals without knowing the basics of plant floor control—including continuous control.

Measurements are extremely important in any automation task. It is really true that you can control well only those things that you can measure—and accuracy and reliability requirements are continuing to increase. Continuous instrumentation is required in many applications throughout automation, although here we call it process instrumentation because the particular type of transmitter packaging discussed is more widely used in process applications.

There are so many measurement principles and variations on those principles that this topic can only scratch the surface of all the available ones, but it hopefully covers the more popular types. Analytical Instrumentation and Control Valves are more applicable to process applications, although these also are used in environmental and other applications in a variety of industries. The type of control system documentation discussed here is also more specific to process industries, but many outside processes can learn from the high degree of development of these documentation conventions. The control equipment topic in this category covers what has traditionally been called distributed control systems (DCSs), although that designation no longer has intrinsic meaning.

Control valves are critical components of a control loop in process and utility industries. It has often been demonstrated that, in nearly all process plants, control valve problems are a major cause of poor loop performance. A general knowledge of the impact of the control valve on loop performance is critical to process control.

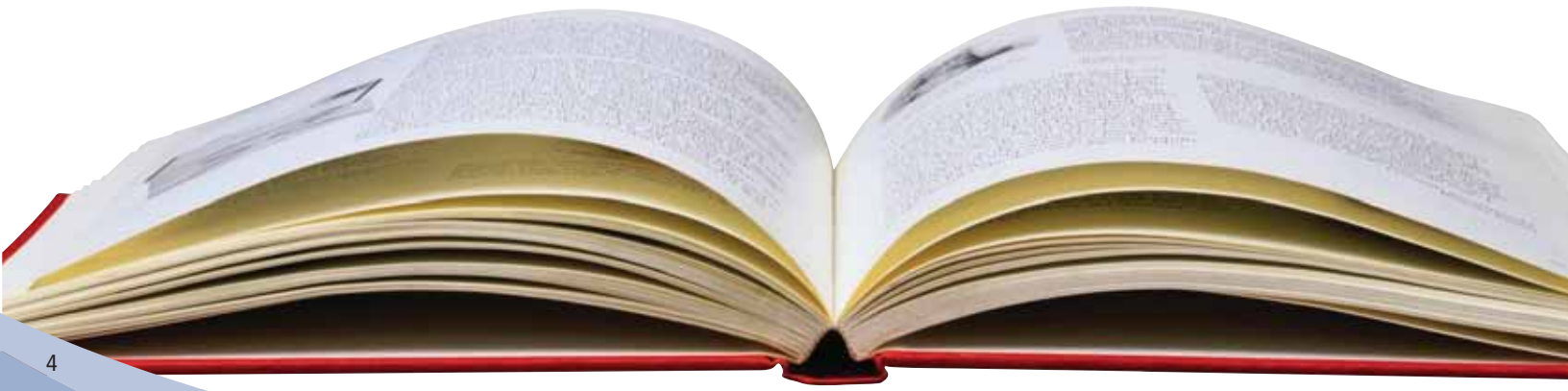
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok



Basic Continuous Control includes analytical instrumentation, process instrumentation, continuous control, control valves, analog communications documentation and control equipment.

Advanced Control Unleashed: Plant Performance Management for Optimum Benefit

Terrence L. Blevins, Gregory K. McMillan, Willy K. Wojsznis, and Michael W. Brown

RAYMOND D. MOLLOY AWARD WINNER

Bridging the gap between theory and practice, this book provides the basis for assessing the benefits of advanced control—covering auto-tuning, model predictive control, optimization, estimators, neural networks, fuzzy control, simulators, expert systems, diagnostics, and performance assessment. The book includes a CD-ROM that provides real DeltaV simulations and displays.



Format: Hardbound Book and CD-ROM – 2003
ISBN: 978-1-55617-815-3
Length: 434 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/advancedcontrol

Advanced pH Measurement and Control, 3rd Edition

Gregory K. McMillan and Robert A. Cameron

This best-selling book is now expanded and updated, providing a clear, concise, and comprehensive view of how to select, install, and maintain electrodes, control valves, and control strategies for pH applications critical for product and water quality in the process industry. The book covers every aspect of system design, including the mixing and reagent piping requirements that are important for a successful application.



Format: Softbound Book – 2005
ISBN: 978-1-934394-43-4
Length: 294 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/pH

Advanced PID Control

Karl J. Åström and Tore Hägglund

RAYMOND D. MOLLOY AWARD WINNER

The authors of the best-selling book PID Controllers: *Theory, Design, and Tuning* once again combine their extensive knowledge in the PID arena to bring you an in-depth look at the world of PID control. This book builds on the basics learned in *PID Controllers*, but augments it through use of advanced control techniques. This book ensures a smooth transition from PID control to more advanced model-based controllers.



Format: Softbound Book – 2006
ISBN: 978-1-55617-942-6
Length: 461 pages; Weight: 2 lbs.
Member: $95.00; List: $112.00
www.isa.org/advancedPID

Advanced Temperature Control

Gregory K. McMillan and C.M. Toarmina

A step-by-step approach for analyzing all of the factors that affect the control of temperature loops, including major sources of measurement errors, this book is packed with information on improving control of temperature loops. The book also focuses on the advantages and disadvantages of new methods for feedback control.



Format: Hardbound Book – 1995
ISBN: 978-1-55617-540-4
Length: 226 pages; Weight: 2 lbs.
Member: $49.00; List: $59.00
www.isa.org/advtempcontrol

Analytical Instrumentation

Robert E. Sherman

RAYMOND D. MOLLOY AWARD WINNER

This treatment of process analytical technology includes theory and real-world hands-on experience. Included are drawings of efficient sample systems. Justifications and organization guidelines on process analyzer systems are presented. The volume describes analyzers from the systems side, looking at implementation issues including justification, purchasing, training, and validation.



Format: Hardbound Book – 1996
ISBN: 978-1-55617-581-7
Length: 728 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/analyticalhandbook

Applying S88: Batch Control from a User's Perspective

Jim H. Parshall and Larry B. Lamb

The release of ANSI/ISA-88.01-1995 led to revolutionary advances in the way factories design, implement, and integrate flexible, modular processes. From designing better processes to combining recipes with equipment control to execution validation test plans, this book offers an introduction to ISA-88 as well as expert insights into applying it. The book's clear, lighthearted style makes it the ideal introduction to ISA-88 for everyone.



Format: Softbound Book – 2000
ISBN: 978-1-55617-703-3
Length: 158 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/applyS88

Automation Applications in Bio-Pharmaceuticals

George Buckbee, P.E. and Joseph Alford, Ph.D., P.E., CAP

This practical guide is ideal for engineers and designers new to the field of bio-pharmaceutical process control. For the experienced automation professional, this book outlines the unique design and application issues for the bio-pharmaceutical industry. For those already familiar with this industry, the book provides specific advice for automating these processes. Each section includes theory, practice, and practical insights into the design and application of automation for the bio-pharmaceutical industry. This book will help the engineer or designer build upon the experience of others, avoid costly design mistakes, and improve process control performance.

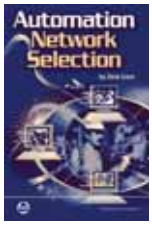


Format: Softbound Book – 2008
ISBN: 978-1-934394-25-0
Length: 250 pages; Weight 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/biopharmaceutical

Automation Network Selection

Dick Caro

Are you trying to make sense of all the different industrial automation networks on the market today? Whether you're a novice industrial network user or someone who simply needs to brush up on the technology, *Automation Network Selection* will help you better understand and select the "right" network for a given application. Industry expert Dick Caro walks you through the various industrial networks (e.g., sensor, fieldbus, control, safety buses, and so on) and then gives you a perspective on the typical applications for industrial automation network technology. The emphasis is on the intended application for each network, rather than on the network protocol that is more typically discussed in most textbooks.



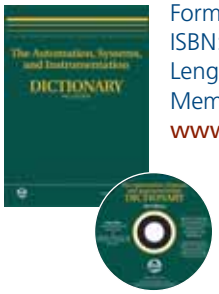
Format: Softbound Book – 2004
ISBN: 978-1-55617-861-0
Length: 161 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/networkselect

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume.

No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Batch Control

Enter the world of batch process control with these in-depth books and learn how to effectively implement the ISA-88 batch series for batch, continuous, or discrete manufacturing environments.

Batch Control Systems — Design, Application, and Implementation, 2nd Edition

William M. Hawkins and Thomas Fisher

This revision of the 1990 work by Thomas Fisher covers an introduction to batch processes; batch control system structures; batch control; batch communications and batch control system design. Hawkins offers a comprehensive analysis of the development and evolution of batch control from the original NAMUR model through the most current publications in the ISA-88 series. Through examples, commentary, analogies and at times, wry humor, the author provides an in-depth philosophical discussion of how batch control and all manufacturing enterprises have been impacted by the work of ISA-88.



Format: Softbound Book – 2006
ISBN: 978-1-55617-967-9
Length: 300 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/batchcontrol

Design Patterns for Flexible Manufacturing

Dennis Brandl

This handy resource defines an effective set of design patterns and rules you should know when applying the widely used ISA-88 industry standards to batch manufacturing (called the S88 design pattern) and continuous and discrete manufacturing (called the NS88 design pattern for non-stop production). This book clearly identifies what elements are defined in the batch series and what elements make up the S88 and NS88 design patterns for flexible manufacturing. The book defines design patterns for control system programming, providing patterns for the organization of programmable logic controller (PLC), digital control system (DCS), and other control system application codes. Whether you are in a batch, continuous, or discrete manufacturing environment, these design patterns can be applied to a wide range of production systems, making systems easier to design and implement.



Format: Softbound Book – 2007
ISBN: 978-1-55617-998-3
Member: $89.00; List: $99.00
Length: 350 pages; Weight: 2 lbs.
www.isa.org/designpatterns

ISA-88 Batch Standards and User Resources

A wealth of information on one CD-ROM, the new *ISA-88 Batch Standards and User Resources* compiles the essential ISA batch systems standards, reference books, and technical papers for the industrial automation and control systems professional.



Format: CD-ROM
ISBN: 978-0-9791330-5-3
Member: $199.00; List: $499.00
www.isa.org/batchresource

Batch Standard Revamps

Batch production industries such as food, beverage, life sciences, and specialty chemical, among others, should see benefits soon from a facelift to the ISA-88 Batch Control series of standards. New working groups will begin cutting into the first two standards in the ISA-88 batch standard series, not only for batch production manufacturers, but for non-batch industries as well. The standard serves as a "solid approach to modular automation, so additional industries could also benefit," said Paul Nowicki, chairman of the ISA-SP88.01 update working group and leader of industry applications in beverage at Rockwell Automation.

The facelift will attempt to clarify issues that have come up over the past 10 years in the ISA-88 series, Nowicki said. "These may include terminology clarifications, more (or clearer) definitions of equipment modules and control modules, a clarification of the types of control defined (coordination, procedural, and basic), and where they fit into the equipment modules," he said.

The reason behind the updates is to, "address some questions manufacturers have had in their attempts to implement systems that follow the standard, resulting in better attainment of the original objectives of the standards to improve batch operations," Nowicki said. One of the key benefits offered in this initiative is modularity, allowing lower engineering and training costs and flexibility for more production capability. Other benefits include the use of standard-based procedure models to meet changing operations requirements, improvements in the consistency and quality of products, and improved control over the manufacturing process.

Because the structure of the batch standard mirrors the structure required for flexible manufacturing, the ISA-88 principles can be extended to many continuous, as well as discrete, applications. The standard is flexible and, depending on the functionality required, can be adapted to many different types of applications. The Make2Pack working group has allowed subgroups of OMAC and the ISA-SP88 committee to work together to find ways to apply these standards-based principles to machine-based control.

But the changes don't come without some challenges. "Users and vendors have developed systems making their best interpretation of the standards," Nowicki said. "Some clarifications on the standard may cause both groups to rethink their solutions to consider the benefits of the large common experience set that the [ISA-SP88] committee represents."

For full *InTech*® story, see **www.isa.org/link/BatchRevamp**

Basic Continuous Control

Automation Unplugged: Pinto's Perspectives, Pointers & Prognostications

Jim Pinto

RAYMOND D. MOLLOY AWARD WINNER

If you have heard industry pundit Jim Pinto speak, or read his barbed writings or laugh-out-loud poems, or subscribe to his popular e-newsletter, you'll enjoy this book. This book is loaded with critical analysis of the changing face of industrial automation; predictions about future automation technology trends—the best of his articles on marketing and distribution; and his highly rated fieldbus commentary.



Format: Softbound Book – 2004
ISBN: 978-1-55617-864-1
Length: 205 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/unplugged

Basic and Advanced Regulatory Control: System Design and Application, 2nd Edition

Harold L. Wade

This revised best-seller teaches the practice of process control for the wet process industries. It stresses the study of real, imperfect processes rather than system theory, and gives guidance on how you can best apply your own experience, intuition, and knowledge of the particular process. The text summarizes the general characteristics of processes and control loops and discusses feedback control and its nuances. The latter part of the book addresses advanced control techniques.



Format: Softbound Book – 2004
ISBN: 978-1-55617-873-3
Length: 372 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/regcontrol

Basic Electricity and Electronics for Control: Fundamentals and Applications, 3rd Edition

Lawrence (Larry) M. Thompson

This class-tested book gives you a familiarity with electricity and electronics as used in the modern world of measurement and control. Integral to the text are procedures performed to make safe and successful measurements of electrical quantities. It will give you a measurement vocabulary along with an understanding of digital and analog meters, bridges, power supplies, solid state circuitry, oscilloscopes, and analog to digital conversions. Where possible, applications are used to illustrate the topics being explained. This book will enhance student success in any industrial or technical school setting where basic technician training is to take place.

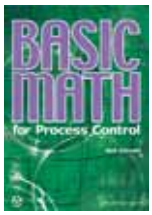


Format: Softbound Book – 2006
ISBN: 978-1-55617-988-4
Length: 383 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/basicelectricity

Basic Math for Process Control

Bob Connell

A practical tutorial on the mathematics essential to the process control field, written by an experienced process control engineer for practicing engineers and students. This quick-and-easy review of the mathematics common to the field, includes chapters on frequency response analysis, transfer functions and block diagrams, and the Z-N approximation.

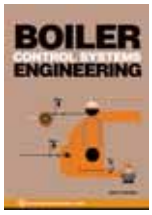


Format: Softbound Book – 2003
ISBN: 978-1-55617-813-9
Length: 200 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/basicmath

Boiler Control Systems Engineering

Jerry Gilman

This book addresses issues to be considered when defining measurement transmitters and specification for transmitters. Final control elements are reviewed as to characteristics and sizing. Engineering details on control systems and the setup of the various control functions are covered with specific examples of boiler control including configuration and tuning.



Format: Softbound Book – 2005
ISBN: 978-1-55617-907-5
Length: 144 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/boilereng

Calibration: A Technician's Guide

Mike Cable

RAYMOND D. MOLLOY AWARD WINNER

This comprehensive review of calibration provides an excellent foundation for understanding principles and applications of the most frequently performed tasks of a technician. Topics addressed include terminology, bench vs. field calibration, loop vs. individual instrument calibration, instrument classification systems, documentation, and specific calibration techniques for temperature, pressure, level, flow, final control, and analytical instrumentation.

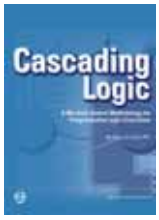


Format: Softbound Book – 2005
ISBN: 978-1-55617-912-9
Length: 273 pages; Weight: 1 lb.
Member: $75.00; List: $85.00
www.isa.org/calibrate

Cascading Logic: A Machine Control Methodology for Programmable Logic Controllers

Gary Kirckof, P.E.

Learn how to create a functional machine control program for industrial equipment that is sequential in nature. The programming methodology starts by breaking the machine into its basic elements. The methodology then shows how to program each element and how to assemble them into a complete machine control program. This book is intended to give you the confidence you need to reach decisions and move forward with certainty. The sequential nature of events will also help operators and maintenance personnel troubleshoot and maintain the equipment after it is put into operation. Ladder logic illustrations useful with any PLC demonstrate each part of the text.



Format: Softbound Book – 2003
ISBN: 978-1-55617-814-6
Length: 206 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/cascadelogic

Order Now
www.isa.org/shopisa
(919) 549-8411

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

Certified Control Systems Technician® (CCST®) Study Guides

ISA

The CCST Study Guides provide assistance in preparing for ISA's CCST Exams, Levels I, II, and III. The guides parallel the structure of each exam, providing you with sample questions and a listing of other sources for study which provide you with many opportunities to test your knowledge. The study guides also include a recommended list of publications you can use to do further study on specific domains.



Level I
Format: Softbound Book – 2006
ISBN: 978-1-55617-573-2
Length: 60 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst1

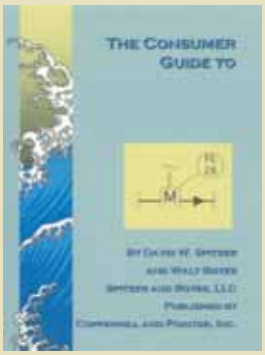


Level II
Format: Softbound Book – 2006
ISBN: 978-1-55617-629-6
Length: 58 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst2



Level III
Format: Softbound Book – 2006
ISBN: 978-1-55617-691-3
Length: 40 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst3

Consumer Guide Series



These consumer guides contain technical information about evaluating and applying instrumentation equipment, including pointers for installation, such as fluid, piping, hydraulic, mounting, and electrical considerations (as applicable). Tables include the types of equipment available and selected features that are available from each supplier, along with their country of origin or source (as applicable).

These guides are indispensable for designers, engineers, owners and end users of instrumentation equipment. See where your instruments rank and use these guides to help you with your next selection/purchase.

The Consumer Guide to Capacitance and Radar Contact Level Gauges

David W. Spitzer

A completely independent guide to the world's capacitance and radar contact level gauges ranked by performance and supplier.

This guide contains technical information about evaluating and applying these level gauges with pointers for installation, such as material, vapor, vessel, sensor, and electrical considerations. Tables include the types of level gauges and selected features that are available from each supplier, along with their country of origin or source.

This guide is indispensable for designers, engineers, owners and end users of level gauges. See where your level gauge ranks and use this guide to help you choose your next capacitance and radar contact level gauge.

Format: Softbound book
ISBN: 1-932095-11-X
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/CRClevelgauges

The Consumer Guide to Fieldbus Network Equipment for Process Control, 3rd Edition

Richard Caro

Format: Softbound book
ISBN: 1-932095-12-8
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/fieldbusguide

The Consumer Guide to Flowmeters Series

David W. Spitzer and Walt Boyes

The first completely independent consumer guides to the world's flowmeters, ranked by performance, by size, and by supplier.

Consumer Guide to Coriolis Mass Flowmeters
Format: Softbound Book – 2002
ISBN: 1-932095-01-2
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/coriolisflow

Consumer Guide to Differential Pressure Flow Transmitters
Format: Softbound Book – 2003
ISBN: 1-932095-03-9
Length: 86 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/dpflow

Consumer Guide to Magnetic Flowmeters 2nd Edition
Format: Softbound Book – 2004
ISBN: 1-932095-04-7
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/magneticflow

Consumer Guide to Ultrasonic and Correlation Flowmeters
Format: Softbound Book – 2004
ISBN: 1-932095-05-5
Length: 100 pages; Weight: 2 lb.
Member: $225.00; List: $250.00
www.isa.org/ultrasonicflow

Consumer Guide to Vortex Shedding and Fluidic Flowmeters
Format: Softbound Book – 2002
ISBN: 1-932095-02-0
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/vortexflow

The Consumer Guide to Industrial pH and ORP Instrumentation

David B. Mills

A completely independent guide to industrial pH and ORP instrumentation ranked by performance and supplier (limited to eight major world-wide manufacturers of pH and ORP instruments).

This guide contains technical information about evaluating and applying pH and ORP instruments with pointers for installation. Tables include the types of pH and ORP instruments and selected features that are available from each supplier, along with their country of origin or source.

This guide is indispensable for designers, engineers, owners and end users of pH and ORP instruments. See where your instrument ranks and use this guide to help you choose your next pH and ORP instruments.

Format: Softbound book
ISBN: 1-932095-09-8
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/pHandORP

Consumer Guide to Non-Contact Level Gauges

David W. Spitzer and Walt Boyes

This is an updated and independent guide to the world's ultrasonic, laser, and radar non-contact level gauges, ranked by performance and by supplier. The guide contains technical information about evaluation and application of level gauges with pointers for installation, such as material, vapor, vessel, sensor, and electrical considerations. Tables include the types of level gauges and selected features that are available from each supplier, along with their country of origin or source. This guide is indispensable for designers, engineers, owners and end users of level gauges. See where your level gauge ranks and use this guide to help you choose your next ultrasonic, laser or radar level gauge.

Format: Softbound Book – 2005
ISBN: 1-932095-07-1
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List; $250.00
www.isa.org/noncontact

Rooftop Maker Solves Burning Problem

New clamp-on ultrasonic flow technology brings temperature control to hot process

Tallant Industries, Inc., a manufacturer of corrugated asphalt roofing panels, was searching for a solution to a burning problem that was causing delays in their process. Tallant's plant in Fredericksburg, Va., produces what they consider a revolutionary roofing product. The corrugated asphalt roofing materials consist of paper impregnated with bitumen for water resistance. A critical component of the company's production is bitumen storage. Bitumen presents a challenge in that high temperatures and its viscous, sticky nature require maintaining tight temperature control throughout the process.

During the process, manufacturers move bitumen from a heated vessel to a smelter where they heat it to 375°F (190°C) just prior to using it for impregnation and coating. They heat the bitumen vessels and smelter via 2-in pipes flowing heat transfer oil at 540°F (282°C). Maintaining the proper flow rate of the transfer oil is critical to getting the heat transfer efficiency they need to heat and maintain bitumen at the right process temperatures.

They typically measure heat transfer oil (HTO) on a 2-in pipe at a temperature higher than 540°F (282°C). The company said a diagnostic tool to measure this HTO circuit has not been available up to this point. The flowmeter they used in the past failed repeatedly because of the high temperature. "This application is extremely problematic for intrusive metering technologies because of the high temperatures, abrasive nature of the liquid, and high flow velocities," said Allan Cowden, Tallant's lead plant engineer. Because the company emphasizes safety, Cowden said they now must avoid potential leakage due to new metering technologies, such as flange connections. As a result, they needed a non-intrusive measurement method for such a highly flammable aggressive medium.

To overcome these limits, the company looked to high-temperature pipe mounting fixtures, which allow clamp-on flow measurement at temperatures of up to 752°F (400°C).

Cowden said the devices allowed the company to verify the problem wasn't in the HTO circuit, but was build-up of coke from the bitumen on the heat transfer coils.

By means of cooling fins, the transducers move away from the measurement point. Engineering of these plates allows areas reflecting heat to be 100 times larger compared to the areas conducting heat. This results in a significant temperature gradient between the pipe wall and the transducer surface area. The drop in temperature is large enough to permit standard temperature rated transducers to see use in high temperature applications exceeding 750°F without reliability or performance degradation.

> "The basic problem in measuring the flow of gases or liquids at high temperatures is survival of the transducers that generate and detect the ultrasonic signals," O'Brien said.

"Flow measurement at high temperatures is problematic by itself," said John O'Brien business development and operations manager with Flexim Instruments LLC in Bohemia, N.Y. High media temperatures (>500°F) have in the past posed problems for non-intrusive ultrasonic manufacturers. Thermal stress in magnetic flowmeters causes ceramic liner cracks. So they're only suitable for measuring up to 356°F (180°C). Vortex shedding meters cover a higher temperature range than magnetic meters, but they are too expensive for large pipe diameters and wouldn't provide the lower flow rate performance this application needs. Even non-intrusive ultrasonic meters, "which are in principle ideal for corrosive and aggressive media," have seen limited use at high temperatures, he said. One reason is the sound coupling gels these meters require have a 482°F (250°C) maximum temperature tolerance. High temperatures accelerate the aging of the ultrasonic transducers' piezo elements and limit their useful operating life.

For full *InTech*® story, go to **www.isa.org/intech/20060402**.

Order Now
www.isa.org/shopisa
(919) 549-8411

Basic Continuous Control

The Condensed Handbook of Measurement and Control, 3rd Edition (with CD-ROM)

N.E. Battikha

RAYMOND D. MOLLOY AWARD WINNER

Selecting and implementing measurement and control devices for process automation applications is made easier with this best-selling reference. This clear and concise handbook provides quick access to ISA symbology, instrument and control valve selection criteria, and conversion guidelines, with new sections on analyzers, programmable electronic systems, alarm and trip systems, control centers, enclosures, and engineering documentation.

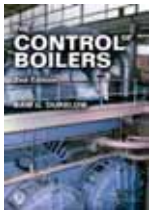


Format: Softbound Book – 2007
ISBN: 978-1-55617-995-2
Length: 441 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/MandCHandbook

The Control of Boilers, 2nd Edition

S.G. Dukelow

This classic resource helps you reduce boiler operating costs through a detailed explanation of all aspects of boiler processes. It presents the basics of boiler control, the interrelationships of the process characteristics, and the dynamics involved, with emphasis on start-up, shut-down, flame monitoring, and safety interlock measures.



Format: Hardbound Book – 1991
ISBN: 978-1-55617-330-1
Length: 412 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/boilercontrol

Control System Engineering Exam Reference Manual: A Practical Study Guide

Bryon Lewis

This practical reference manual, available in downloadable PDF format, is for use in preparing for the Control Systems Engineering (CSE) examination, which covers a broad range of subjects, from the electrical, mechanical, and chemical engineering disciplines. This manual will review the foundations of process control and demonstrate the breadth and width of the CSE exam. Topics include process measurement and calibration, sizing process, control elements, digital logic, analog signals, process safety and shutdown, industrial control networks, NEC and NFPA codes.



Format: PDF on Web – 2007
ISBN: 978-1-934394-22-9
Length: 126 pages
Member: $49.00: List: $59.00
www.isa.org/csereference

Control Systems Engineering (CSE) Study Guide, 4th Edition

This fourth edition, a downloadable PDF File, is for use in preparing for the CSE exam. The manual provides candidates for this professional engineer (PE) license with information about the scope and design of the CSE exam, sample questions, and resources for further study.



Format: PDF on Web – 2003
ISBN: 978-1-55617-881-8
Length: 112 pages
Member: $49.00; List: $59.00
www.isa.org/cseguide

ISA-75 Control Valve Standards and User Resources

Key documents on this new *ISA-75 Control Valve Standards and User Resources* CD-ROM include the ISA-75 series covering all ISA standardization work related to valves. Also included on the CD-ROM are several best-selling books and the chapters "Process Instrumentation" and "Control Valves" from *A Guide to the Automation Body of Knowledge*, 2nd Edition.



Format: CD-ROM
ISBN: 978-0-9792343-3-0
Member: $199.00; List: $599.00
www.isa.org/controlvalveresource

Control System Documentation

Understand and apply the symbols and documentation used to define a modern industrial instrumentation and control system with these practical, easy-to-understand resources.

Control System Documentation: Applying Symbols and Identification, 2nd Edition

Thomas McAvinew and Raymond Mulley

This updated ISA classic provides the symbols and identification commonly used throughout the process industries. It contains sample P&IDs and numerous examples of symbols and tagging concepts. It provides most of the symbols and identifiers that are unique to instrumentation and gives practical examples of their use.



Format: Softbound Book – 2005
ISBN: 978-1-55617-896-2
Length: 231 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/symbolsandID

Instrumentation and Control Systems Documentation

Frederick A. Meier and Clifford A. Meier

RAYMOND D. MOLLOY AWARD WINNER

This book will train you to read, understand, and apply the symbols and documents used to define a modern industrial instrumentation and control system. To help you better understand process documentation today, insight into how, when, and why documents are developed, who develops them, and how they are used is provided. The types of documents discussed include process flow diagrams, piping and instrumentation drawings, instrument lists, specification forms, logic diagrams, installation details, location plans, and loop diagrams.



Format: Softbound Book – 2004
ISBN: 978-1-55617-870-2
Length: 173 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/instdocument

To Show or Not to Show?

One of the challenges you will face is the depiction of third party systems on your P&IDs. If you have an island of equipment furnished by a third party, how much of that equipment should show on your drawing? If the third-party system suppliers have their own P&IDs, do you copy them into your drawing set, or possibly just include their P&ID with your set? As usual, there really is no right answer; each facility is managed differently, each project has a different scope and each stakeholder in the P&IDs has different requirements.

It is not inexpensive to redraw a P&ID within your drawing set, nor is it a particularly good idea to have two drawings that show the same thing—yours, and the system supplier's. The drawings will probably only agree on the day they are checked and issued for use. As soon as someone makes a change, you start to "chase revisions".

One successful and cost-effective approach has been to show the interface points between the vendor's system and your control system—just show the components seen on the operator station. Then, on your drawing, refer to the vendor's P&ID and operating manual for further details.

Source: Instrumentation and Control Systems Documentation

Reading a P&ID

Job Training Systems

Learn how to read and understand Piping and Instrumentation Drawings (P&IDs). After completing the workbook, you should be able to identify symbols and function labels commonly found on P&IDs, describe how system components are related, and trace process stream flow and control loop functions.



Format: Spiralbound Book – 1999
Order No.: RPIDWBOOK
Length: 102 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/readpandid

ISA Documentation Standards and User Resources for Industrial Automation and Control Systems

The new ISA Documentation Standards and User Resources for Industrial Automation and Control Systems CD-ROM offers the widely used ISA documentation standards and technical reports, as well as several best-selling reference books. This comprehensive CD-ROM also includes the Automation, Systems, and Instrumentation Dictionary, as well as chapters on "Process Instrumentation" and "Control System Documentation" from A Guide to the Automation Body of Knowledge, 2nd Edition.



Format: CD-ROM
ISBN: 978-0-9791330-9-1
Member: $249.00; List: $599.00
www.isa.org/documentationresource

Humorous Automation Resources

Enjoy the lighter side of life through humorous stories and cartoons about life as an automation professional in these best-selling classics.



Dispersing Heat Through Conviction: The Funnier Side of Process Control

Gregory K. McMillan

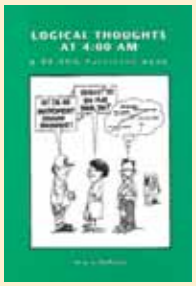


An insider's collection of original satires and jokes from the men and women behind the tanks and valves and digital readouts.

Format: PDF on Web - 2007
ISBN: 978-1-934394-48-9
Length: 134 pages
Member: $15.00; List: $15.00
www.isa.org/funnyprocess

Logical Thoughts at 4:00 A.M.

Gregory K. McMillan and Stan Weiner



Format: Softbound Book
ISBN: 978-1-55617-332-5
Weight: .08 lb.
Member: $20.00;
List: $20.00
www.isa.org/logicalthoughts

How to Become an Instrument Engineer, Part 1.523

Gregory K. McMillan and Stan Weiner



Format: Softbound Book—1994
ISBN: 978-1-55617-520-6
Weight: 1 lb.
Member: $20.00;
List: $20.00
www.isa.org/becomeIE

The Funnier Side of Retirement for Engineers and People of the Technical Persuasion

Gregory K. McMillan and illustrated by Ted Williams

"Engineers...hold on to your Beer!" Here is your chance to have a far-out look (or an up-front stare if your company is headed for reorganization) into what retirement offers for geeks. This humorous book puts the Senior World on your doorstep. It's "Retired Engineers Gone Wild," "Engineering Beyond the Grave," and "Control Engineers in Rest Homes...." Ted Williams' cartoons make this book a graphic exposé. Here is your chance to find out how to have your cake and the enjoyment of eating it too while watching your big screen TV.



Format: Softbound Book – 2008
ISBN: 978-1-934394-28-1
Length: 135 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/retirement

How to Become an Instrument Engineer: The Making of a Prima Donna

Gregory K. McMillan, Stan Weiner

A timeless collection of humorous essays, enjoyed by instrument engineers around the world – making your day more enjoyable one story at a time.



Format: PDF on Web - 1987
ISBN: 978-1-55617-884-9
Length: 148 pages
Member: $20.00; List: $20.00
www.isa.org/primadonna

The Life and Times of an Automation Professional— An Illustrated Guide

Ted Williams, Stan Weiner, and Gregory K. McMillan

Top Five Reasons to Buy This Book

5) Charming anniversary present. This book is a diamond in the rough and says "forever" if dipped in acrylic. Not suitable as a gift for Domestic Companions.
4) Boss-friendly unlike previous books by Stan and Greg. There are very few jokes at the expense of management. In retrospect, this was an oversight on our part, but it does mean the book does not have to be delivered in a plain brown wrapper. Actually, Stan tried to add more management jokes, but Greg seems to be mellowing in anticipation of receiving Medicare.**
3) Get a refund. If this book is no laughing matter, send the original UPC along with a self-addressed stamped envelope to Stan and Greg. They will e-mail you the tracking number for the $0.25 royalty rebate along with a picture of Stan cleaning the pool.
2) Discounts for futuristic "Great Automator" and "Robbie the Robot" action figures. Just show your copy at the next ISA EXPO to your friendly ISA bookstore person. Pose the figures to guard the entrance to your cubicle.*
1) Great preparation for the upcoming "Automation Reality" TV program. Twenty vendors and one "user" are locked in a room with 1000 PowerPoint slides—very similar to a local ISA meeting.

*Action figures may not be available in your lifetime and discounts are not inheritable; however, the editor is working on a plan to make them available as SYMBOLS for your nursing home nightstand.

**In the end, it seems like Ted will slip in a cartoon of a female-type boss trying to convince him that "discrete manufacturing" is more important than his ethics or appearance.

At the current time, this book will not be offered as an Educational Book Cartridge for the Leapster Multimedia Learning System Library. The Leapfrog Company cannot decide whether the book is suitable for pre-kindergarten.

Warning from the Automator General — This book may cause uncontrollable laughter. Do not read this in combination with any other books by Stan and Greg.



Format: Softbound Book – 2006
ISBN: 978-1-55617-957-0
Length: 192 pages; Weight: 1 lb.
Member: $20.00, List: $25.00
www.isa.org/autolife

Control Valve Primer—A User's Guide, 4th Edition

H.D. Baumann

Includes insights on valve sizing, smart (digital) valve positioners, field-based architecture, network system technology, and control loop performance evaluation. The author, a holder of more than 150 patents and author of over a hundred publications in control valve technology, shares his expertise on designing control loops and selecting final control elements. The easy-to-read text provides shortcuts through complex sizing and noise calculation formulas, including for liquids and cavitation, and gives practical advice on how to apply control valves for safety, reduced energy costs, loop stability, and easy maintenance.



Format: Softbound Book – 2008
ISBN: 978-1-934394-50-2
Length: 200 pages; Weight: 2 lbs.
Member: $89.00: List: $99.00
www.isa.org/controlvalveprimer

Control Valves

Guy Borden, Jr., Editor

Providing solutions to problems involving the body assemblies, actuators and accessories of control valves, as well as an overview of valve design and construction, this reference book also includes discussions of applications, safety, troubleshooting, maintenance, testing, standards, valve-related computer programs, and regulators.



Format: Hardbound Book – 1998
ISBN: 978-1-55617-565-7
Length: 739 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/controlvalvehandbook

Electrical Instruments in Hazardous Locations, 4th Edition

Ernest Magison

An encyclopedia of electrical safety, this latest edition features expanded treatment of theory and principles to provide a foundation for addressing some of the questions that arise when no standard or regulation exists.



Format: Softbound Book – 1998
ISBN: 978-0-9792343-1-6
Length: 556 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/hazardlocations

Environmental Instrumentation and Analysis Handbook

Randy D. Down and Jay H. Lehr

A comprehensive resource for information about different technologies and methods to measure and analyze contamination of air, water, and soil that serves as a technical reference in the field of environmental science and engineering. It includes information on instrumentation used for measurement and control of effluents and emissions from industrial facilities that can directly influence the environment, and focuses on applications, making it a practical reference tool.



Format: Hardbound Book – 2004
ISBN: 0-471-46354-X
Length: 1068 pages; Weight: 6 lbs.
Member: $175.00; List: $175.00
www.isa.org/environinst

ISA-77 Fossil Fuel Power Plant Standards and User Resources

This comprehensive *ISA-77 Fossil Fuel Power Plant Standards and User Resources* CD-ROM provides a thorough guide to information about fossil fuel power plants. Key documents focus on practices and procedures related to instrumentation and controls utilized in fossil fuel power generating stations. Also included on the CD-ROM are ISA best sellers, *Control of Boilers*, 2nd Edition and *SCADA: Supervisory Control and Data Acquisition*, 3rd Edition.



Format: CD-ROM
ISBN: 978-0-9792343-4-7
Member: $199.00; List: $599.00
www.isa.org/fossilfuelresource

Order Now
www.isa.org/shopisa
(919) 549-8411

Fieldbuses for Process Control: Engineering, Operation, and Maintenance

Jonas Berge



This implementation-oriented book provides a clear and concise presentation of how to apply fieldbuses for process control. Based on experience collected from end-users in a wide range of industries around the world, it provides "how-to" information for all phases of the system lifecycle. The book covers the three leading process fieldbus technologies: HART, FOUNDATION™ Fieldbus, and PROFIBUS-PA. It covers both the field-level and the Ethernet-based host-level networking. The text also addresses concerns and solutions for interoperability, integration and migration as well as availability and safety. A chapter on benefits helps you justify business advantages to management. The final chapter provides an in-depth explanation of how these fieldbus technologies work. It is ideal for both organized training courses and for self-study and will remain a handy reference when configuring and troubleshooting systems.



Format: Softbound Book – 2002
ISBN: 978-1-55617-904-4
Length: 468 pages; Weight: 2 lbs.
Member: $99.00; List: $109.00
www.isa.org/fieldbuses

Flow Measurement, 2nd Edition

D.W. Spitzer, Editor

The industry's most authoritative handbook on flow measurement—newly updated with added chapters on laminar flow elements, critical flowmeters, statistics for measurement, laboratory primary standards, and uncertainty in flow measurement, and substantial updates to 15 other chapters. A road map to the field of flow measurement, this bestseller discusses strategies for problem solving and puts the whole array of flowmeter types at your disposal. Fundamental considerations such as mixed-phase flow, piping effects, and flow conditioning are examined at length.



Format: Hardbound Book – 2001
ISBN: 978-1-55617-736-1
Length: 830 pages; Weight: 6 lbs.
Member: $129.00; List: $159.00
www.isa.org/flowmeashandbook

Instrument Engineers' Handbooks

This widely used three-volume set provides the automation professional with in-depth coverage on the tools and applications needed to become successful in today's automation environment.

Whether a novice or experienced professional, you will likely turn to these three handbooks again and again for quick access to specific information.

Instrument Engineers' Handbook, 4th Edition, Vol. 1: Process Measurement and Analysis

Béla G. Lipták

This volume incorporates the latest developments in automation and control, and broadens its outlook to a global perspective. The handbook includes topics such as the detection of flow, level, temperature, pressure, density, viscosity, weight, composition, and safety sensors. It includes contributions from approximately 100 leading industrial and academic professionals. Also covered are sensors, detectors, analyzers, and other measuring devices introduced since publication of the third edition.



Format: Hardbound
Book—2003
ISBN: 0849310830
Length: 1824 pages;
Weight: 7 lbs.
Member: $179.95;
List: $179.95
www.isa.org/iehvol1

Instrument Engineers' Handbook, 4th Edition, Vol. 2: Process Control and Optimization

Béla G Lipták

Béla Lipták's acclaimed "bible" of instrument engineering now enters its fourth edition with fully globalized coverage, an outstanding panel of international contributors, and major additional coverage that reflects the advances made over the last decade since publication of the third edition. Expanded coverage includes descriptions of overseas manufacturers' products and concepts, model-based optimization in control theory, new major inventions and innovations in control valves, and a full chapter devoted to safety.



Format: Hardbound
Book—2005
ISBN: 0849310814
Length: 2464 pages;
Weight: 7 lbs.
Member: $179.95;
List: $179.95
www.isa.org/iehvol2

Instrument Engineers' Handbook, 3rd Edition, Vol. 3: Process Software and Digital Networks

Béla G. Lipták

The third volume in this three-volume set, this best-seller provides an in-depth, state-of-the-art review of existing and evolving digital communications and control systems. Several chapters describe a variety of process-control software packages suited for plant optimization, maintenance, and safety-related applications. Other chapters cover plant design and updating, safety and operations-related logic systems, and the design of integrated workstations and control centers. Appendices provide practical information such as bidders' lists and addresses, materials selection for corrosive services, and more.



Format: Hardbound
Book—2002
ISBN: 0849310822
Length: 864 pages;
Weight: 5 lbs.
Member: $179.95;
List: $179.95
www.isa.org/iehvol3

The Squeeze is Pressure

Reaching the primary standard accuracy requires compensating for a large number of factors

The calibration of pressure instrumentation is one of the most common types of calibration that takes place in laboratories and industrial facilities.

The calibration equipment for moderatley accurate calibrations is realtively inexpensive, rugged, and easy to use.

The wide range of pressures in one facility will probably require more than one type or range of calibration standard. The pressure range from 1 inch of water to 100,000 pounds per square inch, for example, represents a 2.77 million to one ratio of maximum pressure to minimum pressure.

The number of standards needed increases if vacuum and absolute pressure calibrations are required and if various fluids such as water, oil, and gas are required.

Pressure is a derived unit based on the fundamental units of lenght (l), mass (m), and time (t). Earlier sections defined pressure as a force acting on an area (12), defined force as an accelerated mass (m), and defined acceleration as (l/t2).

The accuracy of many of the pressure standards depends on how accurately we can produce, maintain, and measure these fundamental units when manufacturing and using the standards.

Dead Weight Testers

Dead weight testers are the most widely used type of pressure calibrator and are available in a variety of accuracies, fluid types, ranges, and pressure units.

A typical dead weight tester consists of a fluid filled chamber that a hand pump, ram screw, or some external force pressurizes. A vertical cylinder together with a close fitting piston that rotates within the cylinder is part of the chamber wall so the fluid pressure acts on the bottom area of the piston.

A pressure port in the chamber allows the connection of the pressure instrument. The fluid pressure acting on the bottom area of the piston produces an upwards force that lifts the piston. The upward force balances out the downward force produced by the addition of weights to the top of the piston.

For full *InTech*® story, see **www.isa.org/link/PressSqueeze**.

Flow of Industrial Fluids—Theory and Equations

Raymond Mulley, P.E.

This book provides context—both theoretical and practical—to understand the application of flow equipment and devices. Insight is the primary goal. Basic theory and equations of fluid flow are given in an understandable, but not overly simplified, manner. The chapters contain a logical, common sense description of the concepts and the equations of fluid flow. The appendices contain detailed derivations of the equations and detailed explanations of the subject matter of the corresponding chapters. Over 60 illustrations and 20 tables provide a graphical representation of data and ideas and give you speedy access to precise data. Technical caveats found throughout the book also encourage you to pay particular attention to specific topics and thus avoid costly errors or misunderstandings.



Format: Hardbound Book – 2004
ISBN: 0-8493-2767-9
Length: 410 pages; Weight: 3 lbs.
Member: $109.00; List: $119.00
www.isa.org/flowfluid

Fundamentals of Industrial Control, 2nd Edition (with CD-ROM)

D.A. Coggan, Editor

This updated edition of a best-selling classic provides a unique balance between control system theory and effective application, covering every major control system element, from sensors to final control elements, in the context of overall control strategies and system design. This edition provides updates on a range of topics including Internet communications, industrial communications network hardware and software, open systems, wireless networks, enterprise computing, and computer and control system security. Other sections introduce the new generation of distributed controls and digital automation systems, and present migration strategies for expanding and upgrading existing systems. As a bonus, the book comes with the full text and graphics on a CD-ROM.



Format: Hardbound Book and CD-ROM – 2005
ISBN: 978-1-55617-863-4
Length: 893 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/fundamentalcontrol

Fundamentals of Process Control Theory, 3rd Edition (with CD-ROM)

P.W. Murrill

ISA's all-time best-selling book now includes a full text and graphics searchable CD, offering a way for you to teach yourself the complexities of process control theory. This classic has been praised for its clear presentation of the principles of process automation and its excellent overview of advanced control techniques. More than just a reference book, it's a complete course on the subject, with exercises and answers to work through. Both theoretical and practical, this guide allows you to teach yourself the fundamental scientific principles that govern process control, particularly feedback control. Its 15 self-study units provide a solid foundation in theory, as well as a discussion of recent technologies such as computer-integrated manufacturing, statistical process control, and expert systems. New chapters focus on the conceptual framework for an application, offering a practical understanding of the theory, along with specific illustrations on how concepts are implemented.



Format: Hardbound Book and CD-ROM – 2000
ISBN: 978-1-55617-683-8
Length: 344 pages; Weight: 2 lbs.
Member: $92.00; List: $109.00
www.isa.org/theoryfundamentals

Fundamentals of Test Measurement Instrumentation

Keith R. Cheatle

This book discusses the specialized test instrumentation used in R&D laboratories, testing organizations, and industrial maintenance departments. It focuses on the practical application of test instrumentation and emphasizes the importance of creating a "measurement system" that involves components, installation, wiring, and calibration. The design, application and calibration of systems for measuring pressure, temperature, flow, force, displacement, and vibration will also be covered.



Format: Softbound Book – 2006
ISBN: 978-1-55617-914-3
Length: 329 pages; Weight: 1 lb.
Member: $89.00; List: $99.00
www.isa.org/testmeasure

Good Tuning: A Pocket Guide, 2nd Edition

Gregory K. McMillan

Every practicing instrument, process control, and process engineer will want to have this practical and to the point pocket guide on good tuning. It includes step-by-step descriptions of the three best field-proven tuning procedures, a table of typical tuning settings, a summary of valve performance problems, logic diagrams for troubleshooting, and over 70 "rules of thumb".



Format: Softbound Book – 2005
ISBN: 978-1-55617-940-2
Length: 181 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/tuningpocket

ISA-92 Industrial Air Measurement Standards and User Resources

The new *ISA-92 Industrial Air Measurement Standards and User Resources* CD-ROM offers the widely used ISA-92 documents addressing performance requirements for industrial air measurement instrumentation related to health and safety. The CD-ROM also includes the best-selling technical publication, *The Automation, Systems, and Instrumentation Dictionary*, 4th Edition.



Format: CD-ROM
ISBN: 978-0-9792343-2-3
Member: $249.00; List: $599.00
www.isa.org/industrialairresource

Industrial Flow Measurement, 3rd Edition

David W. Spitzer

Designed to help practicing engineers avoid costs associated with misapplication of flowmeters, this newly revised text reviews the important concepts of flow measurement and provides explanations, practical considerations, illustrations, and examples of current flowmeter technology. This book presents a rational procedure for flowmeter selection that is based on factual information. Effective flowmeter selection requires a thorough understanding of flowmeter technology in addition to a practical knowledge of the process and the fluid. This book will help experienced engineers, technicians, and others evaluate the appropriate criteria rationally to arrive at the proper flowmeter selection.

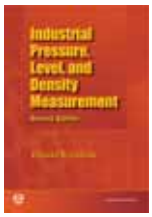


Format: Hardbound Book – 2005
ISBN: 978-1-55617-871-9
Length: 443 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/industrialflowmeas

Industrial Pressure, Level, and Density Measurement, 2nd Edition

Donald R. Gillum

This second edition includes a new chapter covering equipment selection, mounting techniques, and specifications. Other new topics and information include: calibration and re-ranging updates for process calibrators, comparators, and other new test instruments; digital transmitter and communication updates, including HART®, FOUNDATION™ FIELDBUS, wireless transmitters, and multivariable and differential pressure transmitters and applications; added emphasis on non-contact level measurement; advances in hydrostatic tank gauging (HTG); and, improved density sensors and new applications.



Format: Softbound Book – 2008
ISBN: 978-1-934394-34-2
Length: 450 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/pressureleveldensity

Instrumentation Reference Book, 3rd Edition

Walt Boyes

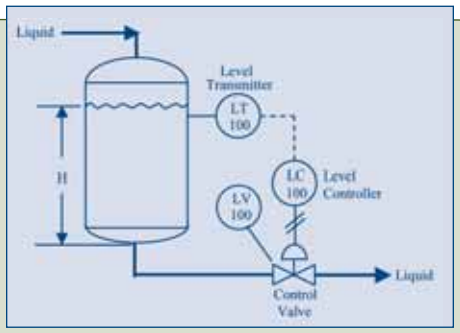
The latest edition of this classic is an authoritative collection covering all aspects of measurement applications and instrumentation. Of direct practical value to instrumentation and control engineers, as well as instrument technicians and users, it is written by more than 40 specialist contributors who address the various applications that may be needed, with reference to practical techniques that are available for the instrumentation or measurement of specific physical quantities or qualities. This edition has been updated to include chapters on the fieldbus standards, reliability, EMC, "virtual instrumentation," fiber optics, smart and intelligent transmitters, and analyzers.



Format: Softbound Book – 2002
ISBN: 0-7506-7123-8
Length: 1062 pages; Weight: 6 lbs.
Member: $180.00; List: $180.00
www.isa.org/instref

Order Now
www.isa.org/shopisa
(919) 549-8411

Measurement and Control Resources

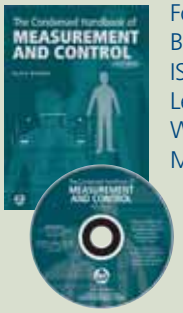


Condensed Handbook of Measurement and Control, 3rd Edition (with CD-ROM)

N.E. Battikha

RAYMOND D. MOLLOY AWARD WINNER

Selecting and implementing measurement and control devices for process automation applications is made easier with this best-selling reference. This clear and concise handbook provides quick access to ISA symbology, instrument and control valve selection criteria, and conversion guidelines, with new sections on analyzers, programmable electronic systems, alarm and trip systems, control centers, enclosures, and engineering documentation.



Format: Softbound Book—2007
ISBN: 978-1-55617-995-2
Length: 441 pages;
Weight: 2 lbs.
Member: $79.00;
List: $89.00
www.isa.org/MandCHandbook

Measurement and Control Basics, 4th Edition (with CD-ROM)

Thomas A. Hughes

Ideal for classroom use or self-study, this revised best-selling book has provided thousands of students, technicians, sales people, and others with a practical introduction to the technologies, systems, and strategies involved in industrial process control. The fourth edition takes the same proven intuitive approach of previous editions. Each chapter begins with basic definitions and mathematical concepts that allow readers to become well-versed in the principles necessary to understand the variables that affect process control systems. New features in the fourth edition include improved coverage of analytical measurement and control, and the addition of sections on the international standard for PLC languages; process visualization; and personal computer-based control systems. The book also includes solutions to exercises that make it more suitable for self-study. A bonus CD-ROM is also included!



Format: Softbound Book—2007
ISBN: 978-1-55617-916-7
Length: 377 pages; Weight: 1 lb.
Member: $95.00; List: $109.00
www.isa.org/mandcbasics

Check out these clear and concise resources for a practical look at the technologies, systems, and strategies involved in process measurement and control.

Improving control

A precision electric actuator is unaffected by valve stiction and can track closed-loop controller demand almost perfectly, without dead time, lag, or overshoot.

A natural response to the increasing economic, environmental, and competitive pressures facing industry is to improve process control performance, and many companies are investing in powerful, state-of-the-art control instrumentation as a result.

Unfortunately, these efforts often suffer from the performance of the final control element, or FCE (defined as the control valve/actuator or damper/actuator assembly).

The positioning performance of the FCE and its importance on process control is often misunderstood, underestimated, or simply ignored.

Updating control instrumentation and design is a wise, but costly, investment. Maximizing the return on the investment is possible, only if the proper actuator couples to a properly applied valve/damper.

Here are the actuation performance issues critical to control. We'll look at why electric actuators designed for precise positioning and continuous modulation duty eliminate a major source of performance limitations and control loop non-linearity.

Of course, this makes the return on a control system investment look good.

The performance issue

The key performance issue of an FCE is its ability to control consistently the process variable in response to the closed loop controller demand, without inducing dead time, lag, or gain variations over the expected control range.

The FCE gain is a function of the valve/damper itself (hereafter referred to as the valve only), while the time-based dynamics (dead time, lag) are largely a function of the actuator performance.

An ideal FCE includes a valve with a constant gain throughout the control range and an actuator that responds to the controller demand perfectly with no effect on the overall process dynamics.

For full *InTech*® story, see **www.isa.org/link/ImproveControl**.

ISA Handbook of Measurement Equations and Tables, 2nd Edition

Jim Strothman

This updated, expanded, and improved version provides hundreds of essential equations and tables to help you select, operate and maintain measurement devices. The second edition adds brand new chapters packed with tables and equations for industrial communications buses, safety, and environmental measurements.



Format: Softbound Book—2006
ISBN: 978-1-55617-946-4
Length: 314 pages;
Weight: 1 lb.
Member: $89.00;
List: $99.00
www.isa.org/measurehandbook

ISA Certified Automation Professional® (CAP®) Job Analysis Study

ISA

The ISA CAP® certification program is designed to assess competence in the automation professionals. The process for identifying competency areas includes a job analysis study, which serves as a blueprint for examination development. For broad content areas and tasks, the study validates importance and criticality to practice. These ratings play an important role in determining the content of the examination. The ISA CAP® job analysis study consists of the following three phases, which are the focus of this report:

- Initial Development and Validation—A panel of experts identified the domains, tasks, knowledge, and skills that are essential elements of the profession of automation.
- Validation Study—A representative sample of 1,500 practicing automation professionals was asked to review and validate the work of the job analysis panel.
- Development of Test Specifications—Based on the ratings gathered from the representative sample of automation professionals, the test specifications for the examination were developed.



Format: PDF on Web – 2004
ISBN: 978-1-55617-903-7
Length: 77 pages
Member: FREE; List: $150.00
www.isa.org/capanalysis

Loop Checking: A Technician's Guide

Harley M. Jeffery

This technician's guide defines loop checking in the broader scope of control loop performance, in addition to the more traditional terms of the plant startup. It discusses general methods and practices that can be applied across many processes/industries. Featured topics include: loop checking basics, factory acceptance testing, wiring and loop checks, performance benchmarking, and sustaining performance.



Format: Softbound Book – 2005
ISBN: 978-1-55617-910-5
Length: 141 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/loopcheck

Maintenance of Process Instrumentation in Nuclear Power Plants

H.M. Hashemian

This book compiles 30 years of practical knowledge gained by the author and his staff in testing the I&C systems of nuclear power plants around the world. It focuses on process temperature and pressure sensors and the verfication of these sensors' calibration and response times. In spite of great advances in electronics, computers, and measurement technologies, important process parameters such as temperature and pressure are still measured with conventional sensing techniques found in RTDs, termocouples, and conventional pressure and differential pressure sensors. Furthermore, no improved technology providing comparable performance is currently on the horizon. Therefore, it is important to understand how these sensors function and the testing techniques for verifying their performance. This book is intended to help provide this understanding.



Format: Hardbound Book – 2007
ISBN: 3-540-33703-2
Length: 304 pages; Weight: 2 lbs.
Member: $129.00; List: $129.00
www.isa.org/nuclearpower

Maintenance of Instruments & Systems, 2nd Edition

L.D. Goettsche, Editor

This handbook provides comprehensive coverage of maintenance requirements for pneumatic and electrical/electronic devices, as well as expanded coverage of DCS systems, analytical instrumentation, fiber optics, and smart instruments. Coverage is organized by device to make finding the information quick and easy. In this edition, special emphasis is given to documentation requirements and safety issues along with updated coverage of newly issued regulations and standards.



Format: Hardbound Book – 2005
ISBN: 978-1-55617-879-5
Length: 452 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/IandSMaintenance

Order Now
www.isa.org/shopisa
(919) 549-8411

Measurement Uncertainty, 4th Edition

Ronald H. Dieck

This book presents engineering students with a comprehensive tutorial of measurement uncertainty methods in a logically categorized and readily utilized format. The new uncertainty technologies embodied in both U.S. and international standards have been incorporated into this edition with a view toward understanding the strengths and weaknesses of both. The book is designed to also serve as a practical desk reference in situations that commonly confront an experimenter. The text presents the basics of the measurement uncertainty model, non-symmetrical systematic standard uncertainties, random standard uncertainties, the use of correlation, curve-fitting problems, and probability plotting, combining results from different test methods, calibration errors, and uncertainty propagation for both independent and dependent error sources.



Format: Softbound Book – 2007
ISBN: 978-1-55617-915-0
Length: 277 pages; Weight: 2 lbs.
Member: $69.00; List: $79.00
www.isa.org/measuncertainty

Mechatronics: An Integrated Approach

Clarence De Silva

This useful reference unifies material from a variety of engineering disciplines, useful to individuals approaching mechatronics from any relevant field. The book offers a strong foundation in such core subjects as dynamic system modeling, electronic components and analysis, mechanical components and analysis, robotics, sensors/transducers and instrumentation, stepper motors, dc and ac motors and drives, hydraulic and pneumatic actuators, fluidics, automatic control, digital processing and hardware, communication and interfacing, software tools, design, and prototyping. Appendices provide additional background on Laplace and Fourier transform techniques, and software tools including MATLAB®, SIMULINK®, and LabVIEW®. The book emphasizes practical situations and applications with numerous worked examples, problems, and exercises. An entire chapter is devoted to practical case studies.



Format: Hardbound Book – 2005
ISBN: 0849312744
Length: 1344 pages; Weight: 5 lbs.
Member: $120.00, List: $120.00
www.isa.org/mechatronics

The Mechatronics Handbook: The Fusion of Mechanics, Electronics, and Information Technology

Robert H. Bishop, Editor

The standard reference, found on the desks of engineering professionals around the world, that defines the state of the art of mechatronics—the fusion of mechanics, electronics, and information technology. The field's leading practitioners and academics provide clear explanations to ensure that non-specialists can understand and use the information.



Format: Hardbound Book – 2002
ISBN: 0-8493-0066-5
Length: 900 pages; Weight: 5 lbs.
Member: $129.95; List: $129.95
www.isa.org/mechatronicshandbook

Model-Based Predictive Control: A Practical Approach

J.A. Rossiter

With its non-theoretical approach to model-based predictive control (MPC), this book is ideal for those who need to understand, concepts of predictive control. The author expands on connections to optimal control and stability, the closed-loop paradigm, robust design in MPC, implementations of MPC using only small online computational burdens, and implicit modeling for predictive control.



Format: Softbound Book – 2003
ISBN: 0849312914
Length: 344 pages; Weight: 2 lbs.
Member: $99.95; List: $99.95
www.isa.org/predictivecontrol

Order Now
www.isa.org/shopisa
(919) 549-8411

Models Unleashed: Virtual Plant and Model Predictive Control Applications

Gregory K. McMillan and Robert A. Cameron

Building on the knowledge and the goals of the best-selling book, *Advanced Control Unleashed*, this portable pocket guide goes beyond theoretical concepts and provides new insight into the implementation practices after the objectives have been defined and the technology decisions have been made. Recognizing the opportunity that models present to capture and exploit plant knowledge, the authors share their expertise on the concepts, procedures, and examples needed to construct and apply different models through the use of "state of the art" software for simulation and model predictive control. Sharing the results of the extensive testing they performed and to extend the utility of *Models Unleashed*, the authors have included a website link that lets you download Microsoft Excel spreadsheets and Powerpoint files.



Format: Softbound Book – 2004
ISBN: 978-1-55617-857-3
Length: 165 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/modelsunleashed

NACE Corrosion Engineer's Reference Book, 3rd Edition

R. Baboian

This handbook contains over 325 tables, graphs, and charts. New sections evaluate corrosion tests and data, and sections on conversion tables and physical and chemical data have been greatly expanded.



Format: Hardbound Book – 2002
ISBN: 1-57590-127-7
Length: 450 pages; Weight: 2 lbs.
Member: $104.00; List: $104.00
www.isa.org/corrosion

New Directions in Bioprocess Modeling and Control: Maximizing Process Analytical Technology Benefits

Michael A. Boudreau and Gregory K. McMillan

If you are a process design, quality control, information systems, or auto-mation engineer in the biopharmaceutical, brewing, or bio-fuel industry, this handy resource will help you define, develop, and apply a virtual plant, model predictive control, first-principle models, neural networks, and multivariate statistical process control. The synergistic knowledge discovery on bench top or pilot plant scale can be ported to industrial scale processes. This learning process is consistent with the intent in the Process Analyzer and Process Control Tools sections of the FDA's Guidance for Industry PAT – *A Framework for Innovative Pharmaceutical Development, Manufacturing and Quality Assurance*.



Format: Softbound Book – 2007
ISBN: 978-1-55617-905-1
Length: 310 pages; Weight: 1 lb.
Member: $89.00, List: $99.00
www.isa.org/bioprocess

PID Controllers: Theory, Design, and Tuning, 2nd Edition

Karl J. Åström and Tore Hägglund

This book examines the auto-tuning and adaptation features of several commercial controllers, including digital PID controllers, and discusses measures for dealing with specific challenges such as reset windup, long process dead times, and oscillatory systems.



Format: Softbound Book – 1995
ISBN: 978-1-55617-516-9
Length: 343 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/pidcontrollers

Basic Continuous Control

Practical Controls: A Guide to Mechanical Systems

Steven R. Calabrese

This book details practical methods of HVAC control, and defines the role of controls in easy-to-understand terms. It explores the various mechanical systems installed in today's facilities and takes a concise look at how each is best controlled using current technology. It covers basic rules and equipment guidelines, pros and cons, and potential mistakes to avoid.



Format: Hardbound Book – 2003
ISBN: 0-88173-447-0
Length: 469 pages; Weight: 2 lbs.
Member: $120.00; List: $120.00
www.isa.org/mechsystems

Practical Thermocouple Thermometry

T.W. Kerlin

This book gives you a resource for selecting, installing, and using devices after only a few days of study. Drawing on original research, along with broad knowledge of other literature in the field, teacher and engineer Kerlin presents a rich, hands-on approach to the subject of thermocouple thermometry, while staying focused on those aspects of the subject most applicable to industry.



Format: Hardbound Book – 1999
ISBN: 978-1-55617-644-9
Length: 126 pages; Weight: 2 lbs.
Member: $59.00; List: $69.00
www.isa.org/thermometry

Programmable Controllers, 4th Edition

Thomas A. Hughes

Ideal for self-study, this book discusses all phases of programmable controller applications from systems design and programming to installation, maintenance, and start-up. Used as a resource by thousands of technicians and engineers, this applications-based book provides a clear and concise presentation of the fundamental principles of programmable controllers for process and machine control. Increased coverage of all five standard PLC programming languages (Ladder Diagram, Function Block Diagram, Sequential Function Chart, Instruction List, and Structured Text) and the addition of numerous programming applications and examples help to clearly explain each programming language.

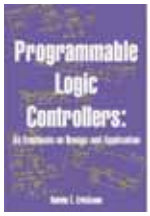


Format: Hardbound Book – 2005
ISBN: 978-1-55617-899-3
Length: 369 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/PLC

Programmable Logic Controllers: An Emphasis on Design and Application

Kelvin T. Erickson

This book presents a comprehensive treatment of programmable logic controllers (PLCs) with an emphasis on program design. The text stresses an organized approach to developing PLC programs — "Given a set of operational specifications, how does one develop the PLC program?" This book develops the design process: the tasks involved, breaking the program into manageable pieces, standard code for the various parts, and handling the sequential parts of the problem. The book also contains example problems demonstrating good design practice and culminates in two full-length case studies.



Format: Hardbound Book – 2007
ISBN: 0-9766259-0-3
Length: 1460 pages; Weight: 5 lbs.
Member: $100.00; List: $100.00
www.isa.org/programmable

Process Analyzer Sample-Conditioning System Technology

Robert E. Sherman

This resource provides both novice and experienced technologists with the technical background necessary to choose sample conditioning system components that will allow the process analyzer system to function reliably with minimal maintenance.



Format: Hardbound Book – 2001
ISBN: 0-471-29364-4
Length: 648 pages; Weight: 5 lbs.
Member: $170.00; List: $170.00
www.isa.org/processanalysis

Process Control: A Primer for the Nonspecialist and the Newcomer, 2nd Edition

George Platt

The source for a basic understanding of process control and instrumentation principles and applications. The elements of process control and measurement are described simply and with examples so that even a non-technical person can understand the essential principles.



Format: Softbound Book – 1998
ISBN: 978-1-55617-633-3
Length: 216 pages; Weight: 1 lb.
Member: $39.00; List: $49.00
www.isa.org/controlprimer

The Quality Calibration Handbook—Developing and Managing a Calibration Program

Jay L. Boucher

Quality calibration systems are the foundation for improving research and development, production, and quality assurance arenas through accurate, reliable, and traceable calibrations. This book is about how to design, implement, maintain, and continuously improve a quality calibration system, with all the required documentation, traceability, and known uncertainty for each and every type of test equipment owned and used by any company. Calibration requirements vary across specific industries, but every organization, large or small, can use the systems described in this book as a foundation for a personalized program.



Format: Hardbound Book – 2007
ISBN: 978-0-87389-704-4
Length: 190 pages; Weight: 1 lb.
Member: $84.00; List: $84.00
www.isa.org/qualitycalibration

Reading a P&ID

Job Training Systems

Learn how to read and understand Piping and Instrumentation Drawings (P&IDs). After completing the workbook, you should be able to identify symbols and function labels commonly found on P&IDs, describe how system components are related, and trace process stream flow and control loop functions.



Format: Spiralbound Book – 1999
Order No.: RPIDWBOOK
Length: 102 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/readpandid

Order Now
www.isa.org/shopisa
(919) 549-8411

Selection Guide for Temperature Instrumentation

Evaluation International

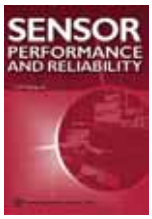
This Selection Guide for Temperature Instrumentation identifies the main categories of temperature measuring instruments widely available in industry, considers the general parameters associated with each instrument type as well as the known pros and cons of each, and discusses general selection criteria. An example instrument specification is also included. If you need to know how to select and use instruments for measuring temperature, this guide introduces the main concepts of temperature measurement, together with terms and definitions. An example instrument specification will be given, and a limited selection of market-leading U.K. manufacturers will be listed.

Format: PDF File
ISBN: RTMPRPT
Length: 141 pages; Weight: 1 lb.
Member: $340.00; List: $340.00
www.isa.org/temperature

Sensor Performance and Reliability

H. M. Hashemian

This book describes several new instrumentation testing, diagnostics, and analysis techniques to verify the reliability, health, and performance of process instrumentation. You will better understand how to objectively assess the accuracy, response time, residual life and other characteristics of installed instrumentation and have a practical means to identify the problems.



Format: Softbound Book – 2005
ISBN: 978-1-55617-897-9
Length: 306 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/senseandmeasure

Software for Automation: Architecture, Integration, and Security

Jonas Berge

This implementation-oriented book provides a clear and concise presentation of how to fully apply software in automation. It provides "how-to" information for all phases of the system lifecycle from configuration, system integration, troubleshooting, and engineering, to administration.



Format: Softbound Book – 2005
ISBN: 978-1-55617-898-6
Length: 325 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/autosoftware

Basic Continuous Control

Start-Up: A Technician's Guide

Diane Harris

This book offers you a clear overview of typical start-up responsibilities for new plants or systems. From the first team meeting to the last round of tuning and loop checking, Harris uses her extensive experience with process control plants to walk you through the issues and skills typically required. Written with the technician in mind, it is a non-mathematical, application-oriented book that details the duties a technician must perform in real-world situations.

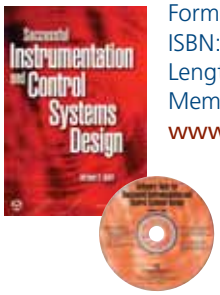


Format: Softbound Book – 2000
ISBN: 978-1-55617-677-7
Length: 217 pages; Weight: 1 lb.
Member: $42.00; List: $52.00
www.isa.org/startup

Successful Instrumentation and Control Systems Design (with CD-ROM)

Michael D. Whitt

This book and CD-ROM combination provides in-depth coverage of the design process, the elements of a successful project, the specific issues to address in a well-designed instrumentation and control system, and the engineering products that enable practical design and successful maintenance.



Format: Softbound Book and CD-ROM – 2004
ISBN: 978-1-55617-992-1
Length: 360 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/iandcdesignCD

Tuning of Industrial Control Systems, 2nd Edition

A.B. Corripio

This book includes numerous examples of tuning, including the effect of hysteresis in flow control loops, averaging and tight level control, cascade control of a jacketed chemical reactor, feedforward control of a heater, and loop interaction and ration control in a blender.



Format: Hardbound Book – 2001
ISBN: 978-1-55617-713-2
Length: 254 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/tuningsystems

Understanding Distributed Processor Systems for Control

Samuel M. Herb

This book addresses the challenges of the process control effort, recognizing that factory automation systems are no longer isolated entities and that the systems and their functions are indeed merging.



Format: Hardbound Book – 1999
ISBN: 978-1-55617-645-6
Length: 727 pages; Weight: 4 lbs.
Member: $89.00; List: $99.00
www.isa.org/distributedcontrol

Variable Speed Drives: Principles and Applications for Energy Cost Savings, 3rd Edition

David W. Spitzer

This venerable classic helps practicing engineers learn about the technical and economic potential of this technology. It presents the electrical, hydraulic, chemical, and instrumentation information necessary to technically evaluate and economically justify variable speed drive (VSD) applications.

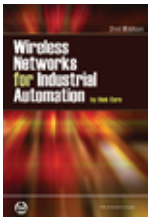


Format: Softbound Book – 2004
ISBN: 978-1-55617-859-7
Length: 204 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/vsd

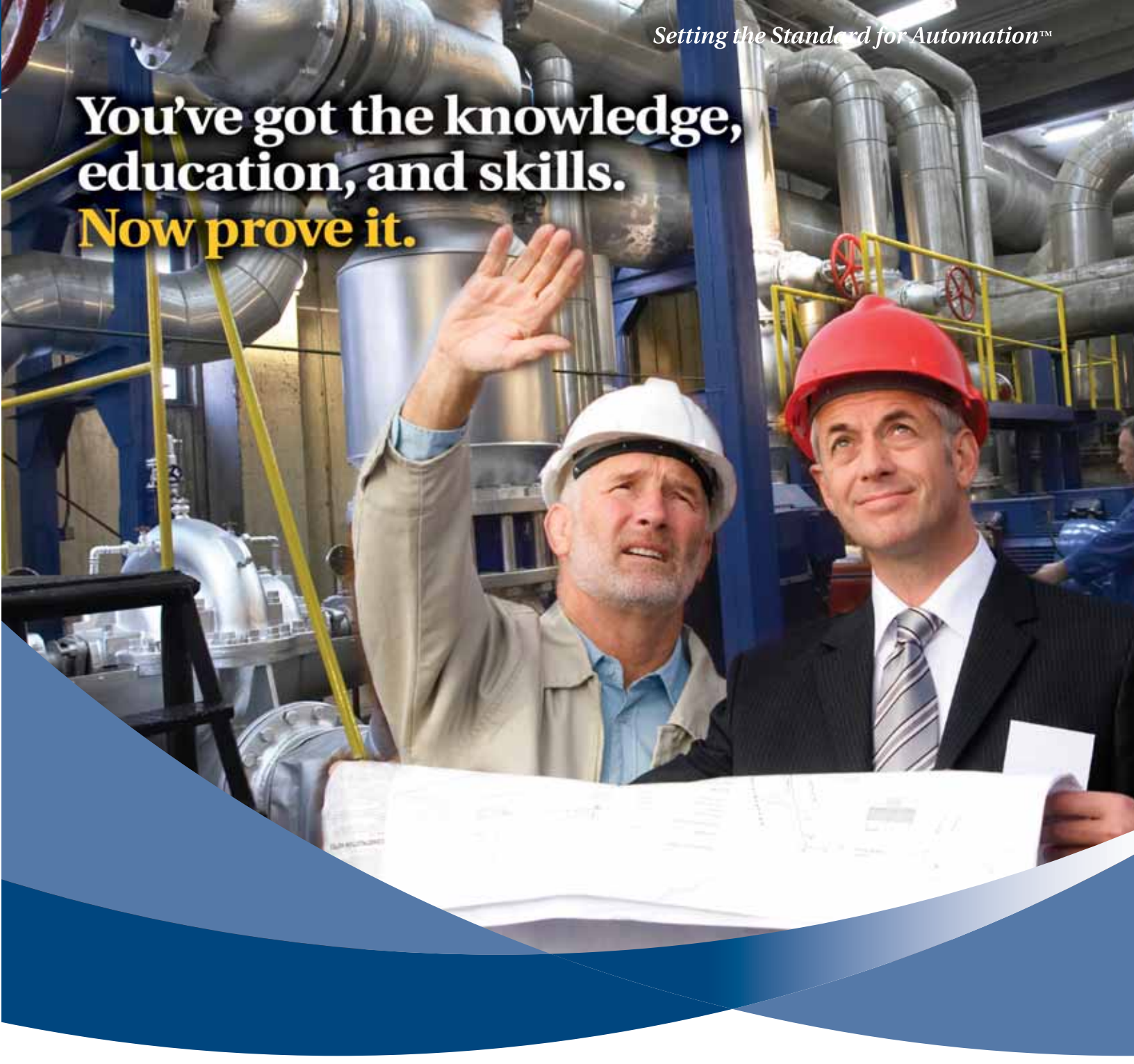
Wireless Networks for Industrial Automation, 3rd Edition

Dick Caro

This updated third edition explores wireless communications from the factory floor and process automation positioning to help engineering professionals make clear decisions on the timing and strategy for implementing wireless networks for automation projects. According to Caro, industrial users must consider cost, privacy, security problems, and the potential for signal loss in plant environments. And, industrial users must have secure communications that never fail. This book is offered in digital format. For a full year, as updates become available, you will receive email notification and be able to download a new PDF file at no charge.



Format: PDF on the Web – 2008
ISBN: 978-1-934394-36-6
Length: 109 pages
Member: $69.00; List: $79.00
www.isa.org/wirelessnetworks



You've got the knowledge, education, and skills. Now prove it.

CAPs are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CCSTs calibrate, troubleshoot, repair and replace instrumentation for measurement and control systems.

Electronic Testing Available.
Get certified when and where it's convenient for you. Hundreds of locations available around the world.

Contact ISA for additional information on any of these programs: **(919) 549-8411**
www.isa.org/certify • info@isa.org

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™

Basic Discrete, Sequencing, and Manufacturing Control

Discrete control used to be considered the opposite of process control, but now the merging of industry needs and the merging of technologies has made the distinction between the two types of control less clear. In fact, it would have made sense to combine the continuous and discrete categories of this book into a single Basic Control category. However the literature is still fairly separate, and this separate categorization still seemed best.

Some of the sensors included in the topic on Discrete Input and Output Devices and General Manufacturing Measurements are continuous, which further confuses the distinction. As with process instrumentation, this topic only scratches the surface of the available types. The topics on Motor and Drive Control and on Motion Control are not "discrete" either, but they fit with PLCs and manufacturing control. Variable Speed Drives are also used in process control.

Motion control is an unknown area to many who work in other areas of automation. However, the tremendous increase in automated equipment is rapidly increasing the range of applications of motion controllers.

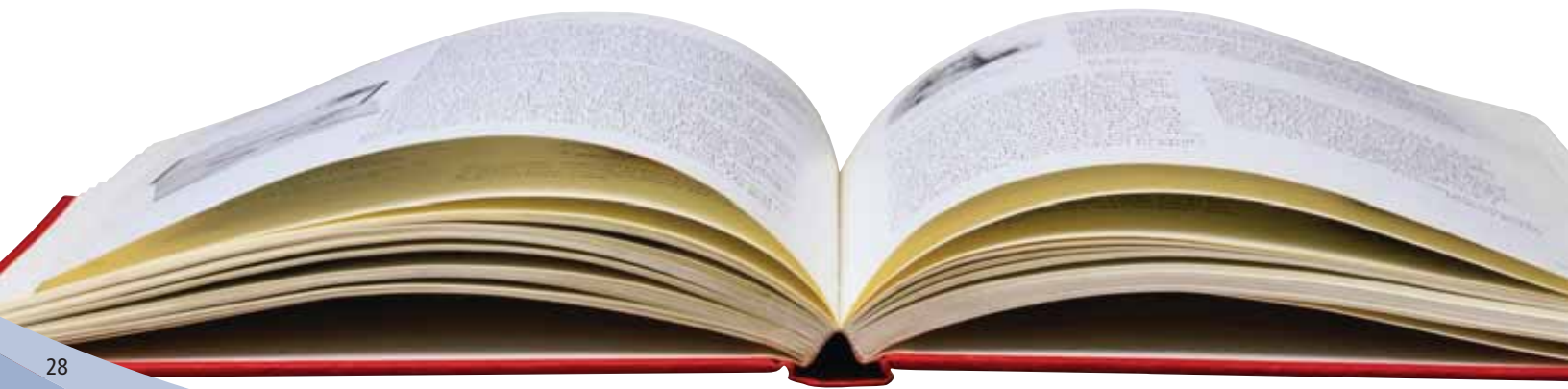
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok



Basic Discrete, Sequencing, and Manufacturing Control includes discrete input and output devices, discrete and sequencing control, as well as motor and drive control.

Advanced Control Unleashed: Plant Performance Management for Optimum Benefit

Terrence L. Blevins, Gregory K. McMillan, Willy K. Wojsznis, and Michael W. Brown

RAYMOND D. MOLLOY AWARD WINNER

Bridging the gap between theory and practice, this book provides the basis for assessing the benefits of advanced control—covering auto-tuning, model predictive control, optimization, estimators, neural networks, fuzzy control, simulators, expert systems, diagnostics, and performance assessment. The book includes a CD-ROM that provides real DeltaV simulations and displays.



Format: Hardbound Book and CD-ROM – 2003
ISBN: 978-1-55617-815-3
Length: 434 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/advancedcontrol

Automation Applications in Bio-Pharmaceuticals

George Buckbee, P.E. and Joseph Alford, Ph.D., P.E., CAP

This practical guide is ideal for engineers and designers new to the field of bio-pharmaceutical process control. For the experienced automation professional, this book outlines the unique design and application issues for the bio-pharmaceutical industry. For those already familiar with this industry, the book provides specific advice for automating these processes. Each section includes theory, practice, and practical insights into the design and application of automation for the bio-pharmaceutical industry. This book will help the engineer or designer build upon the experience of others, avoid costly design mistakes, and improve process control performance.



Format: Softbound Book – 2008
ISBN: 978-1-934394-25-0
Length: 250 pages; Weight 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/biopharmaceutical

Automation Network Selection

Dick Caro

Are you trying to make sense of all the different industrial automation networks on the market today? Whether you're a novice industrial network user or someone who simply needs to brush up on the technology, *Automation Network Selection* will help you better understand and select the "right" network for a given application. Industry expert Dick Caro walks you through the various industrial networks (e.g., sensor, fieldbus, control, safety buses, and so on) and then gives you a perspective on the typical applications for industrial automation network technology. The emphasis is on the intended application for each network, rather than on the network protocol that is more typically discussed in most textbooks.



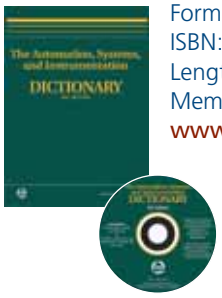
Format: Softbound Book – 2004
ISBN: 978-1-55617-861-0
Length: 161 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/networkselect

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume.

No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Order Now
www.isa.org/shopisa
(919) 549-8411

Automation Unplugged: Pinto's Perspectives, Pointers & Prognostications

Jim Pinto

RAYMOND D. MOLLOY AWARD WINNER

If you have heard industry pundit Jim Pinto speak, or read his barbed writings or laugh-out-loud poems, or subscribe to his popular e-newsletter, you'll enjoy this book. This book is loaded with critical analysis of the changing face of industrial automation; predictions about future automation technology trends—the best of his articles on marketing and distribution; and his highly rated fieldbus commentary.



Format: Softbound Book – 2004
ISBN: 978-1-55617-864-1
Length: 205 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/unplugged

Bottom-Line Automation, 2nd Edition

Peter G. Martin

This newly revised edition helps today's manufacturing companies face the challenges of a global marketplace where every part of the operation must become more efficient to enhance the bottom line. Based on the results and conclusions of an applied research project of process manufacturing operations directed by the author, *Bottom-Line Automation* presents an overview of more than 30 years of industry trends, pointing out the strengths and pitfalls of each. The book also explains how manufacturers selected automation technology suppliers for state-of-the-art technological features in their systems, rather than for the improvements the technology could offer the manufacturing operation. New topics in the book discuss the economic optimization of industrial assets.

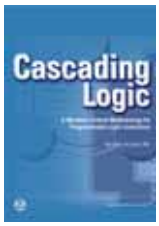


Format: Hardbound Book – 2006
ISBN: 978-1-55617-962-4
Length: 234 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/bottomline

Cascading Logic: A Machine Control Methodology for Programmable Logic Controllers

Gary Kirckof, P.E.

Learn how to create a functional machine control program for industrial equipment that is sequential in nature. The programming methodology starts by breaking the machine into its basic elements. The methodology then shows how to program each element and how to assemble them into a complete machine control program. This book is intended to give you the confidence you need to reach decisions and move forward with certainty. The sequential nature of events will also help operators and maintenance personnel troubleshoot and maintain the equipment after it is put into operation. Ladder logic illustrations useful with any PLC demonstrate each part of the text.



Format: Softbound Book – 2003
ISBN: 978-1-55617-814-6
Length: 206 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/cascadelogic

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

Control System Engineering Exam Reference Manual: A Practical Study Guide

Bryon Lewis

This practical reference manual, available in downloadable PDF format, is for use in preparing for the Control Systems Engineering (CSE) examination, which covers a broad range of subjects from the electrical, mechanical, and chemical engineering disciplines. This manual will review the foundations of process control and demonstrate the breadth and width of the CSE exam. Topics include process measurement and calibration, sizing process, control elements, digital logic, analog signals, process safety and shutdown, industrial control networks, NEC and NFPA codes.



Format: PDF on Web – 2007
ISBN: 978-1-934394-22-9
Length: 126 pages
Member: $49.00; List: $59.00
www.isa.org/csereference

Control Systems Engineering (CSE) Study Guide, 4th Edition

This fourth edition, a downloadable PDF File, is for use in preparing for the CSE exam. The manual provides candidates for this professional engineer (PE) license with information about the scope and design of the CSE exam, sample questions, and resources for further study.



Format: PDF on Web – 2003
ISBN: 978-1-55617-881-8
Length: 112 pages
Member: $49.00; List: $59.00
www.isa.org/cseguide

Order Now
www.isa.org/shopisa
(919) 549-8411

Design Patterns for Flexible Manufacturing

Dennis Brandl

This handy resource defines an effective set of design patterns and rules you should know when applying the widely used ISA-88 industry standards to batch manufacturing (called the S88 design pattern) and continuous and discrete manufacturing (called the NS88 design pattern for non-stop production). The author, who is the chair of the ISA-SP88 committee, developed these patterns and subsequent rules while applying the batch series in several projects. This book clearly identifies what elements are defined in the batch series and what elements make up the S88 and NS88 design patterns for flexible manufacturing. The book defines design patterns for control system programming, providing patterns for the organization of programmable logic controller (PLC), digital control system (DCS), and other control system application code. Whether you are in a batch, continuous, or discrete manufacturing environment, these design patterns can be applied to a wide range of production systems, making systems easier to design and implement.



Format: Softbound Book – 2007
ISBN: 978-1-55617-998-3
Length: 350 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/designpatterns

Fundamentals of Industrial Control, 2nd Edition (with CD-ROM)

D.A. Coggan, Editor

This updated edition of a best-selling classic provides a unique balance between control system theory and effective application, covering every major control system element, from sensors to final control elements, in the context of overall control strategies and system design. This edition provides updates on a range of topics including Internet communications, industrial communications network hardware and software, open systems, wireless networks, enterprise computing, and computer and control system security. Other sections introduce the new generation of distributed controls and digital automation systems, and present migration strategies for expanding and upgrading existing systems. As a bonus, the book comes with the full text and graphics on a CD-ROM.



Format: Hardbound Book and CD-ROM – 2005
ISBN: 978-1-55617-863-4
Length: 893 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/fundamentalcontrol

A Guide to Biotechnology Law and Business

Robert A. Bohrer

This book is a guide to the "big picture" of biotechnology law and business, presenting and analyzing the issues that arise throughout the life cycle of a biotechnology product. The book focuses primarily on the pharmaceutical field from research, through development, into the marketplace. This book aims to provide guidance by providing a practical and integrated discussion in the development and manufacturing of biotechnology products. This book will prove useful to biotech executives who want a concise introduction to the most fundamental issues in biotechnology law and business.



Format: Hardbound Book – 2007
ISBN: 978-1-59460-087-6
Length: 341 pages; Weight: 2 lb.
Member: $60; List: $60
www.isa.org/biotechnology

Human Machine Interface Design for Process Control Applications

Jean-Yves Fiset

Users and designers of industrial control and monitoring systems (e.g., distributed control, supervisory control and data acquisition, and stand-alone) will benefit from this book's easy-to-use—yet effective—method on how to configure, design, and validate human-machine interfaces (HMIs). It discusses the overall HMI design process and how that process relates to system design. It also details design methods, principles and rules for individuals or groups of displays, as well as the integration of software-based and hardwired HMIs. This book will help guide you on the design of HMIs for other, less common, yet important, components, such as expert systems and other electronically-displayed operating procedures. Using the knowledge contained in this book, you can determine how to configure or design a whole new set of displays for a system or how to enhance specific elements of an existing or planned HMI.



Format: Softbound Book - 2008
ISBN: 978-1-934394-35-9
Length: 250 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/HMI

IEC 61499 Function Blocks for Embedded and Distributed Control Systems Design

Valeriy Vyatkin

IEC 61499 is the newly adopted standard for distributed control systems and follows on from the IEC 61131 standard for programmable logic controllers (PLC). This book is a practical guide for component-based development of distributed embedded and control systems as proposed by the new international standard. Each chapter is designed as an independent study unit, making the book ideal for use in university courses, industrial training, or self-study. Working knowledge of the IEC 61499 standard can be achieved in approximately 10 to 15 learning hours. For the control, automation, or software engineer and the embedded systems developer, this book provides concrete directions on how to specify and implement a distributed system according to the IEC 61499 standard and how to create an IEC 61499-compliant control device.



Format: Softbound Book – 2007
ISBN: 978-0-9792343-0-9
Length: 300 pages; Weight 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/iec61499

Industrial Data Communications, 4th Edition

Lawrence (Larry) M. Thompson

Updates in networking and data communications are occurring rapidly. This new edition includes information on 100 MBps and 1000 MBps Ethernet, RIP and OSPF router technologies, OLE for Process Control (OPC), ActiveX, and .NET™, virtual private networks, and more. The author also provides a historical overview, and discusses networking and communication foundations; physical layer and data link standards; LANs; network operating systems and LAN management; industrial networks and their applications; and WANs. A complete glossary and index make this book an excellent desk reference.



Format: Softbound Book – 2007
ISBN: 978-1-934394-24-3
Length: 260 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/datacom

Order Now
www.isa.org/shopisa
(919) 549-8411

Industrial Ethernet, 2nd Edition

Perry S. Marshall and John S. Rinaldi

This revised best-seller is a convenient installation, trouble-shooting, and reference tool on one of the hottest topics in automation and process control—Ethernet and TCP/IP industrial protocols. You will quickly gain a solid grasp of Ethernet basics, the constraints of the industrial environment, and the specialized requirements of machine control.



Format: Softbound Book – 2005
ISBN: 978-1-55617-892-4
Length: 129 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/ethernet

Industrial Ethernet on the Plant Floor: A Planning and Installation Guide

Robert Lounsbury

This book outlines the select components to layout, install, test, certify, and troubleshoot a network system. It describes the fundamentals of designing industrial physical layers, network architectures, and components. The author provides the reader with the basics of noise—how to mitigate and abate it through installation techniques and selection of components that can provide a level of performance needed in a hostile industrial environment. The book also covers environmental considerations, Ethernet security, and certification.



Format: Softbound Book – 2008
ISBN: 978-1-934394-26-7
Length: 150 pages; Weight: 1.0 lb.
Member: $59.00; List: $69.00
www.isa.org/industrialethernet

Industrial Network Security

David J. Teumim

Process control systems for chemical plants and refineries; SCADA networks for utilities like electric, natural gas, water and wastewater; factory automation systems for discrete manufacturing—all of these areas need protection from hackers, disgruntled employees, and even cyberterrorists. This book leverages the technical information ISA has developed in its training courses and conferences, includes a brief tutorial on the basics of IT cybersecurity for the uninitiated, and takes a look at the many different organizations contributing to this area.



Format: Softbound Book – 2005
ISBN: 978-1-55617-874-0
Length: 144 pages; Weight: 1 lb.
Member: $54.00; List: $59.00
www.isa.org/networksecurity

ISA Handbook of Measurement Equations and Tables, 2nd Edition

Jim Strothman

This updated, expanded, and improved version provides hundreds of essential equations and tables to help you select, operate, and maintain measurement devices. The second edition adds new chapters packed with tables and equations for industrial communications buses, safety, and environmental measurements. Tables and equations have been added to all of the previous edition's chapters covering units of measurement, pressure, flow, temperature, level, humidity, electrical and viscosity measurements.



Format: Softbound Book – 2006
ISBN: 978-1-55617-946-4
Length: 314 pages; Weight: 1 lb.
Member: $89.00; List: $99.00
www.isa.org/measurehandbook

The Life and Times of an Automation Professional — An Illustrated Guide

Ted Williams, Stan Weiner, and Gregory K. McMillan

Top Three Reasons to Buy This Book

3) Get a refund. If this book is no laughing matter, send the original UPC along with a self-addressed stamped envelope to Stan and Greg. They will email you the tracking number for the $0.25 royalty rebate along with a picture of Stan cleaning the pool.
2) Discounts for futuristic "Great Automator" and "Robbie the Robot" action figures. Just show your copy at the next ISA EXPO to your friendly ISA bookstore person. Pose the figures to guard the entrance to your cubicle.*
1) Great preparation for the upcoming "Automation Reality" TV program. Twenty vendors and one "user" are locked in a room with 1000 PowerPoint slides—very similar to a local ISA meeting.

*Action figures may not be available in your lifetime and discounts are not inheritable; however, the editor is working on a plan to make them available as SYMBOLS for your nursing home nightstand.

At the current time, this book will not be offered as an Educational Book Cartridge for the Leapster Multimedia Learning System Library. The Leapfrog Company cannot decide whether the book is suitable for pre-kindergarten.

Warning from the Automator General — This book may cause uncontrollable laughter. Do not read this in combination with any other books by Stan and Greg.



Format: Softbound Book – 2006
ISBN: 978-1-55617-957-0
Length: 192 pages; Weight: 1 lb.
Member: $20.00; List: $25.00
www.isa.org/autolife

Motion Control

When it comes to automated equipment, are you finding yourself in unfamiliar territory? Take the mystery out of Motion Control with these comprehensive handbooks and practical guides.

Mechatronics: An Integrated Approach

Clarence De Silva

This useful reference unifies material from a variety of engineering disciplines, useful to individuals approaching mechatronics from any relevant field. The book offers a strong foundation in such core subjects as dynamic system modeling, electronic components and analysis, mechanical components and analysis, robotics, sensors/ transducers and instrumentation, stepper motors, dc and ac motors and drives, hydraulic and pneumatic actuators, fluidics, automatic control, digital processing and hardware, communication and interfacing, software tools, design, and prototyping. Appendices provide additional background on Laplace and Fourier transform techniques, and software tools including MATLAB®, SIMULINK®, and LabVIEW®. The book emphasizes practical situations and applications with numerous worked examples, problems, and exercises. An entire chapter is devoted to practical case studies.



Format: Hardbound Book—2005
ISBN: 0849312744
Length: 1344 pages; Weight: 5 lbs.
Member: $120.00; List: $120.00
www.isa.org/mechatronics

The Mechatronics Handbook: The Fusion of Mechanics, Electronics, and Information Technology

Robert H. Bishop, Editor

The standard reference, found on the desks of engineering professionals around the world, that defines the state of the art of mechatronics—the fusion of mechanics, electronics, and information technology. The field's leading practitioners and academics provide clear explanations to ensure that non-specialists can understand and use the information.

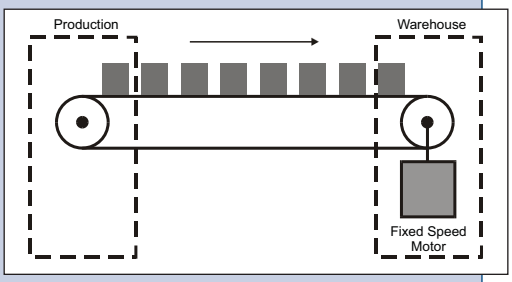


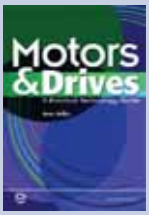
Format: Hardbound Book – 2002
ISBN: 0-8493-0066-5
Length: 900 pages; Weight: 5 lbs.
Member: $129.95; List: $129.95
www.isa.org/mechatronicshandbook

Motors & Drives: A Practical Technology Guide

David Polka

This practical reference introduces engineering concepts of motors and drives in a way that can be easily understood by both engineers unfamiliar with the technology and technicians who are technically literate, but not accustomed to complex theory and mathematics. It features simple explanations, summaries, review questions, glossaries, and reference tables for formulas and conversions. The text explains the principles of DC and variable frequency AC drive technology. It provides an overview of drive components and types of drives, with special emphasis given to common motion control applications for each. It covers DC and AC motor and drive operation, step motors, AC vector motors, brushless servo motors, linear stepper and linear servo motors, drive innovations such as vector drives, PWM stepper and servo drives. Feedback devices such as tachometers, resolvers, and encoders are also addressed as they relate to speed and torque control. Later chapters cover drive systems control methods and the maintenance and troubleshooting of drive systems. Design engineers, automation and control specialists, maintenance technicians, and students will find this to be an invaluable resource both as a tutorial and desk reference.





Format: Softbound Book – 2003
ISBN: 978-1-55617-800-9
Length: 370 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/motorsanddrives

Motors & Drives: A Practical Technology Guide — Instructional Supplement

David Polka

This series of PowerPoint® slides and lab exercises is useful for self-study or as an instructor tool for classroom instruction to understand the concepts of motors and drives. It is a useful companion to the book *Motors & Drives: A Practical Technology Guide.*

Format: Downloadable PowerPoint and Word Files – 2005
ISBN: RMDGUIDE
Member: $69.00; List: $79.00
www.isa.org/manddguide

Loop Checking: A Technician's Guide

Harley M. Jeffery

This technician's guide defines loop checking in the broader scope of control loop performance, in addition to the more traditional terms of the plant startup. It discusses general methods and practices that can be applied across many processes/industries. Featured topics include loop checking basics, factory acceptance testing, wiring and loop checks, performance benchmarking, and sustaining performance.



Format: Softbound Book – 2005
ISBN: 978-1-55617-910-5
Length: 141 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/loopcheck

Measurement and Control Basics, 4th Edition (with CD-ROM)

Thomas A. Hughes

Ideal for classroom use or self-study, this newly revised best-selling book has provided thousands of students, technicians, sales people, and others with a practical introduction to the technologies, systems, and strategies involved in industrial process control. The fourth edition takes the same proven intuitive approach of previous editions. Each chapter begins with basic definitions and mathematical concepts that allow readers to become well-versed in the principles necessary to understand the variables that affect process control systems. New features in the fourth edition include improved coverage of analytical measurement and control, and the addition of sections on the international standard for PLC languages; process visualization; and personal computer- based control systems. The book also includes solutions to exercises that make it more suitable for self-study.



Format: Softbound Book – 2007
ISBN: 978-1-55617-916-7
Length: 375 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/mandcbasics

Order Now
www.isa.org/shopisa
(919) 549-8411

Out of the Barn

Dick Morley

This book, authored by PLC inventor Dick Morley, covers floor level control systems; industrial computing: the camel has the tent; software: indispensable exasperation; managing manufacturing: multi-tasking personified; enhancing corporate wealth; global reports; evaluating entrepreneurship; and we have met the enemy: they is us!



Format: Softbound Book – 2003
ISBN: 978-1-55617-822-1
Length: 150 pages; Weight: 1 lb.
Member: $20.00; List: $20.00
www.isa.org/barn

Practical Machinery Management for Process Plants: Major Process Equipment Maintenance and Repair, 2nd Edition, Volume 4

Heinz P. Bloch and Fred K. Geitner

An invaluable source of practical cost-effective maintenance, repair, installation, and field verification procedures for machinery engineers. It is filled with step-by-step instructions and quick-reference checklists that describe preventive and predictive maintenance for major process units such as vertical, horizontal, reciprocating, and liquid ring vacuum pumps, fans and blowers, compressors, turboexpanders, turbines, and more.



Format: Hardbound Book – 1997
ISBN: 0-88415-663-X
Length: 700 pages; Weight: 1 lb.
Member: $130.00; List: $130.00
www.isa.org/practicalmachinery

Programmable Controllers, 4th Edition

Thomas A. Hughes

Ideal for self-study, this book discusses all phases of programmable controller applications from systems design and programming to installation, maintenance, and start-up. Increased coverage of all five standard PLC programming languages (Ladder Diagram, Function Block Diagram, Sequential Function Chart, Instruction List, and Structured Text) and the addition of numerous programming applications and examples clearly explain each programming language.



Format: Hardbound Book – 2005
ISBN: 978-1-55617-899-3
Length: 369 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/PLC

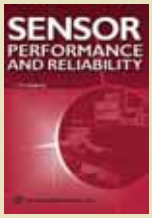
Sensor Technology

Sensor Performance and Reliability

H. M. Hashemian

This book describes several new instrumentation testing, diagnostics, and analysis techniques to verify the reliability, health, and performance of process instrumentation. You will better understand how to objectively assess the accuracy, response time, residual life and other characteristics of installed instrumentation and have a practical means to identify the problems.

Look to these comprehensive resources for self-study purposes and a practical understanding of the challenges faced with the latest sensing technology.



Format: Softbound Book – 2005
ISBN: 978-1-55617-897-9
Length: 306 pages; Weight: 1 lb.
Member: $89.00; List: $99.00
www.isa.org/senseandmeasure

Sensor Selection Guide: Optimizing Manufacturing and Processes, 2nd Edition

Paul W. Chapman

This one-of-its-kind book offers a self-study course addressing all the facets of sensor applications including a sensors application guide. This guide can be used to direct the reader to a family of sensors best suited to be used in specific sensor applications. Filled with examples of applying various materials technologies, this book also provides a wealth of information from theoretical and practical perspectives. This book also includes descriptions of sensors, sensor technologies, sensor manufacturing, artificial intelligence, and much more.



Format: Softbound Book – 2006
ISBN: 978-1-55617-987-7
Length: 168 pages: Weight 1 lb.
Member: $89.00; List: $99.00
www.isa.org/sensorguide

Sensor Technology Handbook

Jon Wilson

Today's smart sensors, wireless sensors, and microtechnologies are revolutionizing sensor design and applications. This volume is a comprehensive sensor reference guide. A CD-ROM accompanies the volume.



Format: Hardbound Book and CD-ROM – 2005
ISBN: 0-7506-7729-5
Length: 704 pages; Weight: 4 lbs.
Member: $120.00; List: $120.00
www.isa.org/sensorhandbook

Temperature sensor diagnostics

Troubleshooting old technology devices takes high-tech tilt

Here is a review of practical techniques for in-situ diagnostics of problems in temperature measurements with resistance temperature detectors (RTDs) and thermocouples in industrial processes. These are diagnostic techniques based on the Loop Current Step Response (LCSR) method for the following applications:

1. How to verify the attachment of temperature sensors to solid surfaces in aerospace applications;
2. Detecting secondary junctions in thermocouples as installed in operating processes;
3. Remote testing to identify reverse-connected thermocouples;
4. Verifying adequate insertion of RTDs and thermocouples in thermowells;
5. Separating temperature sensor problems from cable problems;
6. Using RTDs to verify water level in pipes and vessels;
7. In-situ detection of gross inhomogeneity in thermocouples.

For full *InTech®* story, see **www.isa.org/link/TempSD**

COMMERCIALLY AVAILABLE SENSOR TECHNOLOGIES

A. Basic electromechanical

B. Semiconductor-based
   1. Piezoresistive
   2. Magnetoresistive
   3. Hall effect
   4. Inductive
   5. Thin-film microstructure
      a. Mechanical
      b. Thermal
   6. RTDs (resistance temperature devices)
   7. Gallium arsenide

C. Optoelectronic
   1. Photoelectric
   2. Optoelectronic
   3. Fiberoptic
   4. Vision Sensors
   5. Laser Sensors
   6. VCSEL Sensors

D. Piezoelectric
   1. Solid state
   2. Polymer

E. Biochemical

F. Electrochemical

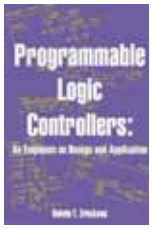
G. Organic/enzymatic

Source: *Sensor Selection Guide: Optimizing Manufacturing and Processes, 2nd Edition* by Paul W. Chapman

Programmable Logic Controllers: An Emphasis on Design and Application

Kelvin T. Erickson

This book presents a comprehensive treatment of programmable logic controllers (PLCs) with an emphasis on program design. The text stresses an organized approach to developing PLC programs — "Given a set of operational specifications, how does one develop the PLC program?" This book develops the design process: the tasks involved, breaking the program into manageable pieces, standard code for the various parts, and handling the sequential parts of the problem. The book also contains example problems demonstrating good design practice and culminates in two full-length case studies.



Format: Hardbound Book – 2007
ISBN: 0-9766259-0-3
Length: 1460 pages; Weight: 5 lbs.
Member: $100.00; List: $100.00
www.isa.org/programmable

Safety Instrumented Systems Verification — Practical Probabilistic Calculations

William M. Goble and Harry L. Cheddie

RAYMOND D. MOLLOY AWARD WINNER

This book clearly explains how to do probabilistic calculations to accomplish safety integrity level (SIL) verification for safety systems. Starting with a description of the safety lifecycle, the authors show where and how SIL verification fits into the key activities from conceptual design through commissioning.



Format: Softbound Book – 2005
ISBN: 978-1-55617-909-9
Length: 382 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/sisverify

SCADA: Supervisory Control and Data Acquisition, 3rd Edition

Stuart A. Boyer

RAYMOND D. MOLLOY AWARD WINNER

This clear and concise reference book offers overviews of SCADA's component technologies, as well as details necessary to understand the big picture. After completing its 14 self-study units, you will be conversant with SCADA nomenclature and architecture, understand the basic technology of the system's building blocks, understand its limitations, understand how it can benefit particular operations, and have a basis for selecting appropriate SCADA technologies for your operational requirements.



Format: Softbound Book – 2004
ISBN: 978-1-55617-877-1
Length: 219 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/scada

Selection Guide for Temperature Instrumentation

Evaluation International

This Selection Guide for Temperature Instrumentation identifies the main categories of temperature measuring instruments widely available in industry, considers the general parameters associated with each instrument type as well as the known pros and cons of each, and discusses general selection criteria. An example instrument specification is also included. If you need to know how to select and use instruments for measuring temperature, this guide introduces the main concepts of temperature measurement, together with terms and definitions. An example instrument specification will be given, and a limited selection of market-leading U.K. manufacturers will be listed.

Format: PDF File
ISBN: RTMPRPT
Length: 141 pages; Weight: 1 lb.
Member: $340.00; List: $340.00
www.isa.org/temperature

Order Now
www.isa.org/shopisa
(919) 549-8411

Basic Discrete, Sequencing, and Manufacturing Control

Software for Automation: Architecture, Integration, and Security

Jonas Berge

This implementation-oriented book provides a clear and concise presentation of how to fully apply software in automation. It provides "how-to" information for all phases of the system lifecycle from configuration, system integration, troubleshooting, and engineering to administration.



Format: Softbound Book – 2005
ISBN: 978-1-55617-898-6
Length: 325 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/autosoftware

Tuning of Industrial Control Systems, 2nd Edition

A.B. Corripio

This book includes numerous examples of tuning, including the effect of hysteresis in flow control loops, averaging and tight level control, cascade control of a jacketed chemical reactor, feedforward control of a heater, and loop interaction and ration control in a blender.



Format: Hardbound Book – 2001
ISBN: 978-1-55617-713-2
Length: 254 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/tuningsystems

Understanding Distributed Processor Systems for Control

Samuel M. Herb

This book addresses the challenges of the process control effort, recognizing that factory automation systems are no longer isolated entities and that the systems and their functions are indeed merging.



Format: Hardbound Book – 1999
ISBN: 978-1-55617-645-6
Length: 727 pages; Weight: 4 lbs.
Member: $89.00; List: $98.00
www.isa.org/distributedcontrol

Variable Speed Drives: Principles and Applications for Energy Cost Savings, 3rd Edition

David W. Spitzer

This venerable classic helps practicing engineers learn about the technical and economic potential of this technology. It presents the electrical, hydraulic, chemical, and instrumentation information necessary to technically evaluate and economically justify variable speed drive (VSD) applications.

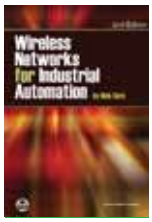


Format: Softbound Book – 2004
ISBN: 978-1-55617-859-7
Length: 204 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/vsd

Wireless Networks for Industrial Automation, 3rd Edition

Dick Caro

This updated third edition explores wireless communications from the factory floor and process automation positioning to help engineering professionals make clear decisions on the timing and strategy for implementing wireless networks for automation projects. According to Caro, industrial users must consider cost, privacy, security problems, and the potential for signal loss in plant environments. And, industrial users must have secure communications that never fail. This book is offered in digital format. For a full year, as updates become available, you will receive email notification and be able to download a new PDF file at no charge.



Format: PDF on the Web – 2008
ISBN: 978-1-934394-36-6
Length: 109 pages
Member: $69.00; List: $79.00
www.isa.org/wirelessnetworks

Order Now
www.isa.org/shopisa
(919) 549-8411



Setting the Standard for Automation™

Free Web Seminars from ISA

Get the ISA Advantage

Enhance your career with ISA membership AND have free access to our library of 40+ pre-recorded online web seminars, valued at $225 each. ISA's web seminars, led by leading experts in the field, cover topics like network security, safety, ethernet, fieldbus, and much more.

To learn more about this benefit and many others, visit **www.isa.org/memberbenefits**.

Take advantage of this Member benefit today!

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™

Advanced Control

Not all the topics in this category are really advanced control, but they are in some sense above and beyond basic plant floor controls. The first topic, Process Modeling, is not really control at all, but modeling is becoming increasingly important in automation as systems become so complex that control design, checkout of configuration, and operator training all need to use process models.

The neural network type of modeling seems so academic to some people who meet it for the first time that it hardly seems worth a second glance. However, the successful application of this technology to virtual or soft sensors for things too complex to easily measure online, including stack gas composition, makes this technology something that every automation professional needs to understand.

Advanced Process Control, the second topic, is truly advanced control. Over time, what we used to call "advanced" becomes commonplace and some new things take on the advanced name. A few years ago, feedforward control was considered "advanced," but today we simply group that with the regulatory control in the Continuous Control topic. The model reference, model predictive control, and fuzzy logic are generally considered advanced today, but since these are built-in capabilities in current-generation DCS controllers, these will become so common that, in a few years, we will consider them part of basic control. Of this group, model predictive control is the most widely used and should be in the knowledge base of all automation professionals, whether or not they work in the process area.

Batch is one of the success stories of ISA standardization work, with the widespread acceptance and benefits of the ANSI/ISA88 Batch Control series of standards. In this sense, "batch" does not mean just a batch process where the steps are sequenced. Rather, batch in the ISA88 context means the use of recipes that are separated from the equipment capabilities. A general knowledge of the batch standard is fundamental to automation knowledge.

Environmental is not exactly an automation discipline, but it is so important in automation that it has been given its own topic. Monitoring of emissions is a very important task for automation professionals so that subject has been given strong emphasis.

Some years ago, heating, ventilating, and air-conditioning controls (HVAC) were considered too commercial and low cost to be a real part of true automation. However, today Building Automation is a rapidly growing technical field that we increasingly need to treat as a part of automation. The sophistication of controls and communications in this area now parallel the sophistication in any other area of automation.

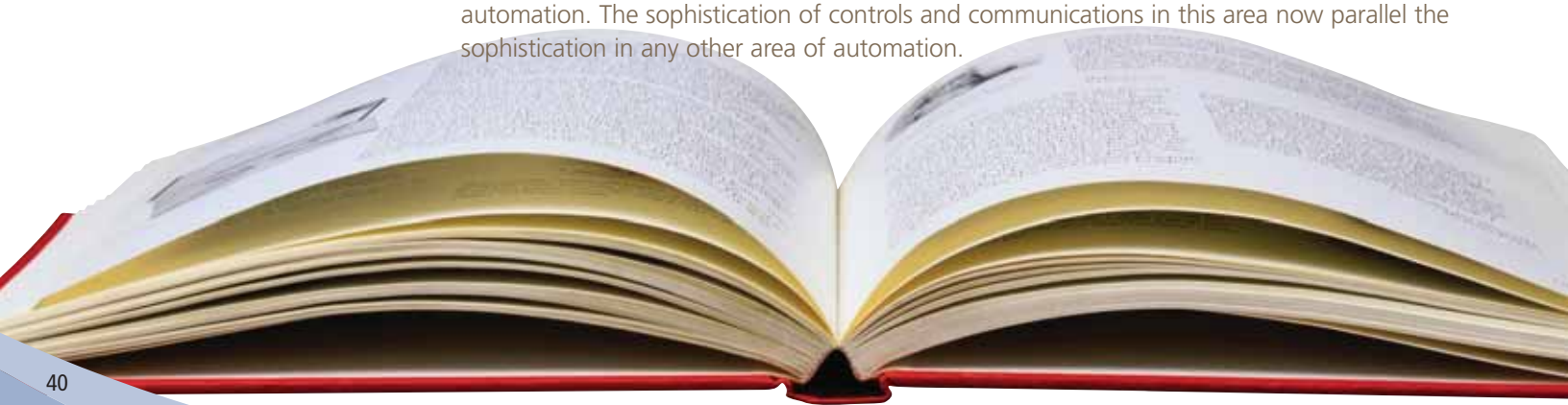
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok



Advanced Control includes process modeling, batch control, building automation, environmental issues, and the like.

Advanced Control Unleashed: Plant Performance Management for Optimum Benefit

Terrence L. Blevins, Gregory K. McMillan, Willy K. Wojsznis, and Michael W. Brown

RAYMOND D. MOLLOY AWARD WINNER

Bridging the gap between theory and practice, this soon-to-be classic provides the basis for assessing the benefits of advanced control. It covers auto-tuning, model predictive control, optimization, estimators, neural networks, fuzzy control, simulators, expert systems, diagnostics, and performance assessment. As an added bonus, the book includes a CD-ROM that provides real DeltaV simulations and displays. Configuration and case files are supplied for a hands-on experience, and PowerPoint files suitable for lectures on each unit are also included.



Format: Hardbound Book and CD-ROM – 2003
ISBN: 978-1-55617-815-3
Length: 434 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/advancedcontrol

Advanced pH Measurement and Control, 3rd Edition

Gregory K. McMillan and Robert A. Cameron

This best-selling book is now expanded and updated, providing a clear, concise, and comprehensive view of how to select, install, and maintain electrodes, control valves, and control strategies for pH applications critical for product and water quality in the process industry. The book covers every aspect of system design, including the mixing and reagent piping requirements that are important for a successful application.



Format: Hardbound Book – 2005
ISBN: 978-1-55617-851-1
Length: 294 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/pH

Advanced PID Control

Karl J. Åström and Tore Hägglund

RAYMOND D. MOLLOY AWARD WINNER

The authors of the best-selling book *PID Controllers: Theory, Design, and Tuning* once again combine their extensive knowledge in the PID arena to bring you an in-depth look at the world of PID control. This book builds on the basics learned in *PID Controllers*, but augments it through use of advanced control techniques. This book ensures a smooth transition from PID control to more advanced model-based controllers.

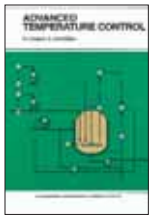


Format: Softbound Book – 2006
ISBN: 978-1-55617-942-6
Length: 461 pages; Weight: 2 lbs.
Member: $95.00; List: $112.00
www.isa.org/advancedPID

Advanced Temperature Control

Gregory K. McMillan and C.M. Toarmina

A step-by-step approach for analyzing all of the factors that affect the control of temperature loops, including major sources of measurement errors, this book is packed with information on improving control of temperature loops. The book also focuses on the advantages and disadvantages of new methods for feedback control.



Format: Hardbound Book – 1995
ISBN: 978-1-55617-540-4
Length: 226 pages; Weight: 2 lbs.
Member: $49.00; List: $59.00
www.isa.org/advtempcontrol

Analytical Instrumentation

Robert E. Sherman

This treatment of process analytical technology includes theory and real-world, hands-on experience. Included are drawings of efficient sample systems. Justifications and organization guidelines on process analyzer systems are presented. The volume describes analyzers from the systems side, looking at implementation issues including justification, purchasing, training, and validation.



Format: Hardbound Book – 1996
ISBN: 978-1-55617-581-7
Length: 728 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/analyticalhandbook

Applying S88: Batch Control from a User's Perspective

Jim H. Parshall and Larry B. Lamb

The release of ANSI/ISA-88.01-1995 led to revolutionary advances in the way factories design, implement, and integrate flexible, modular processes. From designing better processes to combining recipes with equipment control to execution validation test plans, this book offers an introduction to ISA-88 as well as expert insights into applying it. The book's clear, lighthearted style makes it the ideal introduction to ISA-88 for everyone.



Format: Softbound Book – 2000
ISBN: 978-1-55617-703-3
Length: 158 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/applyS88

Basic and Advanced Regulatory Control: System Design and Application, 2nd Edition

Harold L. Wade

This best-seller teaches the practice of process control for the wet process industries. It stresses the study of real, imperfect processes rather than system theory, and gives guidance on how you can best apply your own experience, intuition, and knowledge of the particular process. The text summarizes the general characteristics of processes and control loops and discusses feedback control and its nuances. The latter part of the book addresses advanced control techniques.



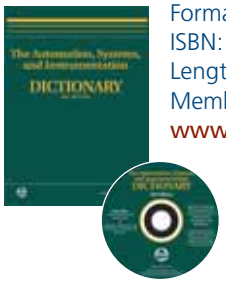
Format: Softbound Book – 2004
ISBN: 978-1-55617-873-3
Length: 372 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/regcontrol

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume.

No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Batch Control Systems—Design, Application, and Implementation, 2nd Edition

William M. Hawkins and Thomas Fisher

This revision of the 1990 work by Thomas Fisher covers an introduction to batch processes; batch control system structures; batch control; batch communications and batch control system design. Hawkins offers a comprehensive analysis of the development and evolution of batch control from the original NAMUR model through the most current publications in the ISA-88 series. Through examples, commentary, analogies and at times, wry humor, the author provides an in-depth philosophical discussion of how batch control and all manufacturing enterprises have been impacted by the work of ISA-88.



Format: Softbound Book – 2006
ISBN: 978-1-55617-967-9
Length: 300 pages; Weight: 1 lb.
Member: $89.00; List: $99.00
www.isa.org/batchcontrol

Order Now
www.isa.org/shopisa
(919) 549-8411

Boiler Control Resources

If you're new to the field of boiler control systems or a veteran wanting new insights, these boiler control resources will help guide you in the right direction.

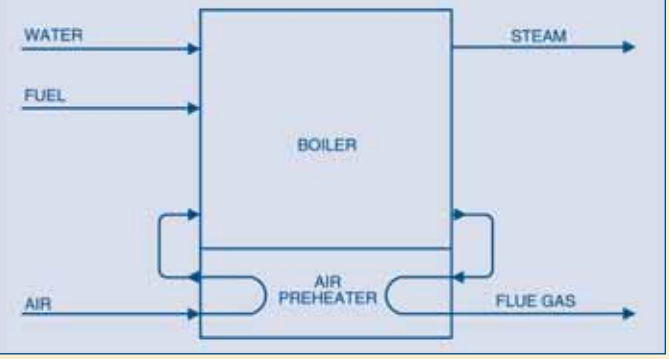
Boiler Control Systems Engineering

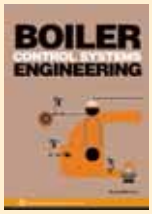
Jerry Gilman

This book addresses issues to be considered when defining measurement transmitters and specification for transmitters. Final control elements are reviewed as to characteristics and sizing. Engineering details on control systems and the setup of the various control functions are covered with specific examples of boiler control including configuration and tuning.

Air Preheater Purpose

Preheat combustion air and absorb additional heat from flue gas.



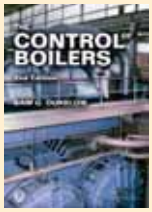


Format: Softbound Book – 2005
ISBN: 978-1-55617-907-5
Length: 144 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/boilereng

The Control of Boilers, 2nd Edition

S.G. Dukelow

This classic resource helps you reduce boiler operating costs through a detailed explanation of all aspects of boiler processes. It presents the basics of boiler control, the interrelationships of the process characteristics, and the dynamics involved, with emphasis on start-up, shut-down, flame monitoring, and safety interlock measures.



Format: Hardbound Book – 1991
ISBN: 978-1-55617-330-1
Length: 412 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/boilercontrol

What's in your Tank?

Users seek answers to expensive corrosion puzzles in power, paper industries

Where there's steam there's water, and where there's water there's probably corrosion. In the power industry, it is a persistent phenomenon that is difficult to control. In the paper industry, harsh chemicals are the corrosion culprit. Preventing corrosion doesn't call for a one-size-fits-all solution. Suppliers, integrators, and users agree, corrosion prevention means billions of dollars a year, and they offer strategies and techniques for their varying corrosion conundrums.

"If we didn't have reliable instrumentation to tell whether we were at the low levels, we'd take the conservative approach and treat to eliminate the oxygen, which means we'd spend the $100k a year" to avoid destroying the boiler pressure system.

Along with corrosion come deposition, scaling, and deposits that fall out of water and collect on condensers and heat exchangers. Corrosion can occur more rapidly as these impurities concentrate in under-deposit corrosion (deposits that stick to the metal surfaces inside these systems). "We always talk about corrosion and deposition together, and in water chemistry, it's difficult to control," said Brian LaBelle, power industry manager at Emerson Process Management in Irvine, Calif. In a power plant, it can change fast because large quantities make up steam losses.

These power plants boil tons of water to turn turbines to contain steam. As the steam goes away, they need to replenish with makeup water. Ground and city water have elements that, left untreated, will cause corrosion. Chemicals such as ammonia added to water to raise pH levels can create a slight alkaline environment to add protective coating on metallic surfaces that come in contact with water.

The power industry is responding to this by monitoring dissolved oxygen, LaBelle said. Companies use sensors sensitive to parts per billion (ppb) levels of oxygen to help control the pH, but adding ammonia to raise pH can create another corrosion problem. Using deaerators to get rid of oxygen raises water temperatures, which introduces oxygen scavengers such as hydrazine.

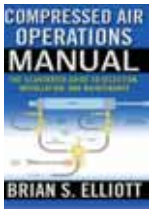
Some avant-garde ways to measure dissolved oxygen are sensitive and less expensive sensors with self-depleting designs, also called membrane-based amperometric sensors. The self-depleting design means there is oxygen inside the sensor that it has to consume. "When you put these things together for the first time before you put them into service, there's atmospheric oxygen to consume," LaBelle said. Some of the more recent innovations are the sensors that deplete themselves rapidly of internal oxygen. "Every time you take the sensor out to clean or calibrate, you expose it to atmospheric oxygen, more than what you use in a power plant. The faster it can consume oxygen in air during maintenance, the better the sensor," he said.

For full *InTech*® story, see **www.isa.org/link/InTank**.

Compressed Air Operations

Brian S. Elliott

This easy-to-understand manual provides you with the background information necessary to make informed, cost-effective decisions regarding compressed air requirements, whether installing, maintaining and/or improving existing systems. *Compressed Air Operations Manual* is essential reading for anyone requiring a concise working knowledge of compressed air systems, support equipment, and pneumatic applications. The book features such topics as compressed air applications; compressor types; the conservation of energy as it applies to compressed air, compression systems, and compressed air dryers; designing or specifying compression systems; and evaluating existing systems. There are two compressed air projects that provide valuable hands-on experience along with tips and techniques that can be applied to both new and existing systems.

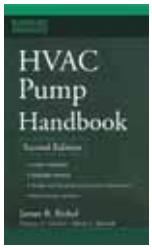


Format: Hardbound Book – 2005
ISBN: 0-07-147526-5
Length: 408 pages; Weight: 1.9 lbs.
Member: $69.95; List: $69.95
www.isa.org/air

HVAC Pump Handbook, 2nd Edition

James B. Rishel, Thomas H. Durkin, Ben L. Kincaid

This second edition equips HVAC designers, engineers, and technicians with concrete methods of achieving efficient operation in pumps for HVAC and water systems using the latest digital electronic technologies. Filled with case studies, current codes, and solved problems, this reference covers HVAC pumps and their performance, the basics of HVAC water systems, pumps for open and closed HVAC cooling systems and hot water systems, pump instrumentation and testing, and much more.



Format: Hardbound Book – 2006
ISBN: 978-0-07-145784-2
Length: 684 pages; Weight: 2 lbs.
Member: $125.00; List: $125.00
www.isa.org/HVAC

Industrial Pressure, Level, and Density Measurement, 2nd Edition

Donald R. Gillum

This second edition includes a new chapter covering equipment selection, mounting techniques, and specifications. Other new topics and information include: calibration and re-ranging updates for process calibrators, comparators, and other new test instruments; digital transmitter and communication updates, including HART®, FOUNDATION™ FIELDBUS, wireless transmitters, and multivariable and differential pressure transmitters and applications; added emphasis on non-contact level measurement; advances in hydrostatic tank gauging (HTG); and, improved density sensors and new applications.



Format: Softbound Book – 2008
ISBN: 978-1-934394-34-2
Length: 450 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/pressureleveldensity

ISA-77 Fossil Fuel Power Plant Standards and User Resources

This comprehensive *ISA-77 Fossil Fuel Power Plant Standards and User Resources* CD-ROM provides a thorough guide to information about fossil fuel power plants. Key documents focus on practices and procedures related to instrumentation and controls utilized in fossil fuel power generating stations. Also included on the CD-ROM are ISA best sellers, *Control of Boilers*, 2nd Edition and *SCADA: Supervisory Control and Data Acquisition*, 3rd Edition.



Format: CD-ROM
ISBN: 978-0-9792343-4-7
Member: $199.00; List: $599.00
www.isa.org/fossilfuelresource

Order Now
www.isa.org/shopisa
(919) 549-8411

Measurement Uncertainty, 4th Edition

Ronald H. Dieck

This book presents engineering students with a comprehensive tutorial of measurement uncertainty methods in a logically categorized and readily utilized format. The new uncertainty technologies embodied in both U.S. and international standards have been incorporated into this edition with a view toward understanding the strengths and weaknesses of both. The book is designed to also serve as a practical desk reference in situations that commonly confront an experimenter. The text presents the basics of the measurement uncertainty model, non-symmetrical systematic standard uncertainties, random standard uncertainties, the use of correlation, curve-fitting problems, and probability plotting, combining results from different test methods, calibration errors, and uncertainty propagation for both independent and dependent error sources.



Format: Softbound Book – 2007
ISBN: 978-1-55617-915-0
Lenght: 279 pages; Weight: 2 lbs.
Member: $69.00; List: $79.00
www.isa.org/measuncertainty

Mechatronics: An Integrated Approach

Clarence De Silva

This useful reference unifies material from a variety of engineering disciplines, useful to individuals approaching mechatronics from any relevant field. The book offers a strong foundation in such core subjects as dynamic system modeling, electronic components and analysis, mechanical components and analysis, robotics, sensors/transducers and instrumentation, stepper motors, dc and ac motors and drives, hydraulic and pneumatic actuators, fluidics, automatic control, digital processing and hardware, communication and interfacing, software tools, design, and prototyping. Appendices provide additional background on Laplace and Fourier transform techniques, and software tools including MATLAB®, SIMULINK®, and LabVIEW®. The book emphasizes practical situations and applications with numerous worked examples, problems, and exercises. An entire chapter is devoted to practical case studies.



Format: Hardbound Book – 2005
ISBN: 0849312744
Length: 1344 pages; Weight: 5 lbs.
Member: $120.00; List: $120.00
www.isa.org/mechatronics

The Mechatronics Handbook: The Fusion of Mechanics, Electronics, and Information Technology

Robert H. Bishop, Editor

The standard reference, found on the desks of engineering professionals around the world, that defines the state of the art of mechatronics—the fusion of mechanics, electronics, and information technology. The field's leading practitioners and academics provide clear explanations to ensure that non-specialists can understand and use the information.



Format: Hardbound Book – 2002
ISBN: 0-8493-0066-5
Length: 900 pages; Weight: 5 lbs.
Member: $129.95; List: $129.95
www.isa.org/mechatronicshandbook

Model-Based Predictive Control: A Practical Approach

J.A. Rossiter

With its non-theoretical approach to model-based predictive control (MPC), this book is ideal for those who need to understand, concepts of predictive control. The author expands on connections to optimal control and stability, the closed-loop paradigm, robust design in MPC, implementations of MPC using only small online computational burdens, and implicit modeling for predictive control.



Format: Softbound Book – 2003
ISBN: 0849312914
Length: 344 pages; Weight: 2 lbs.
Member: $99.95; List: $99.95
www.isa.org/predictivecontrol

Models Unleashed: Virtual Plant and Model Predictive Control Applications

Gregory K. McMillan and Robert A. Cameron

Building on the knowledge and the goals of the best-selling book, *Advanced Control Unleashed,* this portable pocket guide goes beyond theoretical concepts and provides new insight into the implementation practices after the objectives have been defined and the technology decisions have been made. Recognizing the opportunity that models present to capture and exploit plant knowledge, the authors share their expertise on the concepts, procedures, and examples needed to construct and apply different models through the use of "state of the art" software for simulation and model predictive control. Sharing the results of the extensive testing they performed, and to extend the utility of Models Unleashed, the authors have included a website link that lets you download Microsoft Excel spreadsheets and Powerpoint files.



Format: Softbound Book – 2004
ISBN: 978-1-55617-857-3
Length: 165 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/modelsunleashed

New Directions in Bioprocess Modeling and Control: Maximizing Process Analytical Technology Benefits

Michael A. Boudreau and Gregory K. McMillan

If you are a process design, quality control, information systems, or auto-mation engineer in the biopharmaceutical, brewing, or bio-fuel industry, this handy resource will help you define, develop, and apply a virtual plant, model predictive control, first-principle models, neural networks, and multivariate statistical process control. The synergistic knowledge discovery on bench top or pilot plant scale can be ported to industrial scale processes. This learning process is consistent with the intent in the Process Analyzer and Process Control Tools sections of the FDA's Guidance for Industry PAT – *A Framework for Innovative Pharmaceutical Development, Manufacturing and Quality Assurance.*



Format: Softbound Book – 2007
ISBN: 978-1-55617-905-1
Length: 310 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/bioprocess

PID Controllers: Theory, Design, and Tuning, 2nd Edition

Karl J. Åström and Tore Hägglund

This book examines the auto-tuning and adaptation features of several commercial controllers, including digital PID controllers, and discusses measures for dealing with specific challenges such as: reset windup, long process dead times, and oscillatory systems.



Format: Softbound Book – 1995
ISBN: 978-1-55617-516-9
Length: 343 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/pidcontrollers

Reading a P&ID

Job Training Systems

Learn how to read and understand Piping and Instrumentation Drawings (P&IDs). After completing the workbook, you should be able to identify symbols and function labels commonly found on P&IDs, describe how system components are related, and trace process stream flow and control loop functions.



Format: Spiralbound Book – 1999
Order No.: RPIDWBOOK
Length: 102 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/readpandid

Variable Speed Drives: Principles and Applications for Energy Cost Savings, 3rd Edition

David W. Spitzer

This venerable classic helps practicing engineers learn about the technical and economic potential of this technology. It presents the electrical, hydraulic, chemical, and instrumentation information necessary to technically evaluate and economically justify variable speed drive (VSD) applications.



Format: Softbound Book – 2004
ISBN: 978-1-55617-859-7
Length: 204 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/vsd

Setting the Standard for Automation™

Choose Your Specialty!

Want to network with industry leaders?
Need help with your professional needs?
What about your career potential?



Membership in any
of ISA's 17 Technical Divisions
connects you to a group of people around the world
who work in your industry and have the same technical interests.
And for only $9 per Division, why not?

ISA Divisions

Aerospace Industries
Analysis
Automatic Control Systems
Chemical & Petroleum Industries
Computer Technology
Construction & Design
Food & Pharmaceutical Industries
Management
Mining & Metals Industries
Power Industry
Process Measurement & Control
Pulp & Paper Industry
Robotics & Expert Systems
Safety
Telemetry & Communications
Test Measurement
Water & Wastewater Industries

For more information on adding Divisions to your ISA Membership,
log onto **www.isa.org/divisions** or call (919) 549-8411.

Not an ISA Member yet? Join now at **www.isa.org/joinisa**

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™

Reliability, Safety, and Electrical

Alarm Management has become a very important topic in the safety area. The press continues to report plant incidents caused by poorly designed alarms, alarm flooding, and alarms being bypassed. Every automation professional should understand the basic concepts of this topic.

A basic knowledge of Reliability is fundamental to the concepts of safety and safety instrumented systems. Process Safety and Safety Instrumented Systems (SIS) is an increasingly important topic. Safety is important in all industries—but especially in large industrial processes such as petroleum refineries, chemicals and petrochemicals, pulp and paper mills, and food and pharmaceutical manufacturing, for example. Even in areas where the materials being handled are not inherently hazardous, personnel safety and property loss are important concerns. SIS is simple in concept, but requires a lot of engineering to apply well.

The topic on Electrical Installations underscores the reality that the correct installation of electrical equipment is required to design and implement almost any automation system. This is a very large topic with rigid codes; admittedly, we can only scratch the surface here.

While the safety of electrical installations is closely related to correct electrical installations in general, electrical safety is treated separately here to ensure that sufficient emphasis is placed on the safety aspects of these installations.

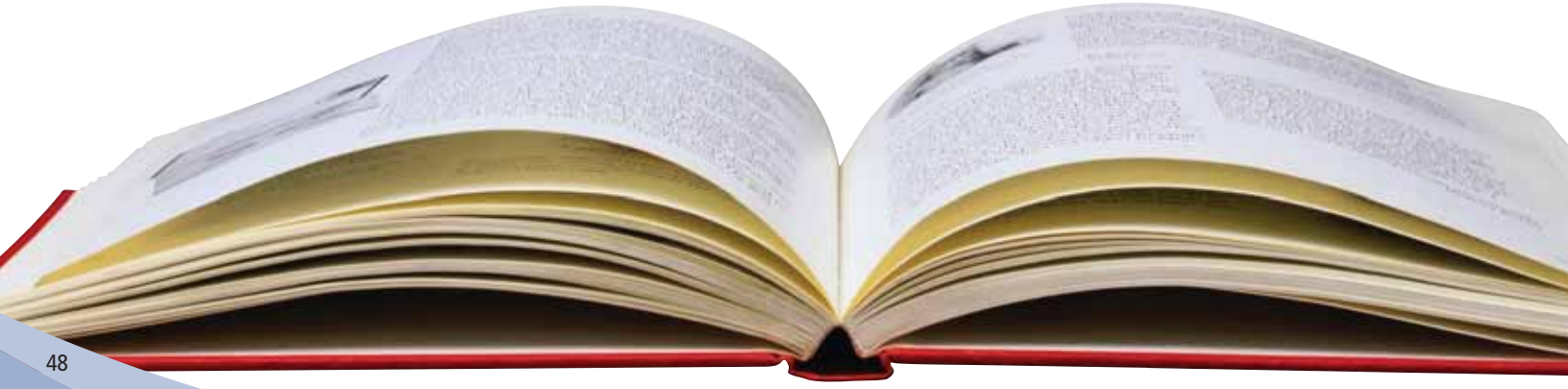
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok

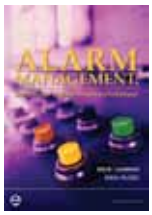


Reliability, Safety, and Electrical includes the areas of alarm management, process safety, electrical installations and electrical safety.

Alarm Management: Seven Effective Methods for Optimum Performance

By Bill R. Hollifield and Eddie Habibi

This comprehensive alarm management resource combines more than a decade of research with the experiences of many alarm management professionals to bring you field-proven methodologies that have helped industrial plants around the world improve their alarm management systems. Learn how properly designed and maintained alarm systems can drastically help improve plant safety, reliability, and profitability. Using seven effective methods, the authors explain how you can improve the performance of your alarm systems, including benchmarking an alarm system performance against industry best practices, developing an alarm philosophy document, rationalizing alarms, and applying real-time alarm management optimization strategies. Whatever the nature of your alarm systems, you will find *Alarm Management: Seven Effective Methods for Optimum Performance* a necessary part of your operational excellence and other business improvement initiatives.



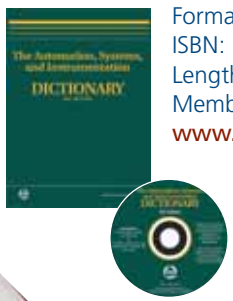
Format: Softbound Book – 2007
ISBN: 978-1-934394-00-7
Length: 178 pages; Weight 1 lb.
Member: $79.00; List: $89.00
www.isa.org/alarmmanagement

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume.

No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Basic Electricity and Electronics for Control: Fundamentals and Applications, 3rd Edition

Lawrence (Larry) M. Thompson

This class-tested book gives you a familiarity with electricity and electronics as used in the modern world of measurement and control. Integral to the text are procedures performed to make safe and successful measurements of electrical quantities. It will give you a measurement vocabulary along with an understanding of digital and analog meters, bridges, power supplies, solid state circuitry, oscilloscopes, and analog to digital conversions. Where possible, applications are used to illustrate the topics being explained. This book will enhance student success in any industrial or technical school setting where basic technician training is to take place.



Format: Softbound Book – 2006
ISBN: 978-1-55617-988-4
Length: 383 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/basicelectricity

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

Certified Industrial Maintenance Mechanic® (CIMM®) Study Guide

ISA

This study guide contains a list of the CIMM® knowledge areas and tasks, 50 review questions and answers complete with justifications and references.



Format: Softbound Book – 2006
ISBN: 978-1-55617-901-3
Length: 48 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/cimmguide

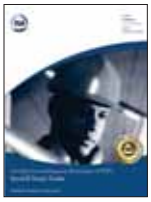
Certified Control Systems Technician® (CCST®) Study Guides

ISA

The CCST® Study Guides provide assistance in preparing for ISA's CCST® Exams, Levels I, II, and III. The guides parallel the structure of each exam, providing you with sample questions and a listing of other sources for study which provide you with many opportunities to test your knowledge. The study guides also include a recommended list of publications you can use to do further study on specific domains.



Level I
Format: Softbound Book – 2006
ISBN: 978-1-55617-573-2
Length: 60 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst1



Level II
Format: Softbound Book – 2006
ISBN: 978-1-55617-629-6
Length: 58 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst2



Level III
Format: Softbound Book – 2006
ISBN: 978-1-55617-691-3
Length: 40 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst3

Control System Power and Grounding Better Practice

David Brown, David Harold, and Roger Hope

Control system power and grounding is possibly the single most important element to ensure that a control system doesn't experience unidentified "gremlins" throughout its life. This resource, when used in conjunction with control system manufacturer installation documentation, can help you expect robust, reliable control system installation, one that remains free of "phantom" problems caused by power and grounding glitches.



Format: Softbound Book – 2004
ISBN: 0-7506-7826-7
Length: 128 pages; Weight: 1 lb.
Member: $49.95; List: $49.95
www.isa.org/grounding

Control Systems Safety Evaluation & Reliability, 2nd Edition

William M. Goble

This book provides a collection of tools to help the control engineer evaluate the safety and reliability of automated systems. Fault Tree Analysis (FTA), Reliability Block Diagrams (RBD), Failure Modes and Effects Analysis (FMEA) and Markov modeling methods are described with many examples. The key issues including component failure modes, on-line diagnostics, common cause, software reliability and operational safety are discussed along with design rules for building better systems. Safety Instrumented Systems (SIS) analysis techniques needed to meet new regulations are covered from sensor to final element.



Format: Softbound Book – 1998
ISBN: 978-1-55617-996-9
Length: 513 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/safetyevaluation

Electrical Instruments in Hazardous Locations, 4th Edition

Ernest Magison

An encyclopedia of electrical safety, this latest edition features expanded treatment of theory and principles to provide a foundation for addressing some of the questions that arise when no standard or regulation exists.



Format: Softbound Book – 1998
ISBN: 0-9792343-1-6
Length: 556 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/hazardlocations

ISA-92 Industrial Air Measurement Standards and User Resources

The new *ISA-92 Industrial Air Measurement Standards and User Resources* CD-ROM offers the widely used ISA-92 documents addressing performance requirements for industrial air measurement instrumentation related to health and safety. The CD-ROM also includes the best-selling technical publication, *The Automation, Systems, and Instrumentation Dictionary*, 4th Edition.



Format: CD-ROM
ISBN: 978-0-9792343-2-3
Member: $249.00; List: $599.00
www.isa.org/industrialairresource

Instrument Engineers' Handbook, 3rd Edition, Vol. 3: Process Software and Digital Networks

Béla G. Lipták

The third volume in this three-volume set, this best-seller provides an in-depth, state-of-the-art review of existing and evolving digital communications and control systems. Several chapters describe a variety of process-control software packages suited for plant optimization, maintenance, and safety-related applications. Other chapters cover plant design and updating, safety and operations-related logic systems, and the design of integrated workstations and control centers.



Format: Hardbound Book – 2002
ISBN: 0849310822
Length: 864 pages; Weight: 5 lbs.
Member: $179.95; List: $179.95
www.isa.org/iehvol3

Measurement Uncurtainty, 4th Edition

Ronald H. Dieck

This book presents engineering students with a comprehensive tutorial of measurement uncertainty methods in a logically categorized and readily utilized format. The new uncertainty technologies embodied in both U.S. and international standards have been incorporated into this edition with a view toward understanding the strengths and weaknesses of both. The text presents the basics of the measurement uncertainty model, non-symmetrical systematic standard uncertainties, random standard uncertainties, the use of correlation, curve-fitting problems, and probability plotting, combining results from different test methods, calibration errors, and uncertainty propagation for both independent and dependent error sources.



ISBN: 978-1-55617-915-0
Length: 277 pages; Weight: 2 lbs.
Member: $69.00; List: $79.00
www.isa.org/measuncertainty

NACE Corrosion Engineer's Reference Book, 3rd Edition

R. Baboian

This handbook contains over 325 tables, graphs, and charts. New sections evaluate corrosion tests and data, and sections on conversion tables and physical and chemical data have been greatly expanded.



Format: Hardbound Book – 2002
ISBN: 1-57590-127-7
Length: 450 pages; Weight: 2 lbs.
Member: $104.00; List: $104.00
www.isa.org/corrosion

ISA-84 Process Safety Standards and User Resources

The new *ISA-84 Process Safety Standards and User Resources* CD-ROM provides the benchmark ISA-84 standards and technical reports. The CD-ROM also includes notable reference books written by industry subject matter experts. Other features include chapters on "Process Safety and Safety Instrumented Systems" and "Safe Use and Application of Electrical Apparatus" from *A Guide to the Automation Body of Knowledge*, 2nd Edition.



Format: CD-ROM
ISBN: 978-0-9791330-7-7
Member: $299.00; List: $1199.00
www.isa.org/safetyresource

Reliability Centered Maintenance

Neil B. Bloom

This book introduces innovative approaches to simplify the RCM process such as single vs. multiple failure analysis, hidden failures analysis, potentially critical components analysis, run-to-failure and the difference between redundant, standby, and backup functions. Included are real-life examples of flawed preventive maintenance programs and how they led to disasters that could have easily been avoided. Also illustrated in detail, with real-life examples, is the step-by-step process for developing, implementing, and maintaining a premier classical RCM program. Senior and middle management, supervisors, and craftsmen/technicians responsible for plant safety and reliability will find this book to be invaluable as a means for establishing a first-class preventive maintenance program.



Format: Hardbound Book – 2006
ISBN: 978-0-07-146069-9
Length: 291 pages; Weight: 2 lbs.
Member: $89.95; List: $89.95
www.isa.org/reliabilty

Order Now
www.isa.org/shopisa
(919) 549-8411

Kletz on Safety

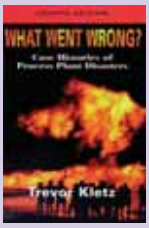


Trevor A. Kletz is widely recognized as an outstanding communicator on safety in the process industries. His work includes studies of human error, accident case histories and investigations, hazards of modifications, and hazards of computer control.

What Went Wrong? Case Studies of Process Plant Disasters, 4th Edition

Trevor A. Kletz

The purpose here is to show what has gone wrong in the past and to suggest how similar incidents might be prevented in the future. The incidents described could occur in many different types of plants and therefore are of widespread interest. Learn from the mistakes of others.



Format: Hardbound Book – 1998
ISBN: 0884159205
Length: 408 pages; Weight 2 lbs.
Member: $78.95; List: $78.95
www.isa.org/wentwrong

"If something looks odd, say so" was just one tip from Trevor Kletz, a veteran plant safety expert with a long list of credentials and honors from his England homeland. "You may embarrass someone, but it's better to do that than to let the plant blow up."

For full *InTech*® story, see www.isa.org/link/AvoidBlowups

Still Going Wrong! Case Histories of Process Plant Disasters and How They Could Have Been Avoided

Trevor A. Kletz

This book follows the revolutionary What Went Wrong?, but focuses on innovations and strategies that Kletz and others have pioneered. It includes how many accidents occur through simple miscommunication and how changing procedures, rather than design, can make a plant safer.



Format: Hardbound Book – 2003
ISBN: 0750677090
Length: 250 pages; Weight 2 lbs.
Member: $59.95; List: $59.95
www.isa.org/stillwrong

An Engineer's View of Human Error, 3rd Edition

Trevor A. Kletz

This third edition includes more examples of accidents caused by the various types of human error. It suggests to engineers how they might approach this problem, and does so by describing accidents, which at first sight seem to be the result of human error.

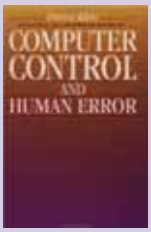


Format: Hardbound Book – 2001
ISBN: 852954301
Length: 281 pages; Weight 2 lbs.
Member: $67.00; List: $67.00
www.isa.org/engineerview

Computer Control & Human Error

Trevor A. Kletz

The author examines some of the unforeseen incidents that have occurred in computer-controlled process plants, and suggests how such incidents can be made less likely in the future. The text examines how Hazop studies can be used to detect hazards in computer-controlled systems and how software errors arise. Some incidents which have occurred in computer-controlled plants are described, and find out how to stop such things happening again.

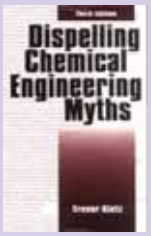


Format: Hardbound Book – 1995
ISBN: 0884152693
Length: 132 pages; Weight 2 lbs.
Member: $69.95; List: $69.95
www.isa.org/humanerror

Dispelling Chemical Engineering Myths

Trevor A. Kletz

Trevor Kletz sheds light on nearly 100 statements of doubtful accuracy that are widely accepted among chemical engineers. The myths address technology, management, toxicology and the environment. Included in each myth is a description of why it is wrong.



Format: Hardbound Book – 1996
ISBN: 1560324384
Length: 244 pages; Weight 2 lbs.
Member: $89.00; List: $89.00
www.isa.org/dispelling

Hazop and Hazan

Trevor A. Kletz

Now in its fourth edition, this book provides the basics of applying hazard and operability study (Hazop) and hazard analysis (Hazan). Substantially revised and upated, this classic, easily accessible introduction to the subject provides essential reading for everyone who needs to know about process safety — from undergraduates through to working engineers — providing plenty of practical advice from the author and more examples of Hazop and Hazan applications.



Format: Hardbound Book – 1998
ISBN: 0852954212
Length: 232 pages; Weight 2 lbs.
Member: $99.00; List: $99.00
www.isa.org/hazop

Learning from Accidents

Trevor A. Kletz

By analyzing accidents that have occurred, we are better able to learn from them; therefore, we should be better able to prevent them from happening again. More information is provided on accident investigations including building collapsing, invisible hazards, railway signal passes at dangerous places, a colliery explosion, computer failure, and much more. This edition emphasizes cause and prevention rather than human interest or cleaning up.



Format: Hardbound Book – 2001
ISBN: 075064883X
Length: 352 pages; Weight 2 lbs.
Member: $75.00; List: $75.00
www.isa.org/accidents

Process Plants: A Handbook for Inherently Safer Designs, 2nd Edition

Trevor A. Kletz

The concept of the "friendly" plants and plant design is the focus of this text. By designing friendlier plants, human error or equipment failure will not have the serious effects on safety, output or efficiency that older, less modern plants would have. In the second edition of this illustrated text, the guide has been enhanced to safe plant design, featuring increased coverage of design procedures.



Format: Hardbound Book – 1998
ISBN: 1560326190
Length: 224 pages; Weight 2 lbs.
Member: $89.00; List: $89.00
www.isa.org/saferdesigns

Safety Equipment Reliability Handbook, 3rd Edition

exida.com

The *Safety Equipment Reliability Handbook*, 3rd Edition, is the ultimate reference book for any safety engineer involved in Conceptual Design and Safety Integrity Level verification. It provides detailed reliability data for both specific manufacturer products, as well as generic equipment items. Many companies have standardized their reliability evaluations on reliability data published by exida. Due to the vastly expanded number of equipment items, the third edition is published as a set or in three separate volumes:

- Volume 01 - Sensors
- Volume 02 - Logic Solvers and Interface Modules
- Volume 03 - Final Elements

The *Safety Equipment Reliability Handbook* provides the ultimate reference source.

Format: Hardbound Book – 2007
ISBN: 978-0-9727234-9-7 (three-volume set)
Weight: 7 lbs.
978-0-9727234-3-5 (Vol. 01)
978-0-9727234-4-2 (Vol. 02)
978-0-9727234-5-9 (Vol. 03)
Three-volume set: $1350.00; Each volume: $495.00
www.isa.org/safetyreliability

Safety Instrumented Systems: Design, Analysis, and Justification, 2nd Edition

Paul Gruhn and Harry L. Cheddie



This book addresses the increased realization that today's engineering systems—and the computers used to control them—are capable of large-scale destruction. When even a single accident could be disastrous, the luxury of learning from experience no longer exists. This book is a practical how-to text on the analysis, design, application, and installation of safety instrumented systems (SIS).



Format: Softbound Book – 2006
ISBN: 978-1-55617-956-3
Length: 306 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/sisdesign

Safety Instrumented Systems Verification—Practical Probabilistic Calculations

William M. Goble and Harry L. Cheddie



This book clearly explains how to do probabilistic calculations to accomplish safety integrity level (SIL) verification for safety systems. Starting with a description of the safety lifecycle, the authors show where and how SIL verification fits into the key activities from conceptual design through commissioning. The book not only explains the theory and methods for doing the calculations, the authors also provide many examples from the chemical, petrochemical, power, and oil & gas industries.



Format: Softbound Book – 2005
ISBN: 978-1-55617-909-9
Length: 382 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/sisverify

Safety Integrity Level Selection: Systematic Methods Including Layer of Protection Analysis

Edward M. Marszal and Eric W. Scharpf

This book describes a systematic method for selecting safety integrity levels (SILs) for safety instrumented systems (SIS). This method emphasizes accounting for existing layers of protection and ensures that the maximum return on risk reduction investments is achieved. This reference demonstrates the application of quantitative risk analysis techniques and tools to the problem of safety integrity level selection. They distill essential theory into a format that the average control systems engineer can quickly apply. It is a focused process for applying simple, yet powerful, tools of quantitative risk analysis specifically to the problem of selecting safety integrity levels for safety instrumented systems.



Format: Hardbound Book – 2002
ISBN: 978-1-55617-777-4
Length: 248 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/safetyintegrity

ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems

The new *ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems* CD-ROM contains critical industry technical reports developed by world-renowned cyber security and automation systems experts, as well as associated reference books and resource papers.



Format: CD-ROM
ISBN: 978-0-9791330-6-0
Member: $79.00; List: $399.00
www.isa.org/securityresource



Setting the Standard for Automation™

Free Technical Papers from ISA

Get the ISA Advantage

Enhance your career with ISA membership AND have free access to over 2,500 technical papers from past ISA conferences and symposia. A search system lets you find papers by keyword, event, date, or author to locate the technical information you seek.

To learn more about this benefit and many others, visit **www.isa.org/memberbenefits**

Take advantage of this Member benefit today!

- Standards
- Certification
- Education & Training
- Publishing
- Conferences & Exhibits

ISA™

Integration and Software

Integrating systems and communications is now fundamental to automation. While some who work in a specific area of automation may have been able to avoid a good understanding of these topics, that isolation is rapidly coming to an end.

In fact, integration and communications are such an important part of automation that, in some companies, automation responsibility has been turned over to the information technology (IT) departments. While that may solve the integration issue, it usually does not deal with the unique real-time and security issues in automation, and it totally ignores the plant floor issues.

What is really needed is a good coupling of IT know-how with a broad knowledge of plant floor automation—either by having IT systems specialists learn plant floor controls, or by having automation professionals learn more about integration, or both.

Functionality and integration of the shorter time frame operating systems with both plant floor controls and with company business systems is called by several names. However, manufacturing execution systems (MES) is the most common.

The concepts of what functions to do and where and when data flows occur, has been wrestled with for a couple of decades since the computer integrated manufacturing (CIM) work of the 1980s. It has continued with the tireless work led by Theodore "Ted" J. Williams, the now retired Purdue University professor and 1968 ISA President. Now, with the ISA-95—Manufacturing Enterprise Systems Standards series, real standardization has arrived in this area. The ISA-95 standards have been adopted by some of the biggest names in manufacturing and business systems. While a large percentage of automation professionals do not know really what MES is, this topic, like integration in general, cannot be ignored any longer.

No topic is hotter today than Network Security—including the Internet. Any automation professional who is working in any type of integration must pay attention to the security of the systems. Operator Interface, Data Management, and other types of software are also now basic topics for automation professionals, and they fit in this category better than anywhere else. Packaged automation software that is open with respect to OPC covers a significant portion of the needs of automation professionals; custom software is still needed in some cases. That custom software must be carefully designed and programmed to perform well and be easily maintained.

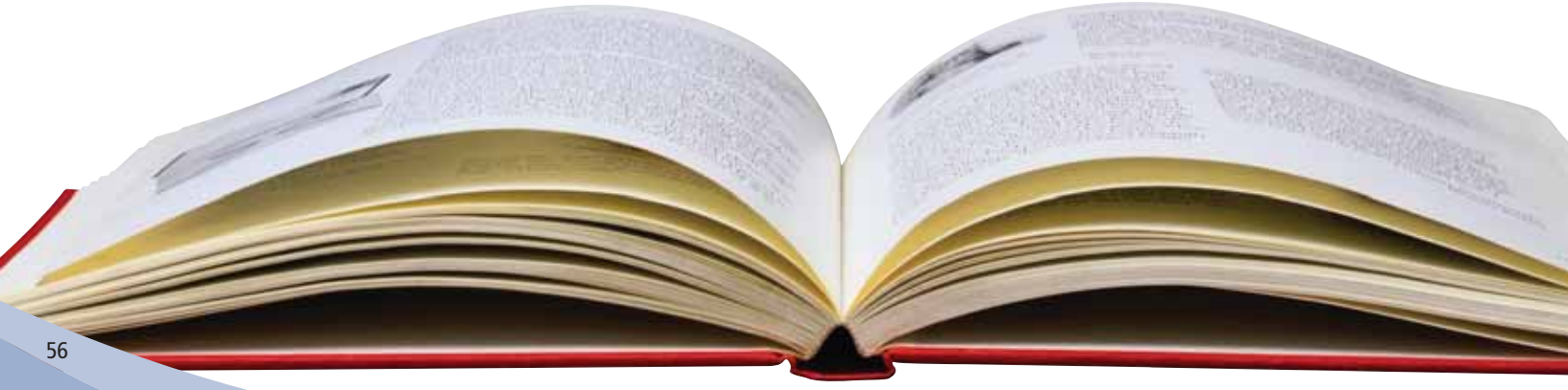
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok



Integration Software includes digital communications, industrial networks, MES integration, network security, software, data management, and operator interfaces.

Advanced Control Unleashed: Plant Performance Management for Optimum Benefit

Terrence L. Blevins, Gregory K. McMillan, Willy K. Wojsznis, and Michael W. Brown

Bridging the gap between theory and practice, this soon-to-be classic provides the basis for assessing the benefits of advanced control—covering auto-tuning, model predictive control, optimization, estimators, neural networks, fuzzy control, simulators, expert systems, diagnostics, and performance assessment. As an added bonus, the book includes a CD-ROM that provides real DeltaV simulations and displays. Configuration and case files are supplied for a hands-on experience, and PowerPoint files suitable for lectures on each unit are also included.



Format: Hardbound Book and CD-ROM – 2003
ISBN: 978-1-55617-815-3
Length: 434 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/advancedcontrol

Automation Network Selection

Dick Caro

Are you trying to make sense of all the different industrial automation networks on the market today? Whether you're a novice industrial network user or someone who simply needs to brush up on the technology, *Automation Network Selection* will help you better understand and select the "right" network for a given application. Industry expert Dick Caro walks you through the various industrial networks (e.g., sensor, fieldbus, control, safety buses, and so on) and then gives you a perspective on the typical applications for industrial automation network technology. The emphasis is on the intended application for each network, rather than on the network protocol that is more typically discussed in most textbooks.

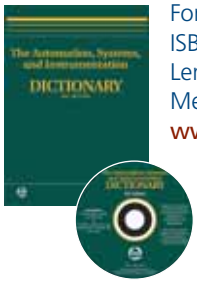


Format: Softbound Book – 2004
ISBN: 978-1-55617-861-0
Length: 161 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/networkselect

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume. No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Automation Unplugged: Pinto's Perspectives, Pointers & Prognostications

Jim Pinto



If you have heard industry pundit Jim Pinto speak, or read his barbed writings or laugh-out-loud poems, or subscribe to his popular e-newsletter, you'll enjoy this book. This book is loaded with critical analysis of the changing face of industrial automation; predictions about future automation technology trends—the best of his articles on marketing and distribution; and his highly rated fieldbus commentary.



Format: Softbound Book – 2004
ISBN: 978-1-55617-864-1
Length: 205 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/unplugged

Order Now
www.isa.org/shopisa
(919) 549-8411

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

The Consumer Guide to Fieldbus Network Equipment for Process Control, 2nd Edition

Richard H. Caro

This independent consumer guide to fieldbus network equipment will help you select a control system architecture compatible with your needs. This guide contains technical information about evaluating fieldbus network equipment, including pointers for architecture and installation considerations.



Format: Softbound Book – 2004
ISBN: 1-932095-06-3
Length: 100 pages; Weight: 1 lb.
Member: $225.00; List: $250.00
www.isa.org/fieldbusguide

Control System Documentation: Applying Symbols and Identification, 2nd Edition

Thomas McAvinew and Raymond Mulley

This updated ISA classic provides the symbols and identification commonly used throughout the process industries. It contains sample P&IDs and numerous examples of symbols and tagging concepts. It provides most of the symbols and identifiers that are unique to instrumentation and gives practical examples of their use.



Format: Softbound Book – 2005
ISBN: 978-1-55617-896-2
Length: 231 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/symbolsandID

Electronic Security for Manufacturing and Control Systems—An Overview

Robert Webb

This series of PowerPoint slides provides an excellent overview of electronic security techniques and standards status for the manufacturing and control systems environment. It is based on the author's experience with ISA's SP99 committee and is a useful tool for both self-study and to acquire familiarity with electronic security systems as applied in an industrial setting.

Format: PowerPoint File – 2005
ISBN: RSECGUIDE
Member: $29.00; List: $39.00
www.isa.org/secguide

Fieldbuses for Process Control: Engineering, Operation, and Maintenance

Jonas Berge



This implementation-oriented book provides a clear and concise presentation of how to apply fieldbuses for process control. Based on experience collected from end-users in a wide range of industries around the world, it provides "how-to" information for all phases of the system lifecycle. The book covers the three leading process fieldbus technologies: HART, FOUNDATION™ Fieldbus, and PROFIBUS-PA. It covers both the field-level and the Ethernet-based host-level networking. The text also addresses concerns and solutions for interoperability, integration and migration as well as availability and safety. A chapter on benefits helps you justify business advantages to management. The final chapter provides an in-depth explanation of how these fieldbus technologies work. It is ideal for both organized training courses and for self-study and will remain a handy reference when configuring and troubleshooting systems.



Format: Softbound Book – 2002
ISBN: 978-1-55617-904-4
Length: 468 pages; Weight: 2 lbs.
Member: $99.00; List: $109.00
www.isa.org/fieldbuses

Order Now
www.isa.org/shopisa
(919) 549-8411

Applying ISA-95

The Hitchhiker's Guide to Manufacturing Operations Management: ISA-95 Best Practices Book 1.0

Charlie Gifford, Editor and Contributing Author

Do you want to dramatically lower total cost of ownership (TCO) for manufacturing IT architectures and manufacturing, as well as reduce supply chain operational costs? The methodologies and technical applications presented in this first annual ISA-95/MESA Best Practices book will help get you started on the right track. This book provides in-depth coverage on how you can apply ISA-95, Enterprise-Control Integration Standard, to help lower TCO of manufacturing operations management (MOM) systems and their enterprise and plant interfaces. It consists of a series of related "how-to" white papers described in the context of ISA-95 models, definitions, and data exchanges. For those who prefer an electronic version of this book, it is also sold separately in CD-ROM format.

Turn to these handy resources for an in-depth look at ISA-95 and learn how to apply this international standard to integrate MES and ERP systems.



Format: Softbound Book – 2007
ISBN: 978-0-9792343-9-2
ISBN: 978-1-9343942-1-2 (CD-ROM format only)
Length: 240 pages; Weight 1 lb.
Member: $95.00; List: $109.00
www.isa.org/isa95guide

The Road to Integration: A Guide to Applying the ISA-95 Standard in Manufacturing

Bianca Scholten

This handy resource explains how to apply ISA-95 in manufacturing enterprise systems (MES) and vertical integration projects, as well as reveals the most important ISA-95 models and terminology. It is ideal for those just starting out who need to get up to speed on ISA-95's background information, but also for those who have some practical experience with ISA-95 and require additional support in carrying out analyses to determine a specific company's MES strategy and to define user requirements. If you develop MES functionality or realize interfaces based on ISA-95, then this guide will help you understand and apply the ISA-95 object models. Most important, the guide explains how to integrate ERP and MES systems using ISA-95. It places ISA-95 in the broader context of modern information exchange technologies and thus offers a complete picture for project managers, consultants, programmers, and information architects who want to integrate ERP and MES systems based on the international standard.



Format: Softbound Book – 2007
ISBN: 978-0-9792343-8-5
Length: 240 pages; Weight 1 lb.
Member: $95.00; List: $109.00
www.isa.org/integration

What would you call IT integration projects that had under a 50% success rate, took more than two years on average, and rarely ever meet the original requirements? The term "failure" would be my top choice, yet this was the state of the art in ERP-to-shop-floor integration projects prior to the use of the ISA-95 Enterprise/Control System Integration standard and B2MML, the WBF Business to Manufacturing Markup Language. These two standards have changed the landscape of ERP-to-shop-floor integration, transforming risky and expensive integration projects into low risk and affordable integration projects. The payoff for these projects can be very large, as manual and error prone processes are replaced with near real-time automated data transfers. Several companies have reduced their final inventory by over 60%, freeing up hundreds in millions of dollars for other investments. ERP systems need massive amounts of data, and much of that data comes from the shop floor. When manual data entry and reconciliation processes with long delays in data entry are replaced with near real-time information, then ERP reports and predictions can be used on actual values instead of estimates.

Source: Dennis Brandl, from his Foreword in The Road to Integration: A Guide to Applying the ISA-95 Standard in Manufacturing *by Bianca Scholten.*

ISA-95 Manufacturing Enterprise Systems Standards and User Resources

The new *ISA-95 Manufacturing Enterprise Systems Standards and User Resources* CD-ROM compiles essential ISA manufacturing enterprise systems standards, notable reference books, chapters on "Automation Benefits and Project Justifications," "Project Management and Execution," and "Manufacturing Execution Systems & Business Integration" from *A Guide to the Automation Body of Knowledge*, 2nd Edition. Also included on the CD-ROM are seven ISA technical papers on manufacturing execution systems.

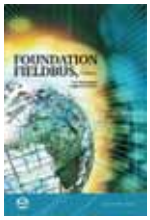


Format: CD-ROM
ISBN: 978-0-9791330-8-4
Member: $159.00; List: $499.00
www.isa.org/mesresource

Foundation Fieldbus, 2nd Edition

Ian Verhappen and Augusto Pereira

This newly revised book, written by control systems engineers with ex-tensive FOUNDATION™ Fieldbus installation experience, provides quick reference information on the FOUNDATION™ Fieldbus H1 protocol, in-stallation tips, and other useful information. Design engineers, control system engineers, and instrumentation technicians will all find this a handy reference when meeting with a vendor or client, and while managing an installation at a job site.



Format: Softbound Book – 2006
ISBN: 978-1-55617-964-8
Length: 125 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/foundation

Fundamentals of Industrial Control, 2nd Edition (with CD-ROM)

D.A. Coggan, Editor

This edition of a best-selling classic provides a unique balance between control system theory and effective application, covering every major control system element, from sensors to final control elements, in the context of overall control strategies and system design. This edition provides updates on a range of topics including Internet communications, industrial communications network hardware and software, open systems, wireless networks, enterprise computing, and computer and control system security. Other sections introduce the new generation of distributed controls and digital automation systems, and present migration strategies for expanding and upgrading existing systems. As a bonus, the book comes with the full text and graphics on a CD-ROM.



Format: Hardbound Book and CD-ROM – 2005
ISBN: 978-1-55617-863-4
Length: 893 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/fundamentalcontrol

Fundamentals of Process Control Theory, 3rd Edition (with CD-ROM)

P.W. Murrill

ISA's all-time best-selling book now includes a full text and graphics searchable CD, offering a way for you to teach yourself the complexities of process control theory. This classic has been praised for its clear presentation of the principles of process automation and its excellent overview of advanced control techniques. More than just a reference book, it's a complete course on the subject, with exercises and answers to work through. Both theoretical and practical, this guide allows you to teach yourself the fundamental scientific principles that govern process control, particularly feedback control. Its 15 self-study units provide a solid foundation in theory, as well as a discussion of recent technologies such as computer-integrated manufacturing, statistical process control, and expert systems. New chapters focus on the conceptual framework for an application, offering a practical understanding of the theory, along with specific illustrations on how concepts are implemented.



Format: Hardbound Book and CD-ROM – 2000
ISBN: 978-1-55617-683-8
Length: 344 pages; Weight: 2 lbs.
Member: $92.00; List: $109.00
www.isa.org/theoryfundamentals

Industrial Data Communications, 4th Edition

L.M. Thompson

Updates in networking and data communications are occurring rapidly. This new edition includes information on 100 MBps and 1000 MBps Ethernet, RIP and OSPF router technologies, OLE for Process Control (OPC), ActiveX, and .NET™, virtual private networks, and more. The author also provides a historical overview, and discusses networking and communication foundations; physical layer and data link standards; LANs; network operating systems and LAN management; industrial networks and their applications; and WANs. A complete glossary and index make this book an excellent desk reference.



Format: Softbound Book – 2007
ISBN: 978-1-934394-24-3
Length: 260 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/datacom

Industrial Ethernet, 2nd Edition

Perry S. Marshall and John S. Rinaldi

This revised best-seller is a convenient installation, troubleshooting, and reference tool on one of the hottest topics in automation and process control. It will help you understand important Ethernet and TCP/IP terminology and provide information about the new industrial protocols. You will quickly gain a solid grasp of Ethernet basics, the constraints of the industrial environment, and the specialized requirements of machine control.



Format: Softbound Book – 2005
ISBN: 978-1-55617-892-4
Length: 129 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/ethernet

Industrial Ethernet on the Plant Floor: A Planning and Installation Guide

Robert Lounsbury

This book outlines the select components to layout, install, test, certify, and troubleshoot a network system. It describes the fundamentals of designing industrial physical layers, network architectures, and components. The author provides the reader with the basics of noise—how to mitigate and abate it through installation techniques and selection of components that can provide a level of performance needed in a hostile industrial environment. The book also covers environmental considerations, Ethernet security, and certification.



Format: Softbound Book – 2008
ISBN: 978-1-934394-26-7
Length: 150 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/industrialethernet

Industrial Network Security

David J. Teumim

Process control systems for chemical plants and refineries; SCADA networks for utilities like electric, natural gas, water and wastewater; factory automation systems for discrete manufacturing — all of these areas need protection from hackers, disgruntled employees, and even cyberterrorists. This book leverages the technical information ISA has developed in its training courses and conferences, includes a brief tutorial on the basics of IT cybersecurity for the uninitiated, and takes a look at the many different organizations contributing to this area.



Format: Softbound Book – 2005
ISBN: 978-1-55617-874-0
Length: 144 pages; Weight: 1 lb.
Member: $54.00; List: $59.00
www.isa.org/networksecurity

Instrument Engineers' Handbook, 3rd Edition, Vol.3: Process Software and Digital Networks

Béla G. Lipták

The third volume in this three-volume set, this best-seller provides an in-depth, state-of-the-art review of existing and evolving digital communications and control systems. Several chapters describe a variety of process-control software packages suited for plant optimization, maintenance, and safety-related applications. Other chapters cover plant design and updating, safety and operations-related logic systems, and the design of integrated workstations and control centers. Appendices provide practical information such as bidders' lists and addresses, materials selection for corrosive services, and more.



Format: Hardbound Book – 2002
ISBN: 0849310822
Length: 864 pages; Weight: 5 lbs.
Member: $179.95; List: $179.95
www.isa.org/iehvol3

Instrumentation and Control Systems Documentation

Frederick A. Meier and Clifford A. Meier



This book will train you to read, understand, and apply the symbols and documents used to define a modern industrial instrumentation and control system. To help you better understand process documentation today, insight into how, when, and why documents are developed, who develops them, and how they are used is provided. The types of documents discussed include process flow diagrams, piping and instrumentation drawings, instrument lists, specification forms, logic diagrams, installation details, location plans, and loop diagrams.



Format: Softbound Book – 2004
ISBN: 978-1-55617-870-2
Length: 173 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/instdocument

Order Now
www.isa.org/shopisa
(919) 549-8411

ISA Documentation Standards and User Resources for Industrial Automation and Control Systems

The new *ISA Documentation Standards and User Resources for Industrial Automation and Control Systems* CD-ROM offers the widely used ISA documentation standards and technical reports, as well as several best-selling reference books. This comprehensive CD-ROM also includes the *Automation, Systems, and Instrumentation Dictionary*, as well as chapters on "Process Instrumentation" and "Control System Documentation" from *A Guide to the Automation Body of Knowledge*, 2nd Edition.

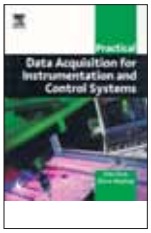


Format: CD-ROM
ISBN: 978-0-9791330-9-1
Member: $249.00; List: $599.00
www.isa.org/documentationresource

Practical Data Acquisition for Instrumentation and Control Systems

John Park and Steve Mackay

This book covers all aspects of the data acquisition system from design and specification to programming, installation, and configuration.

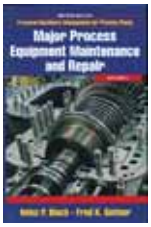


Format: Softbound Book – 2003
ISBN: 0-7506-5796-0
Length: 432 pages; Weight: 3 lbs.
Member: $57.95; List: $57.95
www.isa.org/dataacquisition

Practical Machinery Management for Process Plants: Major Process Equipment Maintenance and Repair, 2nd Edition, Volume 4

Heinz P. Bloch and Fred K. Geitner

An invaluable source of practical cost-effective maintenance, repair, installation, and field verification procedures for machinery engineers. It is filled with step-by-step instructions and quick-reference checklists that describe preventive and predictive maintenance for major process units such as vertical, horizontal, reciprocating, and liquid ring vacuum pumps, fans and blowers, compressors, turboexpanders, turbines, and more.



Format: Hardbound Book – 1997
ISBN: 0-88415-663-X
Length: 700 pages; Weight: 1 lb.
Member: $130.00; List: $130.00
www.isa.org/practicalmachinery

Profibus: A Pocket Guide

Ron Mitchell

Need to get up to speed on one of the leading fieldbuses on the market today? This pocket guide provides a basic overview of PROFIBUS operations, installation and configuration, wiring schemes, troubleshooting, and tips and tricks. It is a simple and convenient reference to enable engineers and technicians to quickly retrieve the essentials for successful understanding and deployment of PROFIBUS. Also useful for engineering management seeking a summary understanding of Profibus. Illustrative figures and technical tips are provided for quick reference.



Format: Softbound Book – 2004
ISBN: 978-1-55617-862-7
Length: 191 pages; Weight: 1 lb.
Member: $39.00; List: $49.00
www.isa.org/profibuspocket

ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems

The new *ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems* CD-ROM contains critical industry technical reports developed by world-renowned cyber security and automation systems experts, as well as associated reference books and resource papers.



Format: CD-ROM
ISBN: 978-0-9791330-6-0
Member: $79.00; List: $399.00
www.isa.org/securityresource

Order Now
www.isa.org/shopisa
(919) 549-8411

SCADA: Supervisory Control and Data Acquisition, 3rd Edition

Stuart A. Boyer

RAYMOND D. MOLLOY AWARD WINNER

This reference book offers overviews of SCADA's component technologies, as well as details necessary to understand the big picture. After completing its 14 self-study units, you will be conversant with SCADA nomenclature and architecture, understand the basic technology of the system's building blocks, understand its limitations, understand how it can benefit particular operations, and have a basis for selecting appropriate SCADA technologies for your operational requirements.



Format: Softbound Book – 2004
ISBN: 978-1-55617-877-1
Length: 219 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/scada

Software for Automation: Architecture, Integration, and Security

Jonas Berge

This implementation-oriented book provides a clear and concise presentation of how to fully apply software in automation. It provides "how-to" information for all phases of the system lifecycle from configuration, system integration, troubleshooting, and engineering to administration.



Format: Softbound Book – 2005
ISBN: 978-1-55617-898-6
Length: 325 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/autosoftware

Successful Instrumentation and Control Systems Design (with CD-ROM)

Michael D. Whitt

This book and CD-ROM combination provides in-depth coverage of the design process, the elements of a successful project, the specific issues to address in a well-designed instrumentation and control system, and the engineering products that enable practical design and successful maintenance.

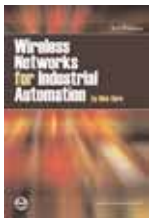


Format: Softbound Book and CD-ROM – 2004
ISBN: 978-1-55617-992-1
Length: 360 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/iandcdesignCD

Wireless Networks for Industrial Automation, 3rd Edition

Dick Caro

This updated third edition explores wireless communications from the factory floor and process automation positioning to help engineering professionals make clear decisions on the timing and strategy for implementing wireless networks for automation projects. According to Caro, industrial users must consider cost, privacy, security problems, and the potential for signal loss in plant environments. And, industrial users must have secure communications that never fail. This book is offered in digital format. For a full year, as updates become available, you will receive email notification and be able to download a new PDF file at no charge.



Format: PDF on the Web – 2008
ISBN: 978-1-934394-36-6
Length: 109 pages
Member: $69.00; List: $79.00
www.isa.org/wirelessnetworks

Deployment and Maintenance

Operator Training continues to increase in importance as systems become more complex, and the operator is expected to do more and more. It sometimes seems that, the more we automate, the more important it is for the operator to understand what to do when that automation system does not function as designed. This topic ties closely with the Modeling topic because simulated plants allow more rigorous operator training.

Automation professionals that only work on engineering projects in the office, and leave the field work to others, may not realize the tremendous amount of work required to get a system operating. Construction staff and plant technicians are doing more and more of the Checkout, System Testing, and Start-up work today, but that only makes it more important that automation professionals understand these topics.

Maintenance, Long-Term Support, and System Management takes a lot of work to do well. The difference in cost and effectiveness between a good maintenance operation and a poor one is easily a factor of two and may be much more. Automation professionals need to understand this area more so that their designs can effectively deal with life-cycle costs.

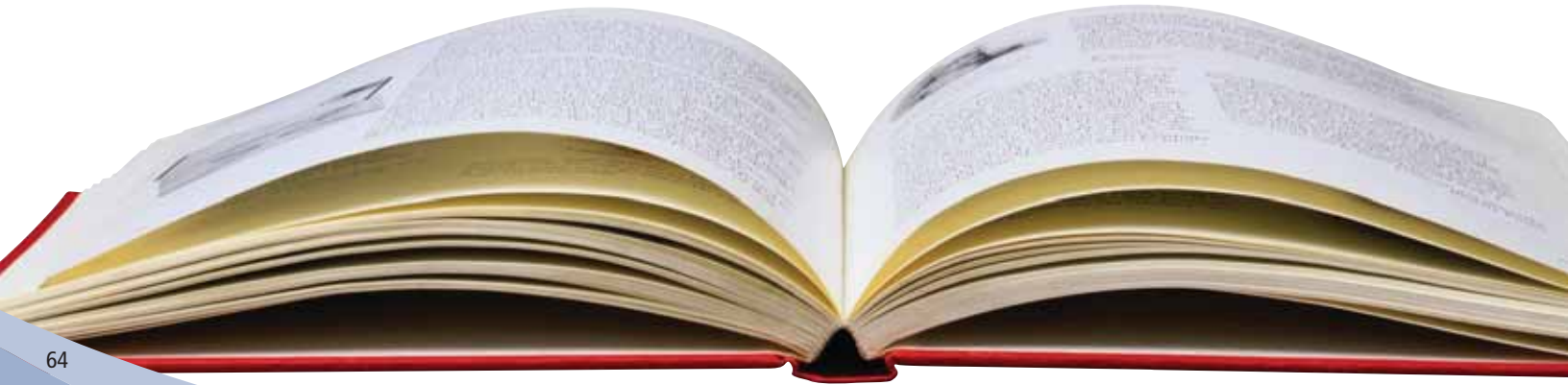
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok

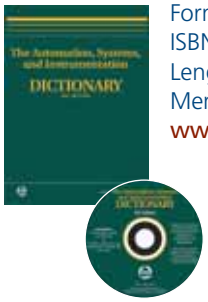


Deployment and Maintenance includes operator training, checkout, system testing and startup, troubleshooting, long-term support, and system management.

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume. No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Batch Control Systems—Design, Application, and Implementation, 2nd Edition

William M. Hawkins and Thomas Fisher

This revision of the 1990 work by Thomas Fisher covers an introduction to batch processes; batch control system structures; batch control; batch communications and batch control system design. Hawkins offers a comprehensive analysis of the development and evolution of batch control from the original NAMUR model through the most current publications in the ISA-88 series. Through examples, commentary, analogies and at times, wry humor, the author provides an in-depth philosophical discussion of how batch control and all manufacturing enterprises have been impacted by the work of ISA-88.

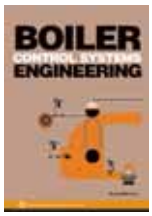


Format: Softbound Book – 2006
ISBN: 978-1-55617-967-9
Length: 300 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/batchcontrol

Boiler Control Systems Engineering

Jerry Gilman

This book deals primarily with water tube boilers with Induced Draft (ID) and Forced Draft (FD) fans or boilers with only a FD fan. Final control elements are reviewed as to characteristics and sizing. Engineering details on control systems and the setup of the various control functions are covered with specific examples of boiler control including configuration and tuning.



Format: Softbound Book – 2005
ISBN: 978-1-55617-907-5
Length: 144 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/boilereng

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

Certified Industrial Maintenance Mechanic® (CIMM®) Study Guide

ISA

This study guide contains a list of the CIMM® knowledge areas and tasks, as well as 50 review questions and answers complete with justifications and references.



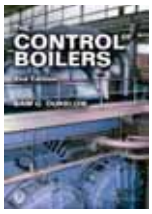
Format: Softbound Book – 2006
ISBN: 978-1-55617-901-3
Length: 48 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/cimmguide

Deployment and Maintenance

The Control of Boilers, 2nd Edition

S.G. Dukelow

This classic resource helps you reduce boiler operating costs through a detailed explanation of all aspects of boiler processes. It presents the basics of boiler control, the interrelationships of the process characteristics, and the dynamics involved, with emphasis on start-up, shut-down, flame monitoring, and safety interlock measures.



Format: Hardbound Book – 1991
ISBN: 978-1-55617-330-1
Length: 412 pages; Weight: 3 lbs.
Member: $89.00; List: $99.00
www.isa.org/boilercontrol

Fieldbuses for Process Control: Engineering, Operation, and Maintenance

Jonas Berge

RAYMOND D. MOLLOY AWARD WINNER

This implementation-oriented book provides a clear and concise presentation of how to apply fieldbuses for process control. Based on experience collected from end-users in a wide range of industries around the world, it provides "how-to" information for all phases of the system lifecycle. The book covers the three leading process fieldbus technologies: HART, FOUNDATION™ Fieldbus, and PROFIBUS-PA. It covers both the field-level and the Ethernet-based host-level networking. The text also addresses concerns and solutions for interoperability, integration and migration as well as availability and safety. A chapter on benefits helps you justify business advantages to management. The final chapter provides an in-depth explanation of how these fieldbus technologies work. It is ideal for both organized training courses and for self-study and will remain a handy reference when configuring and troubleshooting systems.



Format: Softbound Book – 2002
ISBN: 978-1-55617-904-4
Length: 468 pages; Weight: 2 lbs.
Member: $99.00; List: $109.00
www.isa.org/fieldbuses

Human Machine Interface Design for Process Control Applications

Jean-Yves Fiset

Users and designers of industrial control and monitoring systems (e.g., distributed control, supervisory control and data acquisition, and stand-alone) will benefit from this book's easy-to-use—yet effective—method on how to configure, design, and validate human-machine interfaces (HMIs). It discusses the overall HMI design process and how that process relates to system design. It also details design methods, principles and rules for individuals or groups of displays, as well as the integration of software-based and hardwired HMIs. This book will help guide you on the design of HMIs for other, less common, yet important, components, such as expert systems and other electronically-displayed operating procedures. Using the knowledge contained in this book, you can determine how to configure or design a whole new set of displays for a system or how to enhance specific elements of an existing or planned HMI.



Format: Softbound Book – 2008
ISBN: 978-1-934394-35-9
Length: 250 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/HMI

Industrial Data Communications, 4th Edition

Lawrence (Larry) M. Thompson

Updates in networking and data communications are occurring rapidly. This new edition includes information on 100 MBps and 1000 MBps Ethernet, RIP and OSPF router technologies, OLE for Process Control (OPC), ActiveX, and .NET™, virtual private networks, and more. The author also provides a historical overview, and discusses networking and communication foundations; physical layer and data link standards; LANs; network operating systems and LAN management; industrial networks and their applications; and WANs. A complete glossary and index make this book an excellent desk reference.



Format: Softbound Book – 2007
ISBN: 978-1-934394-24-3
Length: 260 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/datacom

Order Now
www.isa.org/shopisa
(919) 549-8411

Technician Series

This technician's series presents technical information in an easy-to-understand format. Each book provides self-study learning objectives, practice questions, answers, and listings of relevant standards.

Calibration: A Technician's Guide

Mike Cable

RAYMOND D. MOLLOY AWARD WINNER

Topics addressed include terminology; bench vs. field calibration; loop vs. individual instrument calibration; instrument classification systems, documentation; and specific calibration techniques for temperature, pressure, level, flow, final control, and analytical instrumentation. An extensive appendix containing sample P&IDs, loop diagrams, sample calibration procedures, and conversion and reference tables serves as a useful reference.



Format: Softbound
Book – 2005
ISBN: 978-1-55617-912-9
Length: 273 pages;
Weight: 1 lb.
Member: $75.00;
List: $85.00
www.isa.org/calibrate

Loop Checking: A Technician's Guide

Harley M. Jeffery

This technician's guide defines loop checking in the broader scope of control loop performance, in addition to the more traditional terms of the plant startup. It discusses general methods and practices that can be applied across many processes/industries. Featured topics include loop checking basics, factory acceptance testing, wiring and loop checks, performance benchmarking, and sustaining performance.



Format: Softbound
Book – 2005
ISBN: 978-1-55617-910-5
Length: 141 pages;
Weight: 1 lb.
Member: $69.00;
List: $79.00
www.isa.org/loopcheck

Start-Up: A Technician's Guide

Diane Harris

When new plants or systems go online, the control systems technician (CST) faces special challenges. The author explores and explains the crucial role of a technician in this process. This book offers you a clear overview of typical start-up responsibilities. From the first team meeting to the last round of tuning and loop checking, Harris uses her extensive experience with process control plants to walk you through the issues and skills typically required. Each chapter includes self-study learning objectives, practice questions and exercises, answers, and listings of relevant standards. Written with the technician in mind, it is an application-oriented book that provides an overview of the scope of duties a technician must perform in real-world situations. Includes over 30 figures and tables, fully indexed.



Format: Softbound Book – 2000
ISBN: 978-1-55617-677-7
Length: 217 pages; Weight: 1 lb.
Member: $42.00; List: $52.00
www.isa.org/startup

Troubleshooting: A Technician's Guide, 2nd Edition

William L. Mostia, Jr., P.E.

Troubleshooting loops and systems is something all technicians must do, but that few truly master. This newly revised edition draws on the author's long experience as an instrument and electrical engineer and his maintenance expertise to provide a detailed look at the skills and knowledge required for troubleshooting. Interspersed with a wealth of practical detail and real-world examples are Mostia's no-nonsense discussions of what a good troubleshooter needs to know. He provides an in-depth discussion of the basic logical framework that underlies all troubleshooting as well as advanced troubleshooting techniques.



Format: Softbound Book – 2006
ISBN: 978-1-55617-963-1
Length: 215 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/troubleshoot

Deployment and Maintenance

Information Technology for Energy Managers

Barney L. Capehart, Ph.D., C.E.M., Editor

This reference prepares energy managers to understand the basic concepts and principles of information technology (IT), so that they can successfully apply this knowledge in their facility — and have the understanding necessary to supervise the IT work of a consultant or vendor.



Format: Hardbound Book – 2004
ISBN: 0-88173-449-7
Length: 426 pages; Weight: 3 lbs.
Member: $145.00; List: $145.00
www.isa.org/itenergy

Maintainability & Maintenance Management, 4th Edition

Joseph D. Patton, Jr.

This book emphasizes the concepts of preventive, predictive, and reliability-based total maintenance, as well as total quality management and continuous improvement. This revised comprehensive version provides an in-depth look at new technological advances and the cost advantages of investing in maintainability. Topics covered in this book include maintainability program planning and management; system engineering of reliability, availability, and maintainability; configuration management; and maintainability process. This book serves as a flexible text and reference for engineers, product planners, designers, and others concerned with the practice of maintainability and maintenance.



Format: Softbound Book – 2005
ISBN: 978-1-55617-944-0
Length: 450 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/maintainability

Maintenance of Instruments & Systems, 2nd Edition

L.D. Goettsche, Editor

This handbook provides comprehensive coverage of maintenance requirements for pneumatic and electrical/electronic devices, as well as expanded coverage of DCS systems, analytical instrumentation, fiber optics, and smart instruments. Coverage is organized by device to make finding the information quick and easy. In this edition, special emphasis is given to documentation requirements and safety issues along with updated coverage of newly issued regulations and standards.



Format: Hardbound Book – 2005
ISBN: 978-1-55617-879-5
Length: 452 pages; Weight: 5 lbs.
Member: $129.00; List: $159.00
www.isa.org/IandSMaintenance

Maintenance of Process Instrumentation in Nuclear Power Plants

H.M. Hashemian

This book compiles 30 years of practical knowledge gained by the author and his staff in testing the I&C systems of nuclear power plants around the world. It focuses on process temperature and pressure sensors and the verification of these sensors' calibration and response times. In spite of great advances in electronics, computers, and measurement technologies, important process parameters such as temperature and pressure are still measured with conventional sensing techniques found in RTDs, thermocouples, and conventional pressure and differential pressure sensors. Furthermore, no improved technology providing comparable performance is currently on the horizon. Therefore, it is important to understand how these sensors function and the testing techniques for verifying their performance. This book is intended to help provide this understanding.



Format: Hardbound Book – 2007
ISBN: 3-540-33703-2
Length: 304 pages; Weight: 2 lbs.
Member: $129.00; List: $129.00
www.isa.org/nuclearpower

Order Now
www.isa.org/shopisa
(919) 549-8411

Plant and Machinery Failure Prevention

A.A. Hattangadi

Maximize productivity by keeping plant and equipment downtime and unscheduled maintenance at a minimum. Constant monitoring and preventative maintenance are not the most effective ways of avoiding common problems in manufacturing equipment. This book is based on the premise of "zero-failure performance," and is packed with expert advice. The author presents solid statistical as well as investigative methods for determining failure in mechanical and electrical equipment components.



Format: Hardbound Book – 2005
ISBN: 978-0-07-145791-0
Length: 458 pages; Weight: 2 lbs.
Member: $99.95; List: $99.95
www.isa.org/plantmachinery

Practical Controls: A Guide to Mechanical Systems

Steven R. Calabrese

This book details practical methods of HVAC control, and defines the role of controls in easy-to-understand terms. It explores the various mechanical systems installed in today's facilities and takes a concise look at how each is best controlled using current technology. It covers basic rules and equipment guidelines, pros and cons, and potential mistakes to avoid.

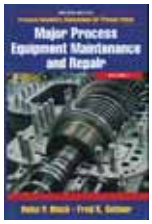


Format: Hardbound Book – 2003
ISBN: 0-88173-447-0
Length: 469 pages; Weight: 2 lbs.
Member: $120.00; List: $120.00
www.isa.org/mechsystems

Practical Machinery Management for Process Plants: Major Process Equipment Maintenance and Repair, 2nd Edition, Volume 4

Heinz P. Bloch and Fred K. Geitner

An invaluable source of practical cost-effective maintenance, repair, installation, and field verification procedures for machinery engineers. It is filled with step-by-step instructions and quick-reference checklists that describe preventive and predictive maintenance for major process units such as vertical, horizontal, reciprocating, and liquid ring vacuum pumps, fans and blowers, compressors, turboexpanders, turbines, and more.



Format: Hardbound Book – 1997
ISBN: 0-88415-663-X
Length: 700 pages; Weight: 1 lb.
Member: $130.00; List: $130.00
www.isa.org/practicalmachinery

Preventive Maintenance, 3rd Edition

J.D. Patton, Jr.

This 3rd edition of a best-selling classic shows how the investment in a preventive maintenance (PM) program repays a company in longer equipment life and smoother operation, planning, and scheduling. It includes enhanced techniques and insights along with new chapters dealing with computerized PM systems. The book addresses major types of maintenance; designing a PM program; nondestructive testing; scheduling; metrics; data issues; and much more.



Format: Softbound Book – 2004
ISBN: 978-1-55617-875-7
Length: 186 pages; Weight: 1 lb.
Member: $69.00; List: $79.00
www.isa.org/preventivemaintenance

The Quality Calibration Handbook – Developing and Managing a Calibration Program

Jay L. Boucher

Quality calibration systems are the foundation for improving research and development, production, and quality assurance arenas through accurate, reliable, and traceable calibrations. This book is about how to design, implement, maintain, and continuously improve a quality calibration system, with all the required documentation, traceability, and known uncertainty for each and every type of test equipment owned and used by any company. Calibration requirements vary across specific industries, but every organization, large or small, can use the systems described in this book as a foundation for a personalized program.



Format: Hardbound Book – 2007
ISBN: 978-0-87389-704-4
Length: 190 pages; Weight: 1.05 lbs.
Member: $84.00; List: $84.00
www.isa.org/qualitycalibration

Reliability Centered Maintenance

Neil B. Bloom

This book introduces innovative approaches to simplify the RCM process such as single vs. multiple failure analysis, hidden failures analysis, potentially critical components analysis, run-to-failure and the difference between redundant, standby, and backup functions. Included are real-life examples of flawed preventive maintenance programs and how they led to disasters that could have easily been avoided. Also illustrated in detail, with real-life examples, is the step-by-step process for developing, implementing, and maintaining a premier classical RCM program. Senior and middle management, supervisors, and craftsmen/technicians responsible for plant safety and reliability will find this book to be invaluable as a means for establishing a first class preventive maintenance program.



Format: Hardbound Book – 2006
ISBN: 978-0-07-146069-9
Length: 291 pages; Weight: 2 lbs.
Member: $89.95; List: $89.95
www.isa.org/reliability

Order Now
www.isa.org/shopisa
(919) 549-8411

Successful Instrumentation and Control Systems Design (with CD-ROM)

Michael D. Whitt

This book and CD-ROM combination provides in-depth coverage of the design process, the elements of a successful project, the specific issues to address in a well-designed instrumentation and control system, and the engineering products that enable practical design and successful maintenance.



Format: Softbound Book and CD-ROM – 2004
ISBN: 978-1-55617-992-1
Length: 360 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/iandcdesignCD

Tuning of Industrial Control Systems, 2nd Edition

A.B. Corripio

Suitable for beginners, this book takes a practical but, systematic approach to tuning. The second edition includes numerous examples of tuning, including the effect of hysteresis in flow control loops, averaging and tight level control, cascade control of a jacketed chemical reactor, feedforward control of a heater, and loop interaction and ration control in a blender. Also included is an introduction to a model reference control and a chemical reactor control example to illustrate it.



Format: Hardbound Book – 2001
ISBN: 978-1-55617-713-2
Length: 254 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/tuningsystems



Setting the Standard for Automation™

Your training needs are covered

ISA Onsite Training brings any ISA course to your location. The expert instruction, training materials, and portable laboratories come directly to you. All you have to do is pick the time and place.

Contact Matt or Joyce at **(919) 549-8411** or **training@isa.org** to do the rest.

Standards

Certification

Education & Training

Publishing

Conferences & Exhibits

ISA™

Work Structure

Automation professionals who work for engineering contractors and system integrators and see the project only after it has been given some level of approval by the end-user management may not realize the months or even years of effort that goes into identifying the scope of a project and justifying its cost. In fact, even plant engineers who are often responsible for doing the justifications often do not realize that good processes exist for identifying benefits from automation.

In the process of developing the CAP® certification program, the first step was to do a job analysis of the work of an automation professional. By focusing on describing the job, it became clear just how big a part project leadership and interpersonal skills are in the work of an automation professional. That effort helped the team realize these topics had to be included in any complete scope of automation.

If the Job Analysis Team had jumped immediately into defining skills, they might have failed to recognize the importance of project and interpersonal skills for automation professionals—whether they are functioning in lead roles or not.

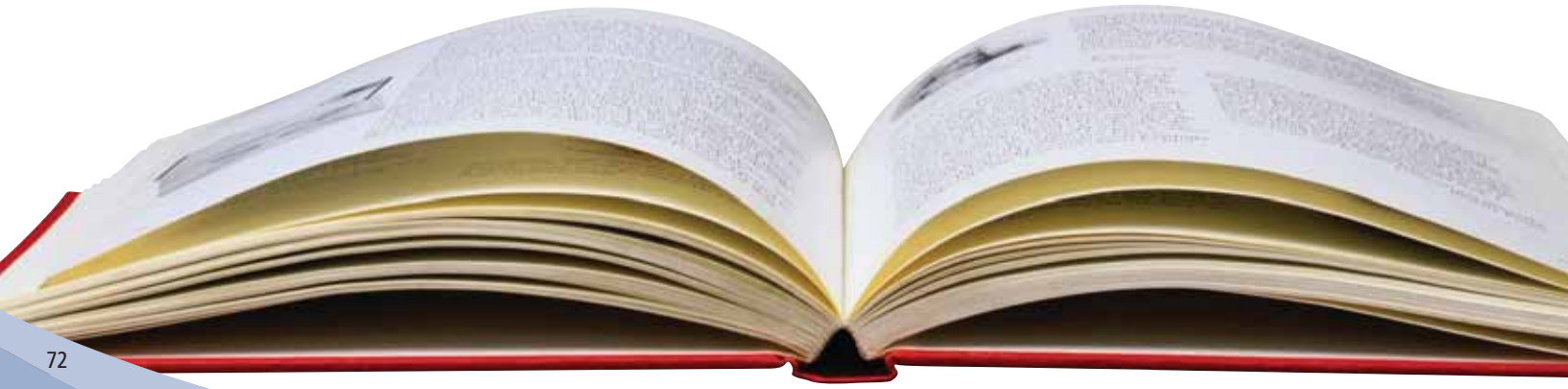
A Guide to the Automation Body of Knowledge, 2nd Edition

Vernon L. Trevathan, Editor

This 2nd edition gives you comprehensive information about all major topics in the broad field of automation. Edited by Vernon Trevathan with contributions from over 30 leading experts from all aspects of automation, this book defines the most important automation concepts and processes. You will turn to this book again and again as the ultimate source on what is encompassed by automation.



Format: Hardbound Book – 2006
ISBN: 978-1-55617-984-6
Length: 506 pages; Weight: 3 lbs.
Member: $109.00; List: $129.00
www.isa.org/autobok



Work Structure includes automation benefits, project justification, project management, and interpersonal skills

Automation and Control Systems Economics, 2nd Edition

Paul G. Friedmann

Covering the economic aspects of automation and process control, this book expands on how the tools of economic and financial analysis can be applied to automation projects. New sections have been added to extend application of economic and financial analysis to control of discontinuous processes, including batch and discrete parts manufacturing. A new glossary has also been added to help define economic terms.



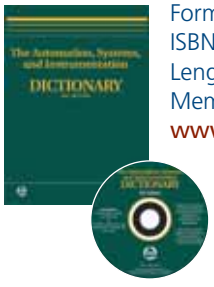
Format: Softbound Book – 2006
ISBN: 978-1-55617-951-8
Length: 182 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/economics

The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM)

ISA

Now available on CD-ROM together with the print volume.

No technical library is complete without this comprehensive dictionary covering the entire vocabulary of automation, systems, and instrumentation. References to relevant ISA and IEC standards are now included throughout, along with illustrations to enhance the definitions of more difficult terms. This reference encompasses not only specialized vocabulary unique to instruments and control systems, but many terms from other engineering disciplines as well. Ideal for engineers, managers, salespeople, technicians, students, and technical writers, this volume is packed with practical information.



Format: Softbound Book – 2003
ISBN: 978-1-55617-778-1
Length: 582 pages; Weight: 2 lbs.
Member: $89.00; List: $109.00
www.isa.org/dictionary

Bottom-Line Automation, 2nd Edition

Peter G. Martin

This newly revised edition helps today's manufacturing companies face the challenges of a global marketplace where every part of the operation must become more efficient to enhance the bottom line. Based on the results and conclusions of an applied research project of process manufacturing operations directed by the author, Bottom-Line Automation presents an overview of more than 30 years of industry trends, pointing out the strengths and pitfalls of each. The book also explains how manufacturers selected automation technology suppliers for state-of-the-art technological features in their systems rather than for the improvements the technology could offer the manufacturing operation. New topics in the book discuss the economic optimization of industrial assets.

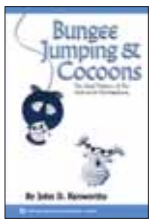


Format: Hardbound Book – 2006
ISBN: 978-1-55617-962-4
Length: 234 pages; Weight: 2 lbs.
Member: $79.00; List: $89.00
www.isa.org/bottomline

Bungee Jumping & Cocoons: The Dual Nature of the Industrial Marketplace

John D. Kenworthy

This book presents a unique tale of two contrasting thoughts: bungee jumping and cocoons. Bungee jumping is "our present culture's predilection for embracing all things extreme," the author explains. Cocoons represent "our willingness to nest within our homes with no need or desire to come out and mingle with the wicked world." How do these dual ideas play out in our society? How do they illuminate our industrial marketplace and e-commerce activities? This book sheds some light on how to get beyond the crossroads we are seemingly at—in terms of industrial maturity, economic transition, and societal metamorphosis.



Format: Softbound Book – 2004
ISBN: 978-1-55617-860-3
Length: 142 pages; Weight: 1 lb.
Member: $20.00; List: $20.00
www.isa.org/bungee

Order Now
www.isa.org/shopisa
(919) 549-8411

Certified Automation Professional® (CAP®) Study Guide

ISA

This is a comprehensive self-study resource for the CAP® certification exam. The study guide contains a list of the CAP® knowledge areas and tasks, 50 review questions and answers complete with justifications and references. The study guide also includes a recommended list of publications you can use to do further study on specific domains.



Format: Softbound Book – 2004
ISBN: 978-1-55617-888-7
Length: 57 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/capguide

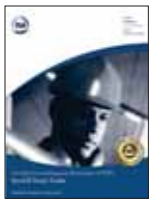
Certified Control Systems Technician® (CCST®) Study Guides

ISA

The CCST Study Guides provide assistance in preparing for ISA's CCST Exams, Levels I, II, and III. The guides parallel the structure of each exam, providing you with sample questions and a listing of other sources for study which provide you with many opportunities to test your knowledge. The study guides also include a recommended list of publications you can use to do further study on specific domains.



Level I
Format: Softbound Book – 2006
ISBN: 978-1-55617-573-2
Length: 60 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst1



Level II
Format: Softbound Book – 2006
ISBN: 978-1-55617-629-6
Length: 58 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst2



Level III
Format: Softbound Book – 2006
ISBN: 978-1-55617-691-3
Length: 40 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/ccst3

Certified Industrial Maintenance Mechanic® (CIMM®) Study Guide

ISA

Study guide contains a list of the CIMM® knowledge areas and tasks, 50 review questions and answers complete with justifications and references.



Format: Softbound Book – 2006
ISBN: 978-1-55617-901-3
Length: 48 pages; Weight: 1 lb.
Member: $29.00; List: $39.00
www.isa.org/cimmguide

Human Factors Methods for Design: Making Systems Human-Centered

Christopher P. Nemeth

This in-depth field guide describes how to create products for human use, including systems, services, and environments. It delivers an overview of human factors and the application of research to product and service development. It examines ways to solve problems in the development process, providing you with the orientation, process, and methods to perform human-centered research.



Format: Hardbound Book – 2004
ISBN: 0415297982
Length: 416 pages; Weight: 2 lbs.
Member: $119.95; List: $119.95
www.isa.org/humanfactors

Pinto on Automation

Automation Unplugged: Pinto's Perspectives, Pointers & Prognostications

Jim Pinto

If you have heard industry pundit Jim Pinto speak, or read his barbed writings or laugh-out-loud poems, or subscribe to his popular e-newsletter, you'll enjoy this book. This book is loaded with critical analysis of the changing face of industrial automation; predictions about future automation technology trends—the best of his articles on marketing and distribution; and his highly rated fieldbus commentary.

Well-known industry analyst, Jim Pinto, shares his thoughts, as well as those of other industry experts, on the future of automation and how to succeed in this ever-changing arena.

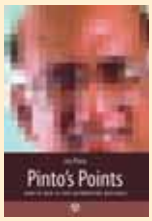


Format: Softbound Book – 2004
ISBN: 978-1-55617-864-1
Length: 205 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/unplugged

Pinto's Points: How to Win in the Automation Business

Jim Pinto

An updated and enhanced collection of ISA's "Pinto's Points" columns, published every week as part of *InTech e-News,* the book covers management topics, globalization, sales and marketing as well as nanotechnology, micro-electronic mechanical systems, robotics and wireless systems. Conveniently organized and with insights from other industry leaders, this book is a must read for every automation professional and a classic for your automation library bookshelf.



Format: Softbound Book – 2006
ISBN: 978-1-55617-953-2
Length: 270 pages; Weight: 1 lb.
Member: $28.00; List: $30.00
www.isa.org/pintopoints

Order Now
www.isa.org/shopisa
(919) 549-8411

Technology treadmill keeps rolling along

By Jim Pinto



Technology keeps accelerating to force change, even on those who are unwilling to cooperate. I never thought techno-geek, gadget-freak Jim Pinto would be in that category.

I have a cellphone that's 2 years old, and Cingular told me I could exchange it for a brand new model at zero cost (provided, of course, that I sign up for another 2 years). Now, I am happy with Cingular (largest provider in the U.S., with the broadest coverage) and I don't expect to change. But I can get a free, new cellphone anyway.

So, I went to take a look. I always go for the latest gadget; but, to tell the truth, I was confused. I didn't want a slimmer phone with a camera and CD music, and I didn't want to browse the Internet everywhere I go with a more bulky cellphone/camera/PDA in my pocket for an additional $40 a month.

Well, I finally decided on a new PC-cellphone, with Windows Mobile 5.0 operating system. I can check my e-mail and browse the web at any time, view picture slide-shows and movies (transferred from a PC) on the screen (which is bigger than an iPod-video screen), and even view Powerpoint presentations, Word and Excel files. But frankly, more than all those things, I enjoy the games, puzzles and solitaire when I'm in a waiting room somewhere.

In 1990, the tech-prophet George Gilder wrote a book called "Life After Television" which declared TV was dead. TV still seems to dominate many American living rooms, with big-screens and HDTV still showing significant market expansion. But, the shift to small screens (iPods and cellphones) is making major impact on TV network plans and projections.

Meanwhile, sales of conventional CDs are declining quickly. Fewer people listen to music via regular, cumbersome CD players. I have a 12-CD player in my car, but changing the discs is too much of a hassle compared with clicking through playlists with thousands of songs on my iPod.

The key change that is occurring is software–hardwired TV, telephones and music-players are giving way to software adaptable "teleputers".

At the AlwaysOn Innovation Summit in Stanford, Calif., tech-guru George Gilder exercised his usual hyperbole: "At the center of the network will be world wide webs of glass and light, and all of the action will move to the edge of the network, ushering in the life after telephony".

Source: *InTech®*

Human Machine Interface Design for Process Control

Jean-Yves Fiset

Users and designers of industrial control and monitoring systems (e.g., distributed control, supervisory control and data acquisition, and stand-alone) will benefit from this book's easy-to-use—yet effective—method on how to configure, design, and validate human-machine interfaces (HMIs). It discusses the overall HMI design process and how that process relates to system design. It also details design methods, principles and rules for individuals or groups of displays, as well as the integration of software-based and hardwired HMIs. This book will help guide you on the design of HMIs for other, less common, yet important, components, such as expert systems and other electronically-displayed operating procedures. Using the knowledge contained in this book, you can determine how to configure or design a whole new set of displays for a system or how to enhance specific elements of an existing or planned HMI.



Format: Softbound Book - 2008
ISBN: 978-1-934394-35-9
Length: 250 pages; Weight: 1 lb.
Member: $79.00; List: $89.00
www.isa.org/HMI

Ideal Enterprise: Managing by the Law of the Sphere

Hans D. Baumann

An entertaining guide on how you can successfully organize a business for maximum efficiency and profits, this book explores strategies for getting starting in business and how to sustain success as your company grows. The easy to read, often witty text covers how to start your own business, how to manage, role of incentives, value of meetings, budgets, computer communications, why smaller is often better in business, mergers, acquisitions and much more.



Format: Hardbound Book – 2002
ISBN: 0-533-14185-0
Length: 220 pages; Weight: 1 lb.
Member: $25.00; List: $25.00
www.isa.org/enterprise

Information Technology for Energy Managers

Barney L. Capehart, Ph.D., C.E.M., Editor

This reference prepares energy managers to understand the basic concepts and principles of information technology (IT), so that they can successfully apply this knowledge in their facility — and have the understanding necessary to supervise the IT work of a consultant or vendor.



Format: Hardbound Book – 2004
ISBN: 0-88173-449-7
Length: 426 pages; Weight: 3 lbs.
Member: $145.00; List: $145.00
www.isa.org/itenergy

ISA Certified Automation Professional® (CAP®) Job Analysis Study

ISA

The ISA CAP® certification program is designed to assess competence in the automation professionals. The process for identifying competency areas includes a job analysis study, which serves as a blueprint for examination development. For both broad content areas and tasks, the study validates importance and criticality to practice. These ratings play an important role in determining the content of the examination. The ISA CAP® job analysis study consisted of the following three phases, which are the focus of this report:

- Initial Development and Validation — A panel of experts identified the domains, tasks, knowledge, and skills that are essential elements of the profession of automation.
- Validation Study — A representative sample of 1,500 practicing automation professionals was asked to review and validate the work of the job analysis panel.
- Development of Test Specifications — Based on the ratings gathered from the representative sample of automation professionals, the test specifications for the examination were developed.



Format: PDF on Web – 2004
ISBN: 978-1-55617-903-7
Length: 77 pages
Member: FREE; List: $150.00
www.isa.org/capanalysis

Order Now
www.isa.org/shopisa
(919) 549-8411

ISA Documentation Standards and User Resources for Industrial Automation and Control Systems

The new *ISA Documentation Standards and User Resources for Industrial Automation and Control Systems* CD-ROM offers the widely used ISA documentation standards and technical reports, as well as several best-selling reference books. This comprehensive CD-ROM also includes the *Automation, Systems, and Instrumentation Dictionary*, as well as chapters on "Process Instrumentation" and "Control System Documentation" from *A Guide to the Automation Body of Knowledge*, 2nd Edition.



Format: CD-ROM
ISBN: 978-0-9791330-9-1
Member: $249.00; List: $599.00
www.isa.org/documentationresource

Maintainability & Maintenance Management, 4th Edition

Joseph D. Patton, Jr.

This book emphasizes the concepts of preventive, predictive, and reliability-based total maintenance, as well as total quality management and continuous improvement. This newly revised comprehensive version provides an in-depth look at new technological advances and the cost ad-vantages of investing in maintainability. Topics covered in this book include maintainability program planning and management; system engineering of reliability, availability, and maintainability; configuration management; and maintainability process. This book serves as a flexible text and reference for engineers, product planners, designers, and others concerned with the practice of maintainability and maintenance.



Format: Softbound Book – 2005
ISBN: 978-1-55617-944-0
Length: 450 pages; Weight: 2 lbs.
Member: $89.00; List: $99.00
www.isa.org/maintainability

Models Unleashed: Virtual Plant and Model Predictive Control Applications

Gregory K. McMillan and Robert A. Cameron

Building on the knowledge and the goals of the best-selling book, *Advanced Control Unleashed*, this portable pocket guide goes beyond theoretical concepts and provides new insight into the implementation practices after the objectives have been defined and the technology decisions have been made. The authors share their expertise on the concepts, procedures, and examples needed to construct and apply different models through the use of "state-of-the-art" software for simulation and model predictive control. Included is a website link that permits downloading Microsoft Excel spreadsheets and Powerpoint files.



Format: Softbound Book – 2004
ISBN: 978-1-55617-857-3
Length: 165 pages; Weight: 1 lb.
Member: $49.00; List: $59.00
www.isa.org/modelsunleashed

Out of the Barn

Dick Morley

This book, authored by PLC inventor Dick Morley, covers floor level control systems; industrial computing: the camel has the tent; software: indispensable exasperation; managing manufacturing: multi-tasking personified; enhancing corporate wealth; global reports; evaluating entrepreneurship; and we have met the enemy: they is us!

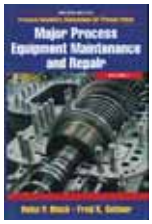


Format: Softbound Book – 2003
ISBN: 978-1-55617-822-1
Length: 150 pages; Weight: 1 lb.
Member: $20.00; List: $20.00
www.isa.org/barn

Practical Machinery Management for Process Plants: Major Process Equipment Maintenance and Repair, 2nd Edition, Volume 4

Heinz P. Bloch and Fred K. Geitner

An invaluable source of practical cost-effective maintenance, repair, installation, and field verification procedures for machinery engineers. It is filled with step-by-step instructions and quick-reference checklists that describe preventive and predictive maintenance for major process units such as vertical, horizontal, reciprocating, and liquid ring vacuum pumps, fans and blowers, compressors, turboexpanders, turbines, and more.



Format: Hardbound Book – 1997
ISBN: 0-88415-663-X
Length: 700 pages; Weight: 1 lb.
Member: $130.00; List: $130.00
www.isa.org/practicalmachinery

Practical Project Management: Learning to Manage the Professional

G.W. Cockrell

This book defines each unique phase of a project and then provides practical knowledge in areas such as budget and cost estimates, contracts, negotiating, team building, scheduling, and choosing project management software. It devotes special attention to project completion and closeout activities.



Format: Softbound Book – 2001
ISBN: 978-1-55617-735-4
Length: 209 pages; Weight: 1 lb.
Member: $59.00; List: $69.00
www.isa.org/projectmanage

Sell More Through Effective Technical Presentations: A Pocket Guide

Paul Gruhn, P.E., C.F.S.E.

This portable pocket guide provides real-life examples on how to give effective technical presentations from a sales perspective. This book covers topics such as overcoming fear, your elevator message, your audience's attitude, your appearance, testing your audience, and much more. This book will be of interest to sales, marketing, and engineering personnel.



Format: Softbound Book – 2003
ISBN: 978-1-55617-839-9
Length: 93 pages; Weight: 1 lb.
Member: $23.00; List: $25.00
www.isa.org/sellmore

Successful Instrumentation and Control Systems Design (with CD-ROM)

Michael D. Whitt

This book and CD-ROM combination provides in-depth coverage of the design process, the elements of a successful project, the specific issues to address in a well-designed instrumentation and control system, and the engineering products that enable practical design and successful maintenance.



Format: Softbound Book and CD-ROM – 2004
ISBN: 978-1-55617-992-1
Length: 360 pages; Weight: 2 lbs.
Member: $95.00; List: $109.00
www.isa.org/iandcdesignCD

Order Now
www.isa.org/shopisa
(919) 549-8411

Setting the Standard for Automation™



ISA CyberU—
The Best Training Right at Your Fingertips

No matter where you spend your time, ISA has training that can go with you—CDs, online courses, and web seminars on the industry's hottest topics.

Call **(919) 549-8411** or visit us online at **www.isa.org/CyberU** to get the training you need, where you need it, when you need it.

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™

Introducing a New CD-ROM Series!

Gleaned from ISA's excellent standards library, pre-eminent publications, and the latest technical papers, these new CDs help busy industrial automation and control professionals increase productivity…and save money. These handy tools offer easy access to a wide variety of materials with on-screen menus and keyword-searchable text, allowing users to quickly view and print information. The new formats provide standards and user resources in key areas such as documentation, process safety, manufacturing enterprise systems, security, and batch control.



ISA Documentation Standards and User Resources for Industrial Automation and Control Systems

This new *ISA* CD-ROM offers the widely used ISA documentation standards and technical reports, and several best-selling reference books. The *Automation, Systems, and Instrumentation Dictionary*, and chapters on "Process Instrumentation" and "Control System Documentation" from *A Guide to the Automation Body of Knowledge*, 2nd Edition are included.



Format: CD-ROM
ISBN: 978-0-9791330-9-1
Member: $249.00; List: $599.00
www.isa.org/documentationresource

ISA-84 Process Safety Standards and User Resources

This new CD-ROM provides the benchmark ISA-84 standards and technical reports. Notable reference books written by industry subject matter experts, chapters on "Process Safety and Safety Instrumented Systems" and "Safe Use and Application of Electrical Apparatus" from *A Guide to the Automation Body of Knowledge*, 2nd Edition are included.



Format: CD-ROM
ISBN: 978-0-9791330-7-7
Member: $299.00; List: $1199.00
www.isa.org/safetyresource

ISA-95 Manufacturing Enterprise Systems Standards and User Resources

This new CD-ROM compiles essential ISA manufacturing enterprise systems standards, notable reference books, chapters on "Automation Benefits and Project Justifications," "Project Management and Execution," and "Manufacturing Execution Systems & Business Integration" from *A Guide to the Automation Body of Knowledge*, 2nd Edition. Also included are seven ISA technical papers on manufacturing execution systems.



Format: CD-ROM
ISBN: 978-0-9791330-8-4
Member: $159.00; List: $499.00
www.isa.org/mesresource

ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems

This new CD-ROM contains critical industry technical reports developed by world-renowned cyber security and automation systems experts, as well as associated reference books and resource papers.



Format: CD-ROM
ISBN: 978-0-9791330-6-0
Member: $79.00; List: $399.00
www.isa.org/securityresource

ISA-88 Batch Standards and User Resources

A wealth of information on one CD-ROM, the new *ISA-88 Batch Standards and User Resources* compiles the essential ISA batch systems standards, reference books, and technical papers for the industrial automation and control systems professional.



Format: CD-ROM
ISBN: 978-0-9791330-5-3
Member: $199.00; List: $499.00
www.isa.org/batchresource

ISA-77 Fossil Fuel Power Plant Standards and User Resources

This comprehensive *ISA-77 Fossil Fuel Power Plant Standards and User Resources* CD-ROM provides a thorough guide to information about fossil fuel power plants. Key documents focus on practices and procedures related to instrumentation and controls utilized in fossil fuel power generating stations. Also included on the CD-ROM are ISA best sellers, *Control of Boilers*, 2nd Edition and *SCADA: Supervisory Control and Data Acquisition*, 3rd Edition.



Format: CD-ROM
ISBN: 978-0-9792343-4-7
Member: $199.00; List: $599.00
www.isa.org/fossilfuelresource

ISA-92 Industrial Air Measurement Standards and User Resources

The new *ISA-92 Industrial Air Measurement Standards and User Resources* CD-ROM offers the widely used ISA-92 documents addressing performance requirements for industrial air measurement instrumentation relted to health and safety. The CD-ROM also includes the best-selling technical publication, *The Automation, Systems, and Instrumentation Dictionary*, 4th Edition.



Format: CD-ROM
ISBN: 978-0-9792343-2-3
Member: $249.00; List: $599.00
www.isa.org/industrialairresource

ISA-75 Control Valve Standards and User Resources

Key documents on this new CD-ROM include the ISA-75 series covering all ISA standardization work related to valves. Also included on the CD-ROM are several best-selling books and the chapters "Process Instrumentation" and "Control Valves" from *A Guide to the Automation Body of Knowledge*, 2nd Edition.



Format: CD-ROM
ISBN: 978-0-9792343-3-0
Member: $199.00; List: $599.00
www.isa.org/controlvalveresource

Industrial Automation Standards

Practical Applications from Industry Experts

As a globally recognized standards-writing organization, ISA develops consensus standards for wide areas of industrial automation, including communications, safety, enterprise-control systems integration, cyber security, wireless systems, batch control, control elements, power plants, and documentation. The standards reflect the experience and insights of automation experts from around the world. Accredited by the American National Standards Institute (ANSI), ISA has published more than 150 standards, recommended practices, and technical reports. To help you better understand ISA's standards committee activities, we have organized them into the following key areas of industrial automation. For more information on all our standards and to order, please visit www.isa.org/standards.

Safety and Security

- ISA-SP12, Electrical Equipment for Hazardous Locations
- ISA-SP18, Instrument Signals and Alarms
- ISA-SP82, Electrical and Electronic Instrumentation
- ISA-SP84, Programmable Electronic Systems for Use in Safety Applications
- ISA-SP91, Criticality Ranking for Instrumentation
- ISA-SP92, Performance Requirements for Industrial Air Measurement Instrumentation Related to Health and Safety
- ISA-SP101, Human-Machine Interface
- ISA-SP99, Manufacturing and Control Systems Security

Documentation

- ISA-SP5, Documentation of Measurement and Control Instruments and Systems
- ISA-SP20, Instrument Specification Forms
- ISA-SP60, Control Centers
- ISA-SP98, Personnel Certification Standards
- ISA-SP104, Electronic Device Description Language (EDDL)

Enterprise

- ISA-SP88, Batch Control Systems
- ISA-SP95, Enterprise/Control Integration Committee
- ISA-SP103, Field Device Tools (FDT)

Industrial Automation Systems (IAS) Communications

- ISA-SP50, Signal Compatibility of Electrical Instruments
- ISA-SP100, Wireless Systems for Automation
- ISA-SP103, Field Device Tools (FDT)
- ISA-SP104, Electronic Device Description Language (EDDL)

Instrumentation

- ISA-SP7, Instrument Air Standards
- ISA-SP37, Measurement Transducers
- ISA-SP52, Environments for Standards Laboratories
- ISA-SP71, Environmental Conditions for Process Measurement and Control
- ISA-SP75, Control Valve Standards
- ISA-SP76, Composition Analyzers
- ISA-SP93, Sealing Technologies
- ISA-SP96, Valve Actuators

Power Plants

- ISA-SP67, Nuclear Power Plant Standards
- ISA-SP77, Fossil Power Plant Standards

Order Now
www.isa.org/shopisa
(919) 549-8411

ISA Standards Library CD – 2008

All the information you need to increase productivity and save money on one convenient CD-ROM! On-screen menus and the keyword-searchable text guide you through the library to quickly view and print information. More than 150 ISA Standards, Recommended Practices, and Technical Reports are included.



Format: ISA Standard CD-ROM
ISBN: 978-1-934394-49-6
Member: $1,495.00; List: $1,695.00
www.isa.org/standardslibrary

ISA-5 P&ID Clip Symbols, Version 2.0

This CD-ROM provides the graphic symbols from the ISA-5 standards as clip art files that can be inserted into P&IDs, loop diagrams, and other documents. The symbols are furnished in .DWG, .WMP, and .DXF file formats usable with Visio, AutoCAD, and CAD software platforms. System Requirements: Windows 95, 98, 2000, or NT; 16 MB RAM; CD-ROM drive, 10MB hard disk space, hardware/software that can use graphics files in the formats listed.



Format: ISA Standard CD-ROM
ISBN: 978-1-55617-684-5; Weight: 1 lb.
Member: $99.00; List: $109.00
www.isa.org/pid

ISA-20 Specification Forms (Excel)

This CD-ROM provides a set of easy-to-use templates that automate the creation of the forms you need in Microsoft® Excel to comply with the ISA-20-1981, Specification Forms for Process Measurement and Control Instruments, Primary Elements and Control Valves. The CD-ROM includes a copy of the ISA-20-1981 standard, 26 "fill-in-the-blank" style templates, and complete instructions for use. System Requirements: Windows 95, 98, 2000, or NT; Microsoft Excel 5.0 or higher; CD-ROM drive.



Format: ISA Standard CD-ROM
ISBN: 978-1-55617-730-9; Weight: 1 lb.
Member: $245.00; List: $295.00
www.isa.org/specformexcel

Instrument Specification Forms: An ISA20 Industry Implementation — Developed for Use in Pharmaceutical and Similar Industry Applications

Easy-to-use Microsoft® Excel templates, based on the popular ISA20 Instrument Specification Forms, extend the coverage and utility of the ISA20 forms to 73 instrument types. Developed by experts working in the pharmaceutical industry, these forms have applicability in other industries as well that deal with similar instrumentation and control systems issues.

These forms, designed to assist the specification writer in presenting key information, are intended to cover the most common instrument types used in the pharmaceutical industry. Blank fields are included to allow users the flexibility to add additional requirements. The forms may be used to specify a single instrument per form, or specify general requirements for a series of instruments. Pre-defined drop-down "pick-lists" for some of the data fields facilitate efficient data entry and data integrity.

This CD-ROM includes 73 "fill-in-the-blank" style templates, in Microsoft® Excel format.



Format: CD-ROM
ISBN: 978-1-934394-32-8
Member: $595.00; List: $695.00
www.isa.org/specformimplement

ISA-TR20.00.01 — Specification Forms in Microsoft® Word Format, 3rd Edition

This new edition includes seven new forms for a total of 78 reusable template forms developed to aid the design, purchase, and manufacture of process measurement and control instrumentation. These expanded specification forms have been modified extensively to cover operating parameters and expand the form content beyond the limitations of the original ISA-20-1981 forms. Many of the forms represent new devices not previously covered by the original forms.



SINGLE USER ONLY. Corporate network license version also available.
Format: ISA Standard CD-ROM
ISBN: 978-1-55617-974-7
Weight: 1 lb.
Member: $595.00; List: $645.00
www.isa.org/specformcd

ISA-84 Process Safety Standards CD-ROM, 4th Edition

The fourth edition of this CD-ROM includes the newly published **ISA-TR84.00.04-2005, Part 1:** Guideline on the Implementation of ANSI/ISA-84.00.01-2004 (IEC 61511 Mod); and **ISA-TR84.00.04-2005, Part 2:** Example Implementation of ANSI/ISA-84.00.01-2004 (IEC 61511 Mod).

Also included are:

ANSI/ISA-84.00.01-2004, Part 1 (IEC 61511-1 Mod) — Functional Safety: Safety Instrumented Systems for the Process Industry Sector — Part 1: Framework, Definitions, System, Hardware and Software Requirements

ANSI/ISA-84.00.01-2004, Part 2 (IEC 61511-2 Mod) — Functional Safety: Safety Instrumented Systems for the Process Industry Sector — Part 2: Guidelines for the Application of ANSI/ISA-84.00.01-2004 Part 1 (IEC 61511-1 Mod) — Informative

ANSI/ISA-84.00.01-2004, Part 3 (IEC 61511-3 Mod) — Functional Safety: Safety Instrumented Systems for the Process Industry Sector — Part 3: Guidance for the Determination of the Required Safety Integrity Levels — Informative

ISA-TR84.00.02-2002 — Safety Instrumented Functions (SIF) Safety Integrity Level (SIL) Evaluation Techniques Parts 1–5

ANSI/ISA-91.00.01-2001 — Identification of Emergency Shutdown Systems and Controls that are Critical to Maintaining Safety in Process Industries

ISA-TR91.00.02-2003 — Criticality Classification Guideline for Instrumentation

Plus, you'll also get 40CFR68 on Chemical Accident Prevention, 29CFR1910.119 concerning Safety Management of Highly Hazardous Chemicals, and selected ISA technical papers covering safety topics.



Format: ISA Standard CD-ROM – 2006
ISBN: 978-1-55617-982-2; Weight: 1 lb.
Member: $835.00; List: $930.00
www.isa.org/safetycd

Setting the Standard for Automation™



All ISA Standards Are Now Free

Get the ISA Advantage

Enhance your career with ISA membership by receiving all ISA standards free. ISA has published 150+ standards, recommended practices, and technical reports. With free technical information at your fingertips, you'll advance your career—and that's the ISA advantage!

To learn more about this benefit and many others, visit **www.isa.org/memberbenefits**

Take advantage of this Member benefit today!

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™



InTech® provides the most thought-provoking and authoritative coverage of automation technologies, applications, and strategies to enhance automation professionals' on-the-job success.

Published in print and on the web, ***InTech*** addresses industry challenges, new technologies, and fundamentals in a practical approach written for engineers, managers, and other automation professionals. ***InTech*** subscribers also receive an e-newsletter with the latest industry news each week.

Articles discuss the most critical issues facing the rapidly changing automation industry, offering a balance of basic and advanced treatment of topics such as

PROCESS AUTOMATION
control systems, instrumentation, sensing, and analysis

FACTORY AUTOMATION
machine control, vision, and discrete sensing

SYSTEM INTEGRATION
hardware, software, networking, retrofits, and components

AUTOMATION IT
software and communications interface

If you're not one of the 70,000 professionals benefiting from *InTech* each month, subscribe now at www.isa.org/intech/qual.

If you already subscribe to *InTech*, then you know the value it brings—encourage your colleagues and friends to subscribe today.

ISA Transactions—A Journal for the Science and Engineering of Measurement and Automation

ISA Transactions aims to be the respected journal of advances and state-of-the-art in the science and engineering of measurement and automation. Research and development personnel from academe and industry in the field of process instrumentation, systems, and automation will be pleased to see that the *ISA Transactions* journal bridges the theory and practice gap. This balance of interests requires simplicity of technique, credible demonstration, fundamental grounding, and connectivity to the state-of-the-art in both theory and practice. Transactions explores the topics of 'measurement', 'automation' and any equipment, techniques and best practices that support them. Optimization, learning systems, gaming strategy development, security, and human interfacing and training are also included.



Print: $59.00 Members
Non-members: Call ISA
www.isa.org/isatransactions

Boiler Sim MS Windows Version

CALA Engineering Services Inc.

This software simulates a single burner, fuel gas fired boiler for training boiler plant operators, and instrumentation and process control personnel. The simulation includes 3 element/1 element drum level control, cross-limited fuel air ratio control and a burner management system that follows the NFPA requirements. The dynamics of the boiler can be easily changed to mimic various sizes of boilers. (Requires 32 MB RAM, 15 MB hard disk space, and MS Windows®.)

Format: CD-ROM – 2002
Order No.: RBOILRSIM
Weight: 1 lb.
Member: $395.00; List: $395.00
www.isa.org/boilersim

FLOWEL—Flow Element Sizing and Documentation Version 4.0 for Windows

Emerson Performance Solution Inc.

FLOWEL 4 makes sizing orifice plates, nozzles, and venturis easier than ever. It performs exact flow rate, exact bore size, rounded bore size, and exact differential pressure calculations for flow elements and critical flow devices. The FLOWEL 4 Optional Package includes built-in fluid property methods to calculate the fluid density of steam and natural gas, methods for determining gas viscosity, isentropic exponent, vapor pressure, and bubblepoint, as well as methods to calculate density. (Requires 32 MB RAM, 100 MB hard disk space, and MS Windows® NT®, ME® 2000, XP® or XP Pro®.

Format: CD-ROM – 2004
Order No.: RFLOWELW
Weight: 1 lb.
Member: $1,895.00; List: $1,895.00
www.isa.org/flowel

FLOWEL 4.0 c/w Fluid Property Package
Format: CD-ROM – 2004
Order No.: RFLOWELFP
Weight: 1 lb.
Member: $2,895.00; List: $2,895.00
www.isa.org/flowelpack

ISA-20—Instrument Forms Plus LITE

Instrument Forms Plus LITE provides ISA-TR20.00.01 Specification Forms in a database management system format that includes tag lists and file utilities. The forms include operating parameter forms to define an instrument's process parameters, and instrument specification forms covering valves and regulator devices, pressure and differential pressure, level, flow, and temperature instruments. Drop-down field edits simplify data entry by offering pre-programmed choices, and user-defined values may also be easily added and edited. An instrument tag list lists the tag numbers along with the assigned specification and parameter form. Print to display, printer or personal document format. Requires Windows 98, XP, 2000 or NT; 128 MB of RAM; and 80 MB of disk space to load program.

Format: CD-ROM
Order No.: RINSTLITE; Weight: 1 lb.
Member: $495.00; List: $495.00
www.isa.org/rinstlite

Order Now
www.isa.org/shopisa
(919) 549-8411

ISA-20 – Instrument Forms Plus Pro

Using Instrument Forms Plus Pro, design engineers can fully document all instrumentation requirements and plant maintenance engineers can document instrument maintenance histories throughout their life cycles. The software provides all the features of Instrument Forms Plus LITE and adds advanced capabilities, including a robust instrument index that defines instrument attributes, drawing references, calibration ranges, associated equipment, systems, links to drawings, and more. Requires Windows 98, XP, 2000 or NT; 128 MB RAM; 80 MB of disk space to load program.

Format: CD-ROM
Order No.: RINSTFMNT; Weight: 1 lb.
Member: $1,395.00; List: $1,395.00
www.isa.org/rinstfmnt

ISA-20 – Instrument Loop Forms Plus Combo

Instrument/Loop Forms Plus Combo combines all features of Instrument Forms Plus Pro with the ability to generate instrument loop diagrams via the Quick Loop interface and in AutoCAD© or AutoCADLT©. With this loop-generating capability, a total software tool is provided for new design or for converting existing process documents into a database electronic format. Includes electrical motor schedule index. Users may produce motor specification data forms and motor control center layout configuration reports, and enter junction box/marshaling cabinet wiring data. Multi-user and network pricing available. Requires Windows 98, XP, 2000 or NT; 128 MB of RAM; and 80 MB of disk space to load program.

Format: CD-ROM
Order No.: RLOOP; Weight: 1 lb.
Member: $5,500.00; List: $5,500.00
www.isa.org/rloop

PC-ControLAB, Release 3

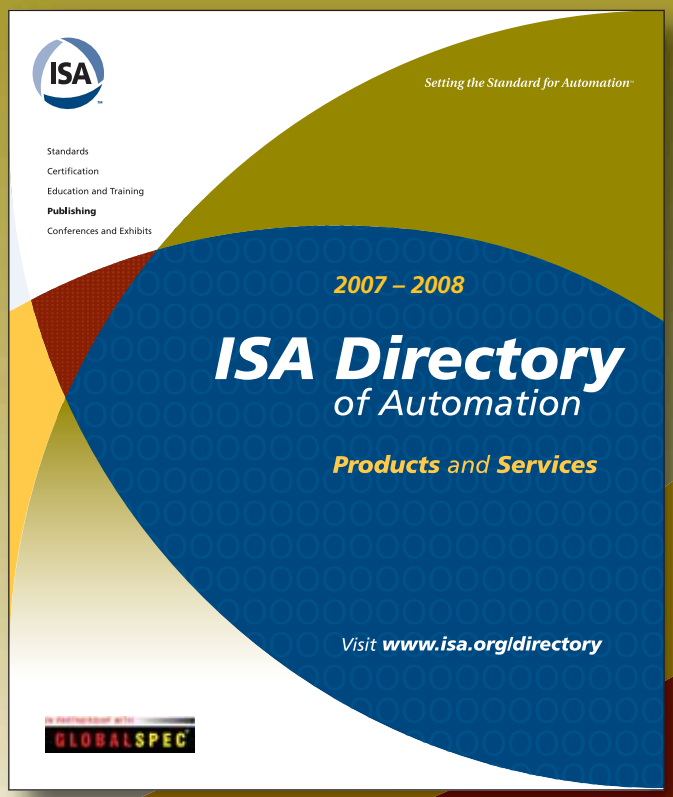
Wade Associates

This MS Windows®-based process control loop simulation program enables the user to practice various controller tuning procedures and view the effect of both set point changes and disturbances. The program provides a choice of control strategies (feedback, cascade, ratio, feedforward, override, decoupling and model-based control) and typical controller configuration options. Predefined process models and process control loops illustrate characteristics of typical temperature, flow, pressure and liquid level control loops, as well as demonstrate real-world problems such as mistuned controllers and sticky valves. Requires Windows 95, 98, ME, 2000, NT or XP and 6 MB free space on the hard disk.

Format: CD-ROM – 2002
Order No.: RPCLAB03; Weight: 1 lb.
Member: $495.00; List: $495.00
Site License Order No.: RPCONSITE
Price: $2,500.00 (Up to 8 users)
www.isa.org/pclab

Order Now
www.isa.org/shopisa
(919) 549-8411

The Only Product Directory Designed Specifically for Automation Professionals



IN PARTNERSHIP WITH GLOBALSPEC®

ISA and GlobalSpec have joined together to place the information you need at your fingertips – when you need it – helping you be more productive. Whether you need to find a specific part number, or search by technical specification, the online ISA Directory of Automation can put you in direct contact with over 90,000 product suppliers.

Try an online search today: **www.isa.org/directory**

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA

Ordering Information

Customer Satisfaction Policies

Your satisfaction with your ISA publication or standard is important to us. If you do not find the product satisfies your needs, please return it within 30 days of purchase — we will refund your purchase price less the cost of shipping and handling. Please return the product in saleable condition. For more details about ISA's return policies, see **www.isa.org/shopisa/about**

Your Privacy

To provide you service, ISA must collect certain information, such as your address, phone number, fax number, e-mail address, and other contact, profile, and billing information. ISA collects this kind of information only by voluntary disclosure directly from you. The information we collect is intended to allow ISA to process your transaction and better serve you. ISA's complete privacy policies can be found at **www.isa.org/lawyer**

Shipping Options

All U.S. book, video, software, and CD orders are shipped via UPS ground service or UPS Overnight service.

Please note the following:

- If you request UPS Overnight service, in order to ship the same day the order is received, your order must be received before 12 PM Eastern Standard Time, Monday through Friday.
- All International orders will ship via DHL Worldwide Express unless otherwise instructed.

International Distributors

If you need local access to ISA publications, check with one of ISA's officially licensed distributors listed at **www.isa.org/shopisa/about**

Shipping and Handling Charges

Shipment Weight	1–5 lbs.	6–11 lbs.	12–18 lbs.	>18 lbs.	Typical Shipping Time
US (UPS) and Canada (USPS)*	$10.00	$12.00	$10.00	$14.00 + $1.00 per item	US: 2–6 days* Canada: 15–30 days
US Overnight UPS	$40.00	$48.00	$10.00	$63.00 + $3.00 per item	See note above
Canada DHL	$23.00	$33.00	$10.00	$45.00 + $3.00 per item	2–4 days
International Surface*	$13.00	$17.00	Not available	Not available	60 days or more
International DHL	$43.00	$63.00	$73.00	$83.00 + $40.00 per item	2–4 days

Payment Options

Credit card payment is the fastest, most efficient method of payment. ISA accepts American Express, Discover, MasterCard, and VISA. Checks, money orders, wire transfers, and purchase orders also are accepted.

Money orders must include $5.00 U.S. for processing. Checks for payment of international orders must include MICR encoding and bank or processing fees, and be drawn on your bank's correspondent NY or other US bank. All wire transfer/bank processing fees are the responsibility of the payer.

Wire Transfer—Local bank fees are the responsibility of the applicant. Wire funds to ISA account #1126294, SunTrust Bank ABA061000104. Our SWIFT/BIC for international wire transfers is SNTRUS3A. Transfer must show applicant's name and address plus reason for payment.

ISA reserves the right to change prices on items in this catalog.

Quantity Discounts (Wholesalers, Bookstores, Agencies and Schools)

5-25 copies	20% off list
26-50 copies	25% off list
51-100 copies	30% off list
+100 copies	Contact ISA for pricing

Custom Ordering

Are you searching for a unique way to promote your company? ISA has the perfect solution for you. These customized books make great gifts for employees or customers and you can also use them as a company training tool.

A few examples of how ISA can customize your book:

Custom Cover—Have a cover changed so that it incorporates your company's colors or trademark

Personalized Title Page—Imprint your company logo or trademark, inscribe with a "compliments of…" text, etc.

Bookmarks—Insert a customized bookmark with your promotional copy or marketing material

Book Band—Customize a removable band of promotional material to wrap around the book

Inserts—"Letters from the President," advertisements etc. can be placed inside the front cover

Shrink Wrapping—Hold all your materials within the book and make distribution easy

For more information on Custom Ordering, contact Chip Lee at (919) 990-9412.

Order Now
www.isa.org/shopisa
(919) 549-8411

Setting the Standard for Automation™

Order Form

Order Online at www.isa.org/shopisa
Call ISA Customer Service at (919) 549-8411
Fax to (919) 549-8288
Mail to ISA, P.O. Box 3561, Durham, NC 27702-3561, USA

Provide priority code **PCSUM8** to speed processing of your order.

1. Customer Information

Please check one: This is my ❑ Home address ❑ Office address

First Name | MI | Family/Last Name (use hyphen if two) | Title

Company | Division

Shipping Address

City | Postal Code | State/Province | Country

( ) Telephone Number | ( ) Fax Number | E-mail address

Check your primary job function: (choose only one)

- ❑ Control Systems Engineering (B)
- ❑ Education (N)
- ❑ Engineering Consulting (AG)
- ❑ Engineering Management (3)
- ❑ Environmental Controls (AI)
- ❑ General or Corporate Management (A)
- ❑ Marketing or Sales (O)
- ❑ Measurement, Testing, Quality or Standards Engineering (I)
- ❑ OEM Product/Systems Engineering (AQ)
- ❑ Plant Engineering, Operations, and Maintenance (E)
- ❑ Plant Information Systems (G)
- ❑ Product Design Engineering, including applied R&D (1)
- ❑ Production Engineering (D)
- ❑ Research & Development (J)
- ❑ Safety Systems Engineering (AJ)
- ❑ Software Engineering (F)
- ❑ Systems Design Engineering (S)
- ❑ Systems Integration (H)
- ❑ Technical or Engineering Support (K)
- ❑ Other (P) _____

Check your industry classification: (choose only one)

- ❑ Chemicals (2800)
- ❑ Electronic & Other Electric Equipment (3600)
- ❑ Food (2000)
- ❑ Industrial Machinery & Equipment, including Computers (3500)
- ❑ Instrumentation, Measurement, Analysis & Control Apparatus (3800)
- ❑ Petroleum Refining & Related Industries (2900)
- ❑ Pharmaceuticals (2830)
- ❑ Primary Metals (3300)
- ❑ Pulp, Paper & Allied Products (2600)
- ❑ Textiles (2200)
- ❑ Tobacco Products (2100)
- ❑ Valves, Fittings, Fabricated Metal Products (3400)
- ❑ Oil and Gas Extraction (1300)
- ❑ Semiconductor and Electronic Components (3670)
- ❑ Transportation (3700)
- ❑ Transportation — Aerospace (3760)
- ❑ Transportation — Automotive (3710)
- ❑ Utilities (4900)
- ❑ Utilities — Electrical (4910)
- ❑ Utilities — Natural Gas (4920)
- ❑ Utilities — Water/Wastewater (4940)
- ❑ Construction (1700)
- ❑ Education (8200)
- ❑ Engineering Services (8710)
- ❑ Instrument & Control Apparatus Sales & Service (5000)
- ❑ Management Consulting (8740)
- ❑ Research, Development & Testing (8730)
- ❑ Systems Integration (7370)
- ❑ Other_____

2. Order Information

Quantity	Title	ISBN / Order Number	Unit Price	Total

Please send me a FREE ❑ ISA Training Catalog ❑ ISA Membership Packet ❑ InTech Subscription **Subtotal:**

3. Member Status

ISA Member #:_____ (Member # required for discount.)

❑ Please renew my ISA and Division membership for 1 year. Annual renewal fee is $85* plus $9 for each Technical Division. Renewal extends membership for an additional year.

❑ Yes, I want to join ISA and the _____ Division(s) and pay the Member price on ISA Publications. ISA Membership fee: $85*, 1 year; $158, 2 years; $225, 3 years; Division fees: $9 per Division per year

Signature:_____Date:_____

*ISA dues include a subscription to *InTech,* for which a nondeductible annual allocation of $8.71 is made for regular members and $2.55 for student members.

4. Payment Summary

Please indicate method of payment:
❑ Check (payable to ISA) ❑ Money Order ❑ Purchase Order #_____
❑ Wire Transfer (see pg. 90)
Chargecards considered pre-paid: ❑ Visa ❑ MasterCard ❑ American Express ❑ Discover
Account #_____ Expire Date:_____
Signature:_____ Date:_____

Order Subtotal.....$ _____
Sales tax.....$ _____
In NC, add 7%, In TX, add 8.25%
Shipping Charges (see pg. 90).....$ _____
Membership or Renewal Fee .....$ _____
Total Amount Due .....$ _____

Index

Title	Member	Non-member
 A Guide to the Automation Body of Knowledge, 2nd Edition pg. 4 www.isa.org/autobok	$109	$129
 A Guide to Biotechnology Law and Business pg. 32 www.isa.org/biotechnology	$60	$60
 Advanced Control Unleashed: Plant Performance Management for Optimum Benefit pg. 5 www.isa.org/advancedcontrol	$89	$99
 Advanced pH Measurement and Control, 3rd Edition pg. 5 www.isa.org/pH	$79	$89
 Advanced PID Control pg. 5 www.isa.org/advancedPID	$95	$112
 Advanced Temperature Control pg. 5 www.isa.org/advtempcontrol	$49	$59
 Alarm Management: Seven Effective Methods for Optimum Performance pg. 49 www.isa.org/alarmmanagement	$79	$89
 An Engineer's View of Human Error, 3rd Edition pg. 52 www.isa.org/engineerview	$67	$67
 Analytical Instrumentation pg. 5 www.isa.org/analyticalhandbook	$129	$159
 Applying S88: Batch Control from a User's Perspective pg. 6 www.isa.org/applyS88	$59	$69
 Automation Applications in Bio-Pharmaceuticals pg. 6 www.isa.org/biopharmaceutical	$89	$99
 Automation and Control Systems Economics, 2nd Edition pg. 73 www.isa.org/economics	$79	$89
 Automation Network Selection pg. 6 www.isa.org/networkselect	$59	$69
 The Automation, Systems, and Instrumentation Dictionary, 4th Edition (with CD-ROM) pg. 6 www.isa.org/dictionary	$82	$99

Title	Member	Non-member
 Automation Unplugged: Pinto's Perspectives, Pointers, & Prognostications pg. 8 www.isa.org/unplugged	$25	$25
 Basic and Advanced Regulatory Control: System Design and Application, 2nd Edition pg. 8 www.isa.org/regcontrol	$89	$99
 Basic Electricity and Electronics for Control: Fundamentals and Application, 3rd Edition pg. 8 www.isa.org/basicelectricity	$79	$89
 Basic Math for Process Control pg. 8 www.isa.org/basicmath	$49	$59
 Batch Control Systems — Design, Application, and Implementation, 2nd Edition pg. 7 www.isa.org/batchcontrol	$89	$99
 Boiler Control Systems Engineering pg. 8 www.isa.org/boilereng	$59	$69
 Boiler Sim MS Windows Version pg. 87 www.isa.org/boilersim	$395.00	$395.00
 Bottom-Line Automation, 2nd Edition pg. 30 www.isa.org/bottomline	$79	$89
 Bungee Jumping & Cocoons: The Dual Nature of the Industrial Marketplace pg. 73 www.isa.org/bungee	$20	$20
 Calibration: A Technician's Guide pg. 9 www.isa.org/calibrate	$75	$85
 Cascading Logic: A Machine Control Methodology for Programmable Logic Controllers pg. 9 www.isa.org/cascadelogic	$69	$79
 Certified Automation Professional (CAP) Study Guide pg. 9 www.isa.org/capguide	$29	$39
 Certified Control Systems Technician (CCST) Study Guide, Level I pg. 9 www.isa.org/ccst1	$29	$39
 Certified Control Systems Technician (CCST) Study Guide, Level II pg. 9 www.isa.org/ccst2	$29	$39

Title	Member	Non-member
 Certified Control Systems Technician (CCST) Study Guide, Level III pg. 9 www.isa.org/ccst3	$29	$39
 Certified Industrial Maintenance Mechanic (CIMM) Study Guide pg. 49 www.isa.org/cimmguide	$29	$39
 Compressed Air Operations pg. 44 www.isa.org/air	$69.95	$69.95
 Computer Control & Human Error pg. 52 www.isa.org/humanerror	$69.95	$69.95
 Condensed Handbook of Measurement and Control, 3rd Edition pg. 12 www.isa.org/MandCHandbook	$95	$109
 The Consumer Guide to Capacitance and Radar Contact Level Gauges pg. 10 www.isa.org/CRClevelgauges	$225	$250
 Consumer Guide to Coriolis Mass Flowmeters pg. 10 www.isa.org/coriolisflow	$225	$250
 Consumer Guide to Differential Pressure Flow Transmitters pg. 10 www.isa.org/dpflow	$225	$250
 Consumer Guide to Fieldbus Network Equipment for Process Control, 3rd Edition pg. 10 www.isa.org/fieldbusguide	$225	$250
 Consumer Guide to Industrial pH and ORP Instrumentation pg. 11 www.isa.org/pHandORP	$225	$250
 Consumer Guide to Magnetic Flowmeters, 2nd Edition pg. 10 www.isa.org/magneticflow	$225	$250
 Consumer Guide to Non-Contact Level Gauges pg. 11 www.isa.org/noncontact	$225	$250
 Consumer Guide to Ultrasonic and Correlation Flowmeters pg. 10 www.isa.org/ultrasonicflow	$225	$250
 Consumer Guide to Vortex Shedding and Fluidic Flowmeters pg. 10 www.isa.org/vortexflow	$225	$250

Title	Member	Non-member
 The Control of Boilers, 2nd Edition pg. 12 www.isa.org/boilercontrol	$89	$99
 Control System Documentation: Applying Symbols and Identification, 2nd Edition pg. 13 www.isa.org/symbolsandID	$79	$89
 Control Systems Engineering (CSE) Study Guide, 4th Edition pg. 12 www.isa.org/cseguide	$49	$59
 Control System Engineering Exam Reference Manual: A Practical Study Guide pg. 12 www.isa.org/csereference	$49	$59
 Control System Power and Grounding Better Practice pg. 50 www.isa.org/grounding	$49.95	$49.95
 Control System Safety Evaluation and Reliability, 2nd Edition pg. 50 www.isa.org/safetyevaluation	$89	$99
 Control Valve Primer, 4th Edition pg. 15 www.isa.org/controlvalveprimer	$89	$99
 Control Valves pg. 15 www.isa.org/controlvalvehandbook	$129	$159
 Design and Development of Medical Electronic Instrumentation: A Practical Perspective of the Design, Construction, and Test of Medical Devices see web www.isa.org/medical	$139.95	$139.95
 Design Patterns for Flexible Manufacturing pg. 7 www.isa.org/designpatterns	$89	$99
 Dispelling Chemical Engineering Myths pg. 52 www.isa.og/dispelling	$89	$89
 Dispersing Heat Through Conviction: The Funnier Side of Process Control pg. 14 www.isa.org/funnyprocess	$15	$15
 Electrical Instruments in Hazardous Locations, 4th Edition pg. 15 www.isa.org/hazardlocations	$89	$99

Index

Title	Member	Non-member
Electronic Security for Manufacturing and Control Systems—An Overview pg. 58 www.isa.org/secguide	$29	$39
Environmental Instrumentation and Analysis Handbook pg. 16 www.isa.org/environinst	$175	$175
Fieldbuses for Process Control: Engineering, Operation, and Maintenance pg. 16 www.isa.org/fieldbuses	$99	$109
FLOWEL — Flow Element Sizing and Documentation Version 4.0 for Windows pg. 87 www.isa.org/flowel	$1,895	$1,895
FLOWEL 4.0 c/w Fluid Property Package pg. 87 www.isa.org/flowelpack	$2,895	$2,895
Flow Measurement, 2nd Edition pg. 16 www.isa.org/flowmeashandbook	$129	$159
Flow of Industrial Fluids—Theory and Equations pg. 18 www.isa.org/flowfluid	$109	$119
Foundation Fieldbus, 2nd Edition pg. 60 www.isa.org/foundation	$49	$59
Fundamentals of Industrial Control, 2nd Edition (with CD-ROM) pg. 18 www.isa.org/fundamentalcontrol	$129	$159
Fundamentals of Process Control Theory, 3rd Edition (with CD-ROM) pg. 18 www.isa.org/theoryfundamentals	$92	$109
Fundamentals of Test Measurement Instrumentation pg. 18 www.isa.org/testmeasure	$89	$99
The Funnier Side of Retirement for Engineers and People of the Technical Persuasion pg. 14 www.isa.org/retirement	$25	$25
Good Tuning: A Pocket Guide, 2nd Edition pg. 19 www.isa.org/ tuningpocket	$49	$59
Guidelines for Analyzing and Managing the Security Vulnerabilities of Fixed Chemical Sites see web www.isa.org/fixedchemical	$89	$89
Hazop and Hazan pg. 53 www.isa.org/hazop	$99	$99
The Hitchhiker's Guide to Manufacturing Operations Management: ISA-95 Best Practices Book 1.0 or CD pg. 59 www.isa.org/isa95guide	$95	$109
How to Become an Instrument Engineer: The Making of a Prima Donna pg. 14 www.isa.org/primadonna	$20	$20
How to Become an Instrument Engineer, Part 1.523 pg. 14 www.isa.org/becomeIE	$20	$20
Human Factors Methods for Design: Making Systems Human-Centered pg. 74 www.isa.org/humanfactors	$119.95	$119.95
Human Machine Interface Design for Process Control Applications pg. 76 www.isa.org/HMI	$79	$89
HVAC Pump Handbook, 2nd Edition pg. 44 www.isa.org/HVAC	$125	$125
Ideal Enterprise: Managing by the Law of the Sphere pg. 76 www.isa.org/enterprise	$25	$25
IEC 61499 Function Blocks for Embedded and Distributed Control Systems Design pg. 32 www.isa.org/iec61499	$79	$89
Industrial Data Communications, 4th Edition pg. 32 www.isa.org/datacom	$89	$99
Industrial Ethernet, 2nd Edition pg. 33 www.isa.org/ethernet	$59	$69
Industrial Ethernet on the Plant Floor: A Planning and Installation Guide pg. 33 www.isa.org/industrialethernet	$59	$69
Industrial Flow Measurement, 3rd Edition pg. 19 www.isa.org/industrialflowmeas	$89	$99

Title	Member	Non-member
 Industrial Network Security pg. 33 www.isa.org/networksecurity	$54	$59
 Industrial Pressure, Level, and Density Measurement, 2nd Edition pg. 19 www.isa.org/pressureleveldensity	$89	$99
 Information Technology for Energy Managers pg. 76 www.isa.org/itenergy	$145	$145
 Information Technology for Manufacturing: Reducing Costs and Expanding Capabilities see web www.isa.org/ITmanufacturing	$69.95	$69.95
 Instrument Engineers' Handbook, 4th Edition, Vol. 1—Process Measurement and Analysis pg. 17 www.isa.org/iehvol1	$179.95	$179.95
 Instrument Engineers' Handbook, 4th Edition, Vol. 2—Process Control and Optimization pg. 17 www.isa.org/iehvol2	$179.95	$179.95
 Instrument Engineers' Handbook, 3rd Edition, Vol. 3—Process Software and Digital Networks pg. 17 www.isa.org/iehvol3	$179.95	$179.95
 Instrument Specification Forms: An ISA20 Industry Implementation pg. 83 www.isa.org/specformimplement	$595	$695
 Instrumentation and Control Systems Documentation pg. 13 www.isa.org/instdocument	$69	$79
 Instrumentation Reference Book, 3rd Edition pg. 19 www.isa.org/instref	$180	$180
 ISA Documentation Standards and User Resources for Industrial Automation and Control Systems pg. 13 www.isa.org/documentationresource	$249	$599
 ISA-5 P&ID Clip Symbols, Version 2.0 pg. 83 www.isa.org/pid	$90	$109
 ISA-20—Instrument Forms Plus LITE pg. 87 www.isa.org/rinstlite	$495	$495

Title	Member	Non-member
 ISA-20 – Instrument Forms Plus Pro pg. 88 www.isa.org/rinstfmnt	$1,395	$1,395
 ISA-20 – Instrument Loop Forms Plus Combo pg. 88 www.isa.org/rloop	$5,500	$5,500
 ISA-20 Specification Forms (Excel) pg. 83 www.isa.org/specformexcel	$245	$295
 ISA-20 Specification Forms (Excel): Developed for Use in Pharmaceutical and Similar Industry Applications pg. 83 www.isa.org/specformimplement	$595	$645
 ISA-TR20.00.01 — Specification Forms in Microsoft® Word Format, 3rd Edition pg. 83 www.isa.org/specformcd	$595	$645
 ISA-84 Process Safety Standards CD-ROM, 4th Edition pg. 84 www.isa.org/safetycd	$835	$930
 ISA-84 Process Safety Standards and User Resources pg. 51 www.isa.org/safetyresource	$299	$1199
 ISA-95 Manufacturing Enterprise Systems Standards and User Resources pg. 59 www.isa.org/mesresource	$159	$499
 ISA-99 Security Guidelines and User Resources for Industrial Automation and Control Systems pg. 54 www.isa.org/securityresource	$79	$399
 ISA-88 Batch Standards and User Resources pg. 7 www.isa.org/batchresource	$199	$499
 ISA-77 Fossil Fuel Power Plant Standards and User Resources pg. 16 www.isa.org/fossilfuelresource	$199	$599
 ISA-92 Industrial Air Measurement Standards and User Resources pg. 50 www.isa.org/industrialairresource	$249	$599
 ISA-75 Control Valve Standards and User Resources pg. 12 www.isa.org/controlvalveresource	$199	$599

Index

Title	Member	Non-member
 ISA Certified Automation Professional® Job Analysis Study pg. 21 www.isa.org/capanalysis	$0	$150
 ISA Certified Industrial Maintenance Mechanic® Job Analysis Study see web www.isa.org/cimmanalysis	$0	$150
 ISA Handbook of Measurement Equations and Tables, 2nd Edition pg. 20 www.isa.org/measurehandbook	$89	$99
 ISA Transactions — A Journal for the Science and Engineering of Measurement and Automation pg. 87 www.isa.org/isatransactions	$59	Call
 ISA Standards Library, 2008 pg. 83 www.isa.org/standardslibrary	$1,495	$1,695
 Learning from Accidents pg. 53 www.isa.org/accidents	$75	$75
 The Life and Times of an Automation Professional—An Illustrated Guide pg. 15 www.isa.org/autolife	$20	$25
 Logical Thoughts at 4:00 A.M. pg. 14 www.isa.org/logicalthoughts	$20	$20
 Loop Checking: A Technician's Guide pg. 21 www.isa.org/loopcheck	$69	$79
 Maintainability & Maintenance Management, 4th Edition pg. 68 www.isa.org/maintainability	$89	$99
 Maintenance of Instruments & Systems, 2nd Edition pg. 68 www.isa.org/IandSMaintenance	$129	$159
 Maintenance of Process Instrumentation in Nuclear Power Plants pg. 21 www.isa.org/nuclearpower	$129	$129
 Measurement and Control Basics, 4th Edition pg. 20 www.isa.org/mandcbasics	$95	$109
 Measurement Uncertainty, 4th Edition pg. 22 www.isa.org/measuncertainty	$69	$79

Title	Member	Non-member
 Mechatronics: An Integrated Approach pg. 22 www.isa.org/mechatronics	$120	$120
 The Mechatronics Handbook: The Fusion of Mechanics, Electronics, and Information Technology pg. 22 www.isa.org/mechatronicshandbook	$129.95	$129.95
 Model-Based Predictive Control: A Practical Approach pg. 22 www.isa.org/predictivecontrol	$99.95	$99.95
 Models Unleashed: Virtual Plant and Model Predictive Control Applications pg. 23 www.isa.org/modelsunleashed	$49	$59
 Motors & Drives: A Practical Technology Guide pg. 34 www.isa.org/motorsanddrives	$89	$99
 Motors & Drives: A Practical Technology Guide—Instructional Supplement pg. 34 www.isa.org/manddguide	$69	$79
 NACE Corrosion Engineer's Reference Book, 3rd Edition pg. 23 www.isa.org/corrosion	$104	$104
 New Directions in Bioprocess Modeling and Control: Maximizing Process Analytical Technology Benefits pg. 23 www.isa.org/bioprocess	$89	$99
 Out of the Barn pg. 35 www.isa.org/barn	$20	$20
 PC-ControLAB, Release 3 pg. 88 www.isa.org/pclab	$495	$495
 PID Controllers: Theory, Design, and Tuning, 2nd Edition pg. 23 www.isa.org/pidcontrollers	$89	$99
 Pinto's Points: How to Win in the Automation Business pg. 75 www.isa.org/pintopoints	$28	$30
 Plant and Machinery Failure Prevention pg. 69 www.isa.org/plantmachinery	$99.95	$99.95

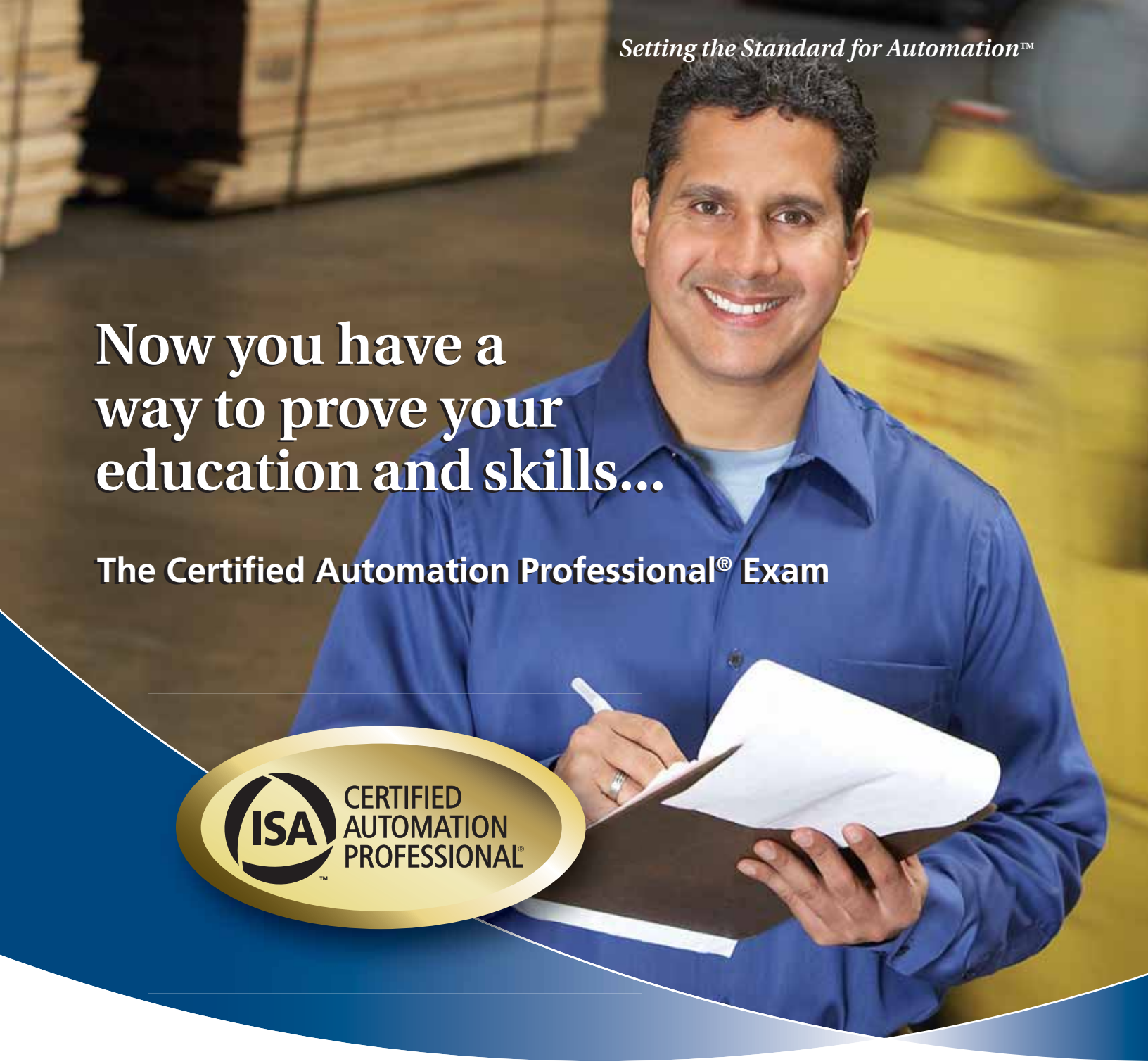
Title	Member	Non-member
 Practical Controls: A Guide to Mechanical Systems pg. 24 www.isa.org/mechsystems	$120	$120
 Practical Data Acquisition for Instrumentation for Control Systems pg. 62 www.isa.org/dataacquisition	$57.95	$57.95
 Practical Machinery Management for Process Plants: Major Process Equipment Maintenance and Repair, 2nd Edition, Volume 4 pg. 62 www.isa.org/practicalmachinery	$130	$130
 Practical Project Management: Learning to Manage the Professional pg. 78 www.isa.org/projectmanage	$59	$69
 Practical Thermocouple Thermometry pg. 24 www.isa.org/thermometry	$59	$69
 Preventive Maintenance, 3rd Edition pg. 69 www.isa.org/preventivemaintenance	$69	$79
 Process Analyzer Sample-Conditioning System Technology pg. 24 www.isa.org/processanalysis	$170	$170
 Process Control: A Primer for the Nonspecialist and the Newcomer, 2nd Edition pg. 24 www.isa.org/controlprimer	$39	$49
 Process Plants: A Handbook for Inherently Safer Designs, 2nd Edition pg. 53 www.isa.org/saferdesigns	$89	$89
 Profibus: A Pocket Guide pg. 62 www.isa.org/profibuspocket	$39	$49
 Programmable Controllers, 4th Edition pg. 24 www.isa.org/PLC	$89	$99
 Programmable Logic Controllers: An Emphasis on Design and Application pg. 24 www.isa.org/programmable	$100	$100
 The Quality Calibration Handbook – Developing and Managing a Calibration Program pg. 25 www.isa.org/qualitycalibration	$84	$84

Title	Member	Non-member
 The Road to Integration: A Guide to Applying the ISA-95 Standard in Manufacturing pg. 59 www.isa.org/integration	$95	$109
 Reliability Centered Maintenance pg. 70 www.isa.org/reliability	$89.95	$89.95
 Reading a P&ID pg. 13 www.isa.org/readpandid	$59	$69
 Safety Equipment Reliability Handbook, 3rd Edition pg. 53 www.isa.org/safetyreliability	$495	$495
For 3-Volume set:	$1350	$1350
 Safety Instrumented Systems: Design, Analysis, and Justification, 2nd Edition pg. 53 www.isa.org/sisdesign	$89	$99
 Safety Instrumented Systems Verification—Practical Probabilistic Calculations pg. 37 www.isa.org/sisverify	$89	$99
 Safety Integrity Level Selection—Systematic Methods Including Layer of Protection Analysis pg. 54 www.isa.org/safetyintegrity	$89	$99
 SCADA: Supervisory Control and Data Acquisition, 3rd Edition pg. 37 www.isa.org/scada	$79	$89
 Selection Guide for Temperature Instrumentation pg. 25 www.isa.org/temperature	$340	$340
 Sell More Through Effective Technical Presentations: A Pocket Guide pg. 78 www.isa.org/sellmore	$23	$25
 Sensor Performance and Reliability pg. 25 www.isa.org/senseandmeasure	$89	$99
 Sensor Selection Guide: Optimizing Manufacturing and Processes, 2nd Edition pg. 36 www.isa.org/sensorguide	$89	$99
 Sensor Technology Handbook pg. 36 www.isa.org/sensorhandbook	$120	$120
 Software for Automation: Architecture, Integration, and Security pg. 38 www.isa.org/autosoftware	$89	$99

Index

	Title	Member	Non-member
	Start-Up: A Technician's Guide pg. 26 www.isa.org/startup	$42	$52
	Still Going Wrong! Case Histories of Process Plant Disasters and How They Could Have Been Avoided pg. 52 www.isa.org/stillwrong	$59.95	$59.95
	Successful Instrumentation and Control Systems Design (with CD-ROM) pg. 26 www.isa.org/iandcdesignCD	$95	$109
	Troubleshooting: A Technician's Guide, 2nd Edition pg. 67 www.isa.org/troubleshoot	$59	$69
	Tuning of Industrial Control Systems, 2nd Edition pg. 26 www.isa.org/tuningsystems	$89	$99
	Understanding Distributed Processor Systems for Control pg. 26 www.isa.org/distributedcontrol	$89	$99
	Variable Speed Drives: Principles and Applications for Energy Cost Savings, 3rd Edition pg. 26 www.isa.org/vsd	$69	$79
	What Went Wrong? Case Studies of Process Plant Disasters, 4th Edition pg. 52 www.isa.org/wentwrong	$78.95	$78.95
	Wireless Networks for Industrial Automation, 3rd Edition pg. 26 www.isa.org/wirelessnetworks	$69	$79

Setting the Standard for Automation™



Now you have a way to prove your education and skills...

The Certified Automation Professional® Exam

ISA CERTIFIED AUTOMATION PROFESSIONAL®

Become an ISA Certified Automation Professional® (CAP®) and prove that you have the knowledge and skills required to be an automation leader.

For more information:
www.isa.org/CAP
CAP@isa.org
(919) 549-8411

Electronic Testing Now Available
Get certified when and where it's convenient for you.

Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

ISA™

New CDs help busy industrial automation and control professionals increase productivity



These handy tools offer easy access to a wide variety of materials with on-screen menus and keyword-searchable text, allowing users to quickly view and print information. The new formats provide **standards and user resources** in key areas such as documentation, process safety, manufacturing enterprise systems, security, and batch control.

Turn to page 70 for more details!

Founded in 1945, ISA (www.isa.org) is a leading, global, nonprofit organization that is setting the standard for automation by helping over 30,000 worldwide members and other professionals solve difficult technical problems, while enhancing their leadership and personal career capabilities. Based in Research Triangle Park, North Carolina, ISA develops standards; certifies industry professionals; provides education and training; publishes books and technical articles; and hosts the largest conference and exhibition for automation professionals in the Western Hemisphere. ISA is the founding sponsor of The Automation Federation (www.automationfederation.org)

Customer Satisfaction Guarantee

Your satisfaction with your ISA publication or standard is important to us. If you do not find the product satisfies your needs, please return it within 30 days of purchase—we will refund your purchase price less the cost of shipping and handling. Please return the product in saleable condition. For more details about ISA's return policies, see www.isa.org/shopisa/about.

40-2020-018

ISA
PO Box 12277
67 Alexander Drive
Research Triangle Park, NC 27709

Nonprofit Org.
U.S. Postage
PAID
Lebanon Junction, KY
Permit #192